CFA LEVEL I / FIXED INCOME

固定收益

中英对照速学与原题详解

Learn the concepts. Work the original questions.

106	8	38
原题 / Original questions	学习章节 / Learning chapters	分步计算题 / Worked calculations

学习顺序：知识点讲解 → 原题示范 → 每个选项解析 → 公式与错题复盘。

Study sequence: concept lesson → original worked example → option-by-option explanation → formula and error review.

覆盖所给题库固定收益全部 106 题。原题英文、数字和选项保留，中文紧随对应内容；仅按知识点重新编排。

Covers all 106 fixed-income questions in the supplied bank. Original English, figures and options are retained, with aligned Chinese translations and a new concept-based order.

> 快速使用：先按两小时建议路线学习，再完成剩余原题。答案紧跟每题，复习时可遮住答案区域。
>
> Fast use: follow the suggested two-hour route, then complete the remaining originals. Answers follow each question; cover the solution area for self-testing.

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资料年份与题目来源身份未注明；本讲义不把它标成已核实的官方题或某一年度完整考纲。

The supplied files do not establish a curriculum year or verified question provenance; the guide does not label them as confirmed official questions or a complete annual syllabus.

资料与编排说明

Sources and editorial approach

题目来源：1000题(20261103-062907).pdf，共 202 页；本讲义只提取其第 69–90 页的“固定收益”部分，原题 Q1–Q106。

Question source: 1000题(20261103-062907).pdf, 202 pages. This guide uses only the Fixed Income section on physical pages 69–90, original Q1–Q106.

答案来源：固定收益(2).pdf，共 106 页。该文件以扫描图像为主，答案第 n 页与原题 Qn 配对。已逐题识别答案并核对；计算题另行复算。

Solution source: 固定收益(2).pdf, 106 pages, predominantly scanned images. Solution page n maps to original Qn. Answers were matched question by question and numerical examples independently recalculated.

年份与来源标签：文件未明确标注教材年份，且没有足够可核验信息确认全部题目属于官方题还是模拟题。因此统一标为“用户提供原题 / 来源类型未核实”。文件名中的日期和 PDF 时间戳不作为教材年份。

Year and provenance: the files do not clearly identify a curriculum year or provide sufficient verification to label every item as official or mock. They are labeled user-provided originals, with provenance unverified. Filename dates and PDF timestamps are not treated as curriculum years.

单元名称：原答案大部分仅标注 Fixed Income，Q106 标注 Fixed-Income Bond Valuation: Prices and Yields。本讲义建立 8 个教学章节，属于学习重组，不冒称原资料已给出官方 Learning Module 编号。

Unit names: most solution pages state only Fixed Income; Q106 states Fixed-Income Bond Valuation: Prices and Yields. The eight chapters are a teaching reorganization, not invented official Learning Module numbers.

保留与增加的内容

What is preserved and added

保留 / Preserved	增加 / Added
全部 106 道英文原题、318 个选项及32个数据表。 All 106 English questions, 318 options and 32 data tables.	逐项中文翻译，公式口径与术语解释。 Aligned Chinese translation, notation and terminology.
原题编号、原答案字母、题库与答案页码。 Original IDs, source answer keys and source page references.	每题对应解析，两个干扰项的排除理由。 Paired solutions and explanations of both distractors.
原题的数字、单位及可识别的措辞。 Original figures, units and identifiable wording.	38题分步计算；15题附资料或条件提示。 Step-by-step calculations for 38 items and source/assumption notes for 15.

遇到原文矛盾、缺失条件或答案排版笔误时，保留原题并给出提示。Q51 表格前孤立的“<”视为排版残留移除；Q35 的英文“3yaers”保留，并译为其明确含义“3 年”。

Contradictions, missing assumptions and solution typos are flagged without silently changing the question. The isolated “<” before Q51’s table is removed as a formatting remnant; Q35’s English “3yaers” is retained and translated as three years.

两小时建议学习路线

Suggested two-hour first pass

目标是先建立解题框架，不是在两小时内熟练全部106题。先看各章知识讲解，再做下列30道原题作为检查；剩余题用于巩固同类规则。

The first pass builds a framework; it does not promise mastery of all 106 questions in two hours. Read the chapter lessons, then use the 30 originals below as checkpoints. Use the rest for consolidation.

阶段 / Stage	建议用时 / Time	内容与原题 / Content and originals
01	20 分钟 / 20 min	基本条款与估值 / Features and pricing Q1, Q63, Q67, Q106, Q2, Q35
02	15 分钟 / 15 min	曲线与远期 / Curves and forwards Q59, Q73, Q79, Q93
03	35 分钟 / 35 min	久期与凸性 / Duration and convexity Q48, Q25, Q55, Q56, Q78, Q4, Q12, Q75
04	20 分钟 / 20 min	实际回报 / Realized return Q66, Q103, Q91, Q99
05	15 分钟 / 15 min	信用与清偿 / Credit and priority Q30, Q51, Q18, Q89
06	15 分钟 / 15 min	证券化 / Securitization Q8, Q45, Q71, Q41

做题时只记录五类错误

Record just five error types

错因 / Error type	复盘动作 / Review action
概念 / Concept	用一句话说清概念差异。 Explain the distinction in one sentence.
公式 / Formula	写出适用条件与每个变量含义。 Write the conditions and variable meanings.
口径 / Convention	检查年化、每期、净价/全价、市值/面值。 Check annual/periodic, clean/full and market/par conventions.
时间 / Timing	画时间轴，重数剩余期数。 Draw the timeline and recount remaining periods.
审题 / Reading	圈出 most likely、closest to 和问题要求。 Identify qualifiers and the exact requested quantity.

复习顺序：先独立作答 → 再看答案 → 解释错误选项 → 遮住解析重做。猜对的题也记入复盘，直到能说明为什么另外两个选项不成立。

Review sequence: answer independently, check the key, explain the distractors and retry with the solution covered. Review guessed-correct answers until you can explain why both alternatives fail.

目录与使用索引

Contents and navigation

章节 Chapter	题数 Count	起始页 Page
读题基础 / Essential reading rules	—	5
01 债券是什么：条款、现金流与交易 Bond features, cash flows and trading	15	6
02 债券估值与收益率：把时间和口径对齐 Valuation and yields: align timing and conventions	14	25
03 利率曲线、远期利率与利差 Yield curves, forward rates and spreads	8	42
04 久期：选对指标，再算敏感度 Duration: choose the measure, then calculate sensitivity	27	53
05 凸性：为什么涨跌不对称 Convexity and asymmetric price changes	10	83
06 持有期回报、再投资风险与久期缺口 Holding-period return, reinvestment risk and duration gap	9	95
07 信用风险、偿债能力与清偿顺序 Credit risk, debt service and payment priority	13	106
08 证券化、抵押贷款与结构化分层 Securitization, mortgages and tranching	10	121
公式速查 Formula reference	—	133
原资料提示汇总 Source and assumption notes	—	136
原题编号、答案与页码 Question-to-answer page index	—	139
复盘与掌握检查 Review and mastery checklist	—	143

每章先给教学内容，再按知识点安排原题；每道题占独立页面，题目、答案、解析与资料提示相互紧邻。纯数字选项由中英文共用，避免重复占行。左侧 PDF 书签可直接跳到章节、题组和单题。

Each chapter begins with a lesson and groups original questions by concept. Each question occupies its own page with its answer, explanation and source notes together. Numerical-only options are shared between languages. PDF bookmarks navigate to chapters, groups and individual questions.

公式中 m 为每年复利或付息次数，y 为与其匹配的名义年率；除非明确写成百分比，代入计算的利率均用小数。上下情景的 delta_y 都指单边幅度。

In formulas, m is the annual compounding or payment frequency and y is the matching nominal annual rate. Use decimal rates in calculations unless a percentage is explicitly shown. Scenario delta_y denotes a one-sided change.

读题基础

Essential reading rules

先掌握 6 条读题规则

Six rules before solving questions

1. 最可能（most likely）、最准确（most accurate）和最接近（closest to）是在现有选项中判断；不要替题目添加未给出的市场条件。

1. Most likely, most accurate and closest to ask you to choose among the supplied options. Do not add market conditions that the question does not provide.

2. 1% = 0.01；1 个基点（bp）= 0.01% = 0.0001。75 bp 应输入 0.0075，10 bp 应输入 0.001。

2. 1% = 0.01 and 1 basis point (bp) = 0.01% = 0.0001. Enter 75 bp as 0.0075 and 10 bp as 0.001.

3. 半年付息时：期数乘 2，年票息金额除以 2，按半年复利报价的名义年 YTM 除以 2。若给出 EAR，半年利率为 (1+EAR)^(1/2)−1。最后报价是否年化，另看题目。

3. Semiannual payment changes three inputs: double the number of periods, halve the annual coupon and halve a nominal annual yield quoted on that basis. If EAR is given instead, the half-year rate is (1+EAR)^(1/2)−1. Check separately whether the requested answer is annualized.

4. “每 100 面值”是价格报价单位；“整笔持仓金额”是货币单位。给出总面值时，按面值/100 把单位价格转换为总市值。

4. A price per 100 par is a quotation unit, whereas a full holding value is a monetary amount. Multiply a unit price by total par/100 to obtain the position value.

5. 阅读表格时先圈变量：YTM 变动、benchmark curve 变动、current yield、clean/full price。这些词决定公式口径。

5. First identify the table’s variables: YTM change, benchmark-curve change, current yield and clean/full price. These words determine the correct formula.

6. 本讲义保留原题编号；练习按知识点重排，所以正文题号不连续。题目数据不补写、不改数，讲解中的算式仅推导原题。

6. Original question numbers are retained. They are non-consecutive in the guide because questions are grouped by concept. No missing question data or options are invented; worked calculations derive from the supplied originals.

你应当能够说出的核心句子

Core statements you should be able to explain

中文 / Chinese	English
利率上升，固定现金流债券价格通常下降。	A rise in yield generally lowers the price of fixed cash flows.
票息率决定付款，YTM 连接价格与付款。	The coupon rate sets payments; YTM links payments to price.
久期看一阶敏感度，凸性补偿弯曲。	Duration captures first-order sensitivity; convexity adds curvature.
持有到期仍可能有票息再投资风险。	Holding to maturity does not eliminate coupon reinvestment risk.
信用损失同时看违约概率和违约损失率。	Credit loss depends on both default probability and loss given default.
证券化分层可以分配损失，也可以分配收款时间。	Securitization tranches can allocate losses or repayment timing.

CHAPTER 01 / 15 ORIGINAL QUESTIONS

债券是什么：条款、现金流与交易

Bond features, cash flows and trading

> 看到一种债券，先说清谁付款、何时付款、付多少、谁拥有选择权。
> Identify who pays, when payments occur, how much is paid and who controls any option.

先做这些原题检验理解：Q1, Q63, Q67, Q61。
Start with these original checkpoints: Q1, Q63, Q67, Q61.

1.1 用现金流理解债券

1.1 Understand a bond through its cash flows

买债券就是今天付出价格，换取未来约定的现金流。发行人是借款人，债券持有人是出借人。面值是合同中的本金金额；市场价格是今天买卖的金额，两者可以不同。
Buying a bond exchanges a price today for contractual future cash flows. The issuer borrows and the bondholder lends. Par value is contractual principal; market price is the amount paid today, which can differ from par.

术语 / Term	含义 / Meaning
面值 / Par value	到期应偿还的本金基数。 The contractual principal amount.
票息率 / Coupon rate	按面值计算的年利息比例。 Annual interest as a percentage of par.
票息 / Coupon payment	每次实际支付的利息。 The cash interest paid on each payment date.
到期期限 / Time to maturity	距离最终本金偿还的时间。 Time until final principal repayment.
收益率 / Yield	把价格与未来现金流联系起来的回报率。 A return measure linking price with future cash flows.

$$C = \frac{F\,c}{m}$$

m 为每年的付息次数。原题 Q106：面值 100、票息率 12%、半年付息，每期票息为 6，共 6 期，最后一期现金流为 106。
m is the number of coupon payments per year. In Q106, par is 100, the annual coupon rate is 12% and payments are semiannual: each coupon is 6, there are six periods, and the final cash flow is 106.

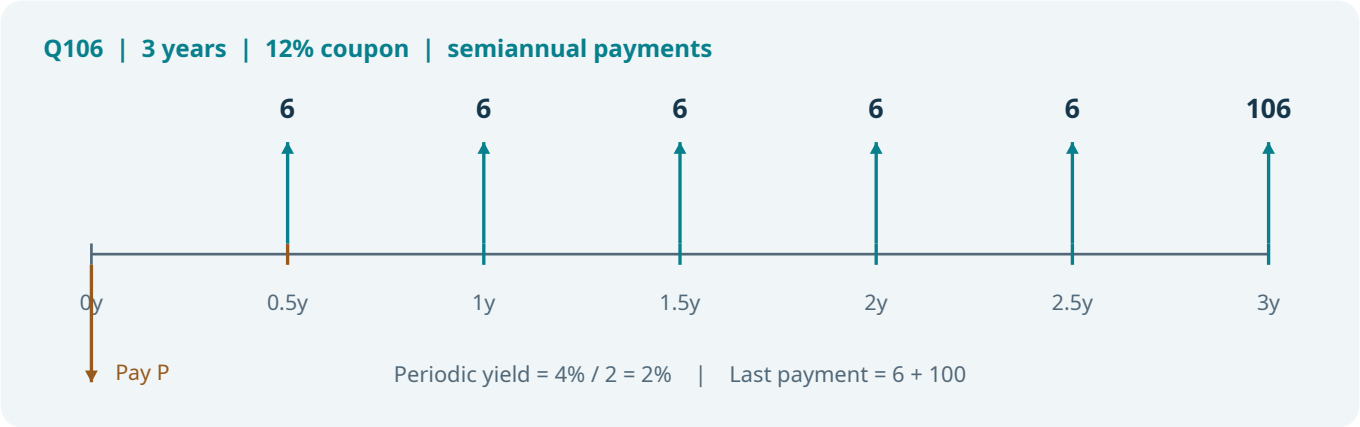


图 1：用原题 Q106 的数字画现金流，先画时间，再填金额。

Figure 1. Cash-flow timeline using the original Q106 data: mark dates before entering amounts.

1.2 四种容易混淆的现金流结构

1.2 Four easily confused cash-flow structures

结构 / Structure	识别方式 / Recognition rule
偿债基金 / Sinking fund	发行人必须按安排逐步偿还部分本金。Q1 The issuer must retire specified portions of principal. Q1
递增票息 / Step-up coupon	票息率按预定规则上升。 The coupon rate rises according to a predetermined schedule.
递延票息 / Deferred coupon	前期不付或延迟支付现金票息。 Cash coupons are delayed or absent during an initial period.
实物支付票息 / Payment-in-kind (PIK)	以增加债务等形式支付利息，而非当前现金。Q6 Interest is paid by adding debt rather than current cash. Q6

偿债基金强调“必须偿还”；可赎回条款强调发行人“可以赎回”。不能因为两者都可能减少未偿债务，就把它们当成同一种条款。

A sinking fund creates a repayment obligation; a call provision gives the issuer an option. Both may reduce outstanding debt, but they are not the same contractual feature.

1.3 谁拥有期权，谁获益

1.3 Identify who benefits from the option

条款 / Provision	持有人、价值与收益率 / Holder, value and yield
可赎回 / Callable	发行人可提前还款；投资者让出权利，其他条件相同通常要求更高收益率。Q61 The issuer may redeem early; investors give up an option and generally require a higher yield. Q61
可回售 / Putable	投资者可按条款卖回发行人；通常增加投资者价值。 The investor may sell the bond back under the contract, which generally adds investor value.
可转换 / Convertible	投资者可换成普通股；转换比率是可换取的股数。Q70 The investor may convert into common shares; the conversion ratio is the number of shares. Q70

$$\text{Conversion ratio} = \frac{\text{Par value}}{\text{Conversion price}}$$

转换价值 = 转换比率 × 当前股价。不要把“股数”“每股转换价格”“可换股票的市值”混为一谈。

Conversion value equals the conversion ratio times the current share price. Distinguish the share count, the conversion price per share and the market value of shares received.

限制性条款（negative covenant）约束发行人不能做什么，例如限制再发行更高顺位债务。肯定性条款（affirmative covenant）要求发行人做什么，例如遵守法律或提供报表。Q49 要识别约束性质。

A negative covenant restricts issuer actions, such as issuing debt that ranks ahead of existing creditors. An affirmative covenant requires actions, such as complying with laws or providing financial statements. Q49 tests the nature of the restriction.

1.4 浮息债：先看重设时点

1.4 Floating-rate notes: locate the reset date

```
Coupon rate for the next period = Reference rate at reset + Quoted margin
```

浮息债 FRN 的合同加点叫 quoted margin；required margin 是市场目前要求的信用或流动性补偿。二者不必相等。Q98。

An FRN contract specifies a quoted margin. The required margin is the spread currently demanded by the market for credit or liquidity risk. They need not be equal. Q98.

> 原题 Q63：12 月付息对应此前 6 月已经确定的利率。因此年化票息率 = 1.95% + 0.45% = 2.40%，选 A。题目问票息率，不是把 2.40% 再除以 2 后的半年现金利息比例。
>
> Original Q63: the December coupon uses the rate fixed in June. The annualized coupon rate is 1.95% + 0.45% = 2.40%, answer A. The question asks for the coupon rate, not the half-year cash-interest fraction.

1.5 净价、全价与市场分类

1.5 Clean price, full price and market classifications

```
Full price = Clean price + Accrued interest
```

净价也叫 clean/flat price；全价也叫 full/dirty/invoice price。Q67 的常规结论是按净价报价、按全价结算。Q68：应计利息由票息和经过的计息时间决定，不由市场折现率决定。

Clean price is also called flat price. Full price is also called dirty or invoice price. Q67 uses the standard convention: quote clean, settle full. In Q68, accrued interest depends on the coupon and elapsed accrual time, not the market discount rate.

品种 / Instrument	速记 / Fast distinction
担保债券 / Covered bond	通常可追索发行人及独立的担保资产池。Q7 Typically provides recourse to the issuer and a segregated cover pool. Q7
商业票据 / Commercial paper	短期企业融资；本题库强调备用信贷额度。Q22 Short-term corporate funding; the supplied question emphasizes a backup credit line. Q22
回购协议 / Repo	出售证券并约定购回，本质上是有担保融资；初始保证金只提供部分保护。Q43 A security sale with agreed repurchase provides collateralized financing; initial margin offers partial protection. Q43

品种 / Instrument	速记 / Fast distinction
欧洲债券 / Eurobond	不表示只能用欧元计价；Q28 的指定答案是记名形式。 The name does not restrict denomination to euros; Q28 specifies registered form as its answer.
外国债券 / Foreign bond	境外发行人进入某本地市场，并以该市场货币发行。Q62 A foreign issuer issues in a domestic market and its domestic currency. Q62
市政债 / Municipal bond	属于非主权政府债券。Q77 A non-sovereign government bond. Q77

先练 Q1、Q63、Q67、Q61。能用自己的话解释“偿债义务 / 重设时点 / 结算金额 / 期权归属”后，再完成本章其余题。

Start with Q1, Q63, Q67 and Q61. Explain the repayment obligation, reset date, settlement amount and option ownership before completing the remaining questions.

现金流和合约条款 / **Cash flows and contractual terms**

Q001 | 原题 / Original question

偿债基金条款：强制分期偿还本金

Sinking fund provision: mandatory scheduled principal retirement

来源：题库 PDF 第 69 页（固收第 1 页）；答案 PDF 第 1 页。

Source: question bank p. 69 (Fixed Income p. 1); solution PDF p. 1.

如果发行人被要求每年偿还债券本金的一个指定部分，该债券最可能：

If an issuer is required to retire a specified portion of the bond's principal each year, the bond most likely:

A. 可由发行人赎回。
is callable.

B. 属于递增票息票据。
is a step-up note.

C. 设有偿债基金条款。
has a sinking fund provision.

正确答案 / Answer: C

关键词是“被要求”及“每年偿还部分本金”。偿债基金条款规定发行人按约定计划逐步偿还债务，而不必等到到期日一次偿还全部本金。

The clues are “required” and “a specified portion each year.” A sinking fund imposes a scheduled retirement of principal, reducing the amount left to repay at final maturity.

错误选项为什么不成立 / Why the other options fail

A. 赎回权给予发行人提前偿还的选择权，通常不是逐年必须履行的义务。

A. A call gives the issuer an option to repay early; it does not by itself require annual principal retirement.

B. 递增票息改变票息率，不能说明本金必须逐年偿还。

B. A step-up note changes the coupon rate, not the required principal repayment schedule.

Q006 | 原题 / Original question

PIK：用增加债务余额支付票息

Payment in kind: interest paid by adding debt

来源：题库 PDF 第 69 页（固收第 1 页）；答案 PDF 第 6 页。

Source: question bank p. 69 (Fixed Income p. 1); solution PDF p. 6.

一只债券允许发行人通过增加现有债券发行额来支付利息，而不是支付现金，这最符合：

A bond that allows the issuer to pay interest in the form of additional amounts of the existing bond issue rather than a cash payment best describes a:

A. 递增票息债券。
step-up coupon bond.

B. 递延票息债券。
deferred coupon bond.

C. 实物付息（PIK）债券。
payment-in-kind coupon bond.

正确答案 / Answer: C

PIK 的核心是利息并非立即以现金支付，而是通过增加债券或债务余额支付。投资者得到更多债权，发行人当期现金支出较少，但未来偿还义务增加。这里的“实物”不是普通商品，而是额外债务权益。

PIK interest is paid by issuing additional debt or increasing the debt balance rather than paying cash immediately. The investor receives a larger debt claim, while the issuer conserves current cash and increases its future repayment obligation. “In kind” here refers to additional debt, not ordinary physical goods.

错误选项为什么不成立 / Why the other options fail

A. 递增票息债券强调票息率按约定提高，不能说明以债代息。

A. A step-up coupon describes scheduled increases in the coupon rate, not payment with additional debt.

B. 递延票息债券强调初期暂停付息、之后开始付息，不等同于增加债务来付息。

B. A deferred coupon postpones early coupon payments; it is distinct from paying interest with additional debt.

Q049 | 原题 / Original question

限制性契约与肯定性契约

Negative and affirmative covenants

来源：题库 PDF 第 77 页（固收第 9 页）；答案 PDF 第 49 页。

Source: question bank p. 77 (Fixed Income p. 9); solution PDF p. 49.

以下哪项最可能是限制性契约？

Which of the following is most likely a negative covenant?

A. 发行人必须遵守全部法律法规。
The issuer must comply with all laws and regulations.

B. 新发行债务的清偿顺序必须次于现有债券持有人债务。
The issuance of new debt must be junior to existing bondholder debt.

C. 新债务与借款人的其他优先债务享有同等地位。
New debt obligations are treated the same as the borrower's other senior debt instruments.

正确答案 / Answer: B

限制性契约约束发行人“不能做什么”或对某些行为设限。要求新债必须处于现有债务之后，等于限制发行同等或更优先债务，保护现有债券持有人。肯定性契约则要求发行人履行某种义务，例如依法经营和维持约定的债务地位。

A negative covenant restricts actions the issuer may take. Requiring new debt to rank below existing debt limits issuance of equal- or higher-priority claims and protects current bondholders. Affirmative covenants require the issuer to fulfill obligations, such as complying with laws or maintaining an agreed status.

错误选项为什么不成立 / Why the other options fail

A. 遵守法律是必须履行的义务，属于肯定性契约。

A. Compliance with laws is a required action, an affirmative covenant.

C. 这是同等受偿地位的承诺，题库归为肯定性契约；它没有禁止发行更优先债务。

C. This is a promise of equal ranking, classified here as affirmative; it is not the restriction on senior issuance described in B.

Q061 | 原题 / Original question

发行人受益的期权需要补偿投资者

Issuer-benefiting options require compensation to investors

来源：题库 PDF 第 80 页（固收第 12 页）；答案 PDF 第 61 页。

Source: question bank p. 80 (Fixed Income p. 12); solution PDF p. 61.

与其他条件相同的无嵌入期权债券相比，投资者对具有以下哪项条款的公司债要求更高收益率？

Compared to an otherwise similar option-free bond, investors require a higher yield for a corporate bond with a:

A. 回售条款。
put provision.

B. 赎回条款。
call provision.

C. 转换条款。
conversion provision.

正确答案 / Answer: B

赎回权属于发行人：当利率下降时，发行人可提前还本再以低息融资，投资者则失去高票息并面临低息再投资。因此，可赎回债券通常需要较高收益率来补偿投资者。回售权和转股权给投资者额外选择权，通常增加债券价值、降低所需收益率。

A call option belongs to the issuer. When rates fall, the issuer may repay early and refinance cheaply, while investors lose high coupons and must reinvest at lower rates. Callable bonds therefore generally require a higher yield. Put and conversion options give investors additional rights, typically increasing bond value and reducing required yield.

错误选项为什么不成立 / Why the other options fail

A. 回售权保护投资者，通常降低其要求的收益率。

A. A put protects investors and generally lowers the yield they require.

C. 转股权给投资者参与股价上涨的机会，通常降低所需收益率。

C. Conversion rights provide equity upside and generally reduce required yield.

Q070 | 原题 / Original question

可转债的三个转换术语

Three conversion terms

来源：题库 PDF 第 82 页（固收第 14 页）；答案 PDF 第 70 页。

Source: question bank p. 82 (Fixed Income p. 14); solution PDF p. 70.

一只可转换债券能够换取的普通股股数称为：

The number of common shares a convertible bond can be converted into is the:

A. 转换比率。
conversion ratio.

B. 转换价格。
conversion price.

C. 转换价值。
conversion value.

正确答案 / Answer: A

转换比率表示每只债券可换成多少股普通股。转换价格是为取得每股股票而隐含支付的价格，通常等于债券面值除以转换比率。转换价值则是当前换成股票能获得的市场价值，等于转换比率乘以当前股价。题目问的是股数，所以选转换比率。

The conversion ratio is the number of common shares obtainable for each bond. The conversion price is the implied price paid per share, generally par value divided by the conversion ratio. Conversion value is the current market value of the shares received, equal to the conversion ratio times the current share price. The question asks for the number of shares, so conversion ratio is correct.

错误选项为什么不成立 / Why the other options fail

B. 转换价格衡量每股价格，不是股数。

B. Conversion price is a price per share, not a share count.

C. 转换价值衡量转换后股票的市场金额，不是股数。

C. Conversion value is the market value of the shares, not their number.

浮息债和报价 / **Floating-rate notes and price conventions**

Q063 | 原题 / Original question

浮息票据：期初重设、期末付息

Floating-rate notes: reset at the start, pay at the end

来源：题库 PDF 第 81 页（固收第 13 页）；答案 PDF 第 63 页。

Source: question bank p. 81 (Fixed Income p. 13); solution PDF p. 63.

一只浮息票据每半年支付一次利息，票息率等于六个月市场参考利率加 45 个基点，付息月份为 6 月和 12 月。如果同年 6 月的六个月市场参考利率为 1.95%，12 月为 2.25%，则该年 12 月支付的票息所使用的票息率最接近多少？

A floating-rate note makes semiannual interest payments and has a coupon rate equal to the six-month market reference rate plus 45 basis points. The interest payments are made in June and December. If the six-month market reference rate was 1.95% in June and 2.25% in December of the same year, the coupon rate paid in December of that year was closest to:

A. 2.40%.

B. 2.55%.

C. 2.70%.

正确答案 / Answer: A

标准浮息票据在计息期开始时确定该期票息，在期末支付。12 月支付的是 6 月至 12 月期间的利息，所以用 6 月参考利率 1.95% 加 0.45%，等于年化票息率 2.40%。12 月的 2.25% 用于下一计息期。题目问票息率，不是半年的实际付息金额。

A standard FRN sets the rate at the start of an accrual period and pays interest at the end. The December payment relates to June-December, so use June's 1.95% reference rate plus 0.45%: an annualized coupon rate of 2.40%. December's 2.25% reference rate sets the next period. The question asks for the coupon rate, not the semiannual cash amount.

1. 选取计息期起点的参考利率。

Use the reference rate at the start of the accrual period.

```
December coupon rate = 1.95% + 0.45% = 2.40%
```

错误选项为什么不成立 / Why the other options fail

B. 错误地将 6 月和 12 月参考利率取平均后再加利差。

B. This incorrectly averages the June and December reference rates before adding the spread.

C. 错误地用付款日的 12 月参考利率确定已经结束计息期的票息。

C. This uses December's payment-date reference rate for an accrual period already completed.

Q098 | 原题 / Original question

浮息债：报价利差是合同加点，要求利差是市场要求

FRNs: quoted margin is contractual; required margin is market-determined

来源：题库 PDF 第 88 页（固收第 20 页）；答案 PDF 第 98 页。

Source: question bank p. 88 (Fixed Income p. 20); solution PDF p. 98.

对于浮动利率票据，高于参考利率的约定收益率利差，最恰当的名称是：

For a floating-rate note, the specified yield spread over the reference rate best defines the:

A. 票息。
coupon

B. 报价利差。
quoted margin.

C. 要求利差。
required margin.

正确答案 / Answer: B

合同事先写定、加到参考利率上的部分叫quoted margin。浮息票息率由参考利率加上报价利差构成；required margin是市场根据当前信用、流动性等条件要求的利差，会变化。

The contractually specified spread added to the reference rate is the quoted margin. The floating coupon rate combines the reference rate and the quoted margin. The required margin is the spread demanded by the market for current credit, liquidity, and other conditions and may change.

错误选项为什么不成立 / Why the other options fail

A. 票息率包含参考利率和加点，并不只指加点本身。

A. The coupon rate includes both the reference rate and the margin, not just the spread.

C. 要求利差由当前市场决定，不是合同预先约定的加点。

C. The required margin is determined by current market conditions rather than fixed in the contract.

Q067 | 原题 / Original question

净价报价，全价结算

Quote clean; settle dirty

来源：题库 PDF 第 82 页（固收第 14 页）；答案 PDF 第 67 页。

Source: question bank p. 82 (Fixed Income p. 14); solution PDF p. 67.

债券通常使用以下哪种方式报价与结算？

Bonds are quoted using the:

A. 按全价报价，按净价结算。
full price and settled using the flat price.

B. 按净价报价，按全价结算。
flat price and settled using the full price.

C. 按“脏价”报价，按发票价格结算。
"dirty" price and settled using the invoice price.

正确答案 / Answer: B

净价又称 clean price 或 flat price，不含应计利息；全价又称 dirty price、full price 或 invoice price，等于净价加应计利息。通常报价展示净价，实际买方支付全价，以补偿卖方在本付息期内已赚取的利息。

Clean price, also called flat price, excludes accrued interest. Full price, dirty price and invoice price refer to the price including accrued interest: full price = clean price + accrued interest. Standard quoting uses the clean price, while settlement pays the full price to compensate the seller for interest accrued during the coupon period.

错误选项为什么不成立 / Why the other options fail

A. 把报价和结算价格的用途颠倒了。

A. This reverses the quoted and settlement prices.

C. dirty price 与 invoice price 是全价的同义词，未正确表述净价报价的惯例。

C. Dirty price and invoice price are both full-price terms; this misses the clean-price quoting convention.

Q068 | 原题 / Original question

应计利息取决于票息和经过的时间

Accrued interest depends on the coupon and elapsed time

来源：题库 PDF 第 82 页（固收第 14 页）；答案 PDF 第 68 页。

Source: question bank p. 82 (Fixed Income p. 14); solution PDF p. 68.

以下哪项不取决于市场贴现率？债券的：

Which of the following does not depend on the market discount rate? A bond's:

A. 净价。
flat price

B. 全价。
full price

C. 应计利息。
accrued interest

正确答案 / Answer: C

应计利息按票息金额、距上一付息日经过的天数以及计息规则计算，而不是用市场收益率贴现得出。全价是未来现金流的现值，会随市场贴现率变化；净价等于全价减应计利息，因此也随市场贴现率变化。

Accrued interest is determined by the coupon, time elapsed since the last coupon date and the day-count convention; it is not calculated by discounting at the market yield. Full price is the present value of future cash flows and changes with the discount rate. Clean price equals full price minus accrued interest and also changes with the discount rate.

错误选项为什么不成立 / Why the other options fail

A. 净价 = 全价 - 应计利息，仍会随贴现率改变。

A. Clean price equals full price minus accrued interest and remains discount-rate sensitive.

B. 全价直接由现金流贴现得到，依赖市场贴现率。

B. Full price is obtained by discounting cash flows and depends directly on the discount rate.

发行市场、担保与融资 / **Markets, security and financing**

Q007 | 原题 / Original question

担保债券的双重追索权

Dual recourse of covered bonds

来源：题库 PDF 第 70 页（固收第 2 页）；答案 PDF 第 7 页。

Source: question bank p. 70 (Fixed Income p. 2); solution PDF p. 7.

发生违约时，担保债券（covered bonds）的投资者最可能有权向下列哪一方追索？

In the event of default, investors in covered bonds most likely have recourse against:

A. 仅发行人。
the issuer only.

B. 仅一个独立划分的资产池。
a segregated pool of assets only.

C. 发行人及一个独立划分的资产池。
both the issuer and a segregated pool of assets.

正确答案 / Answer: C

Covered bond 的考点是双重追索：债券持有人对发行人拥有债权，同时受到专门划分的担保资产池保护。不要仅因为存在资产池，就把它当作只能追索资产池的证券化产品。

The defining feature tested is dual recourse: bondholders have a claim on the issuer as well as protection from a segregated cover pool. The presence of an asset pool does not mean their recourse is limited to that pool.

错误选项为什么不成立 / Why the other options fail

A. 遗漏了专门担保资产池提供的保护。

A. This omits recourse to the segregated cover pool.

B. 遗漏了对发行人的追索权。

B. This omits recourse to the issuer.

Q022 | 原题 / Original question

商业票据与备用银行信贷额度

Commercial paper and backup credit facilities

来源：题库 PDF 第 72 页（固收第 4 页）；答案 PDF 第 22 页。

Source: question bank p. 72 (Fixed Income p. 4); solution PDF p. 22.

下列哪项表述最准确？美国商业票据通常：

Which of the following statements is most accurate? US commercial paper typically:

A. 要求发行人提供抵押品。
requires the issuer to pledge collateral.

B. 要求发行人拥有备用信贷额度。
requires the issuer to have a backup line of credit.

C. 期限从几天到两年不等。
has a maturity ranging from a few days up to two years.

正确答案 / Answer: B

题目强调商业票据的再融资风险：发行人可能无法续发票据，于是需要备用银行信贷额度来支持到期兑付。普通商业票据通常是无担保短期债务；本题所附答案以美国商业票据通常不超过 270 天的期限说明 C 错误。

The question tests refinancing risk: an issuer may be unable to roll over its paper and needs a backup bank credit facility to support repayment. Ordinary commercial paper is typically short-term unsecured debt. The supplied solution explains that the usual U.S. maturity is no more than 270 days, rather than two years.

错误选项为什么不成立 / Why the other options fail

A. 普通商业票据通常无担保；不要把它与资产支持商业票据混淆。

A. Ordinary commercial paper is generally unsecured; do not confuse it with asset-backed commercial paper.

C. 两年超出本题所述美国商业票据的通常期限。

C. Two years exceeds the usual U.S. commercial-paper maturity described in the supplied solution.

Q028 | 原题 / Original question

Eurobond 的名称、发行市场与登记形式

Eurobond terminology, issuance market, and registration

来源：题库 PDF 第 73 页（固收第 5 页）；答案 PDF 第 28 页。

Source: question bank p. 73 (Fixed Income p. 5); solution PDF p. 28.

在欧洲债券（Eurobond）市场发行的债券最可能：

Bonds issued in the Eurobond market are most likely:

A. 只以欧元计价。
denominated only in euros.

B. 采用记名债券形式。
in the form of registered bonds.

C. 在发行人本国的司法管辖区内发行。
issued within the jurisdiction of the issuer's home country.

正确答案 / Answer: B

按本题所附答案，正确项为 B：记名债券形式。Eurobond 中的 “Euro” 不是指一定用欧元，也不是只在欧洲发行；它指国际离岸债券市场。识别题目时，应将计价货币、发行市场和登记形式分开判断。

The supplied answer selects B, registered form. “Euro” in Eurobond does not mean that the bond must be denominated in euros or issued only in Europe; the term refers to the international offshore bond market. Keep denomination, issuance market, and registration form distinct.

错误选项为什么不成立 / Why the other options fail

A. Eurobond 可以使用不同币种，名称不等于欧元债。

A. Eurobonds can be denominated in different currencies; the name does not mean euro-denominated debt.

C. 发行人本国司法管辖区并不是 Eurobond 市场的定义性特征。

C. Issuance within the issuer’s home jurisdiction is not the defining feature of the Eurobond market.

> 原资料提示 / 使用条件：本题按所附答案保留 B。Eurobond 历史上存在无记名形式，登记形式也受市场及规则影响；不要把本题“最可能”的答案记成“所有 Eurobond 必须记名”。
>
> Source note / condition: B is retained from the supplied key. Eurobonds have historically included bearer forms, and registration depends on market practices and rules; do not turn this “most likely” answer into a claim that every Eurobond must be registered.

Q043 | 原题 / Original question

回购中的抵押品与初始保证金

Repo collateral and initial margin

来源：题库 PDF 第 76 页（固收第 8 页）；答案 PDF 第 43 页。

Source: question bank p. 76 (Fixed Income p. 8); solution PDF p. 43.

在回购协议中：

In a repurchase agreement:

A. 只有资金贷出方承担信用风险。
only the lender of funds is exposed to credit risk.

B. 使用高评级主权债作为抵押品便可消除信用风险。
credit risk is eliminated by using highly rated sovereign bonds as collateral.

C. 初始保证金能对抵押品市场价值变化提供部分保护。
the initial margin offers partial protection against changes in the market value of the collateral.

正确答案 / Answer: C

回购把现金借贷与证券转移相结合。抵押品的价值通常高于借出的现金，差额形成缓冲，以部分吸收抵押品价格下跌和处置成本。但抵押品可能跳跌、对手方可能违约，保证金并不能完全消除风险。双方在不同违约情景下都可能承担对手方风险。

A repo combines cash borrowing with a transfer of securities. Collateral is typically worth more than the cash advanced, creating a buffer against collateral price declines and liquidation costs. This margin provides partial protection, not elimination of risk. Collateral can fall abruptly, and either party can face counterparty exposure under different default scenarios.

错误选项为什么不成立 / Why the other options fail

A. 现金贷出方和证券提供方都可能受到对方违约影响。

A. Both the cash lender and the securities provider can be affected by counterparty default.

B. 高评级抵押品降低部分风险，但仍有价格、对手方和处置风险。

B. High-quality collateral mitigates risk but does not remove price, counterparty or liquidation risk.

Q062 | 原题 / Original question

决定基准利率的是计价货币

The denomination currency determines the benchmark rate

来源：题库 PDF 第 80 页（固收第 12 页）；答案 PDF 第 62 页。

Source: question bank p. 80 (Fixed Income p. 12); solution PDF p. 62.

一家美国公司在英国发行并以英镑计价的债券，其价格最可能在以下何种情况下变化？

The price of a bond issued in the United Kingdom by a US-based company and denominated in British pounds most likely changes when:

A. 仅美国利率变化。
US interest rates change only.

B. 仅英国利率变化。
British interest rates change only.

C. 美国和英国利率变化均会影响。
both US and British interest rates change.

正确答案 / Answer: B

本题考查贴现利率与计价货币的对应关系。现金流以英镑支付，应使用英镑利率曲线并加上适当信用利差来贴现；发行人设在美国并不意味着用美元利率贴现英镑现金流。因此按本题设定选择英国利率变化。

This question tests the link between a bond's currency and its discount-rate benchmark. Sterling cash flows are discounted using a sterling curve plus an appropriate credit spread; a US issuer does not turn sterling cash flows into dollar cash flows. Under this currency-benchmark interpretation, the supplied answer is changes in British rates.

错误选项为什么不成立 / Why the other options fail

A. 把发行人所在地误当成现金流计价货币。

A. This confuses the issuer's location with the cash-flow currency.

C. 本题不是问宏观经济的间接联动，而是问该债券所用利率基准。

C. The question targets the applicable rate benchmark, rather than indirect macroeconomic transmission.

> 原资料提示 / 使用条件："only"是本题简化设定。现实中美国利率也可能通过汇率、英国利率、信用利差或发行人经营状况间接影响价格；不能将答案 B 理解为任何环境下都完全不受美国利率影响。
>
> Source note / condition: "Only" reflects the question's simplified framework. In reality, US rates may indirectly affect sterling rates, exchange rates, credit spreads or the issuer's business. Answer B should not be read as complete insulation from US rates in every setting.

Q077 | 原题 / Original question

主权与非主权政府发行人

Sovereign versus non-sovereign government issuers

来源：题库 PDF 第 84 页（固收第 16 页）；答案 PDF 第 77 页。

Source: question bank p. 84 (Fixed Income p. 16); solution PDF p. 77.

美国市政债券最恰当的分类是：

US municipal bonds are best described as:

A. 政府机构债券。
agency bonds.

B. 非主权政府债券。
non-sovereign bonds.

C. 准政府债券。
quasi-government bonds.

正确答案 / Answer: B

美国市政债由州、市及其他地方政府发行，属于一国之内的地方政府融资，因此归为非主权政府债。识别发行主体层级即可：国家中央政府与地方政府的分类不同。

US municipal bonds are issued by states, municipalities, and other local governments. They finance government entities below the sovereign level and are classified as non-sovereign government bonds. Identify the level of the issuer first.

错误选项为什么不成立 / Why the other options fail

A. 机构债强调承担特定公共职能的政府机构，不是市政债的最准确总分类。

A. Agency bonds refer to agencies carrying out particular public functions; this is not the best general classification of municipal bonds.

C. 准政府债通常指政府关联机构发行的债券，不能取代本题的地方政府分类。

C. Quasi-government bonds generally refer to debt of government-related agencies, rather than the local-government category tested here.

CHAPTER 02 / 14 ORIGINAL QUESTIONS

债券估值与收益率：把时间和口径对齐

Valuation and yields: align timing and conventions

> 不先背答案：先确定期数、每期现金流、每期利率，再计算价格或收益率。
> Determine the number of periods, cash flow per period and rate per period before solving for price or yield.

先做这些原题检验理解：Q106, Q2, Q35, Q82。
Start with these original checkpoints: Q106, Q2, Q35, Q82.

2.1 估值只有一条主线：未来现金流折现

2.1 The central valuation rule: discount future cash flows

$$P = \sum_{t=1}^{N} \frac{C}{(1 + y/m)^t} + \frac{F}{(1 + y/m)^N}$$

F 是面值；C 是每期票息；y 是按每年 m 次复利报价的名义年收益率；N = 年数 × m。求和 t 从 1 到 N，票息在每期末收到。
F is par value, C is the coupon per period, y is a nominal annual yield with m compounding periods, and N equals years times m. The sum runs from t = 1 to N, with end-of-period coupons.

收益率是折现率，不是票息率。票息率决定合同付多少钱；收益率决定市场愿意花多少钱购买这些付款。风险或市场利率改变时，票息可以不变，价格却会变。
Yield is the discount rate, not the coupon rate. The coupon rate sets contractual interest; the market yield determines what investors pay for those cash flows. Price can change while the coupon remains fixed.

下表适用于普通固定利率、按面值偿还的债券，并在付息日比较价格（无应计利息）；票息率与 YTM 的频率口径须一致。
The following comparison assumes a conventional fixed-rate bond redeemed at par, valued on a coupon date without accrued interest, and consistent coupon and YTM frequency conventions.

关系 / Relationship	价格 / Price
票息率 > YTM / Coupon rate > YTM	溢价 / Premium: P > F
票息率 = YTM / Coupon rate = YTM	平价 / Par: P = F
票息率 < YTM / Coupon rate < YTM	折价 / Discount: P < F

> 原题 Q106：N = 3 × 2 = 6；C = 100 × 12% / 2 = 6；每期 y = 4% / 2 = 2%。P = 6/(1.02) + … + 106/(1.02)^6 = 122.4057，选 B。把年利率与半年现金流混用会落到干扰项。
> Original Q106: N = 3 x 2 = 6, C = 100 x 12% / 2 = 6, and the periodic yield is 4% / 2 = 2%. P = 6/(1.02) + … + 106/(1.02)^6 = 122.4057, answer B. Mixing annual rates with semiannual cash flows produces a distractor.

2.2 YTM、当期收益率和真实回报不是同一个数

2.2 YTM, current yield and realized return differ

$$\text{Current yield} = \frac{\text{Annual coupon}}{\text{Current price}}$$

当期收益率只考虑票息相对于价格，不考虑到期本金与购买价格的差额，也不计票息再投资。Q23：4.5/85.70 = 5.2509%，选 B。

Current yield measures annual coupon income relative to price. It ignores the difference between redemption value and purchase price and ignores reinvestment. Q23: 4.5/85.70 = 5.2509%, answer B.

YTM 是令约定现金流现值等于价格的内部收益率。实际复合回报要看是否持有至到期、是否违约以及票息以什么利率再投资。第 6 章用原题专门拆解这些条件。

YTM is the internal rate of return equating contractual cash flows with price. Realized compound return depends on the holding horizon, default outcomes and coupon reinvestment rates. Chapter 6 examines these conditions using the original questions.

2.3 复利换算：先保持一年后的财富相同

2.3 Compounding conversion: preserve one-year wealth

$$\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n$$

APR 是名义年率，不是每期利率；EAR 是有效年收益率。不要将“季度收益率”与“按季度复利报价的年收益率”混为一谈。

APR is a nominal annual rate, not a periodic rate. EAR is the effective annual return. A quarterly periodic rate is different from an annual rate quoted with quarterly compounding.

$$EAR = \left(1 + \frac{APR_m}{m}\right)^m - 1$$

Q2：APR4 = 4 × [(1 + 0.08/2)^(1/2) - 1] = 7.9216%，选 C。Q10：EAR = (1 + 0.0366/2)^2 - 1 = 3.6935%，选 B。

Q2: APR4 = 4 x [(1 + 0.08/2)^(1/2) - 1] = 7.9216%, answer C. Q10: EAR = (1 + 0.0366/2)^2 - 1 = 3.6935%, answer B.

Q29 说的是同一经济收益在不同复利频率下的报价转换。负收益率时，提高复利频率使对应的名义年收益率更负。这个结论不能拿来描述“固定名义利率不变时”的另一个问题。

Q29 concerns equivalent yield quotations at different compounding frequencies. For a negative yield, a higher frequency implies a more negative equivalent nominal annual yield. This is distinct from holding the nominal rate fixed.

2.4 即期折现与矩阵估值

2.4 Spot discounting and matrix pricing

$$P = \frac{CF_1}{1+z_1} + \frac{CF_2}{(1+z_2)^2} + \cdots + \frac{CF_n}{(1+z_n)^n}$$

z_t 是对应期限的即期利率。每个现金流用自己的期限利率，不能把最后一期的即期利率用到所有现金流。Q35：4/1.06 + 4/(1.05)^2 + 104/(1.04)^3 = 99.8573，选 B。

z_t is the spot rate for maturity t. Discount each cash flow with its matching spot rate, rather than applying the final spot rate to every cash flow. Q35: 4/1.06 + 4/(1.05)^2 + 104/(1.04)^3 = 99.8573, answer B.

Q14：若按即期曲线算出的无套利价格已知，YTM 就是再找出一个单一利率，使同一组现金流算出同一价格。因此两种现值之和相等。两种利率体系不同，并不意味着价格必须不同。

In Q14, YTM is the single rate that reproduces the no-arbitrage price obtained from spot rates. Both present-value sums therefore equal the same price, despite using different discount-rate representations.

$$y(T) = y_A + \frac{T - T_A}{T_B - T_A}(y_B - y_A)$$

矩阵估值利用可比债券估计缺乏直接报价的债券，包括尚未发行的债券。Q60：3.4% + (5−3)/(8−3) × (5.4%−3.4%) = 4.2%。线性插值是题目给定资料下的估计，并非完整信用定价模型。

Matrix pricing estimates a bond value from comparables, including bonds not yet issued. Q60: 3.4% + (5−3)/(8−3) x (5.4%−3.4%) = 4.2%. Linear interpolation is an estimate based on the supplied information, not a complete credit-pricing model.

2.5 可赎回债与货币市场收益率

2.5 Callable-bond and money-market yields

$$YTW = \min(YTM, YTC_1, YTC_2, \ldots)$$

Q82 需分别用到期、第一次赎回及第二次赎回的日期与偿还金额求收益率，再取最低值。最早赎回不一定给最低收益率；本题答案是第二次赎回收益率。

Q82 requires a separate yield calculation for maturity and each call date using the applicable redemption price. Take the lowest result. The earliest call need not have the lowest yield; this question selects the second-call yield.

$$d = \frac{F - P}{F}\frac{\text{Year basis}}{\text{Days}}$$

贴现率的分母是到期兑付额；以购买成本为分母的是另一种收益率口径。Q104：140,500/5,000,000 × 365/160 = 6.4103%，选 B。题目给 365 天，就不要擅自改为 360。

The discount yield uses redemption value in the denominator. Using purchase cost gives a different yield convention. Q104: 140,500/5,000,000 x 365/160 = 6.4103%, answer B. Preserve the specified 365-day year.

> Q92 是时间陷阱：一年后先把剩余期数从 6 期改为 4 期，再用新的 YTM 估价。债券原本溢价，向面值回归的影响可能超过降息带来的涨价；本题价格实际下降约 0.57。
>
> Q92 is a timing trap: after one year, reduce the remaining periods from six to four before using the new YTM. For a premium bond, movement toward par can outweigh the price benefit from a lower yield; its price actually falls by about 0.57.

现金流折现与定价 / Discounting cash flows and pricing

Q106 | 原题 / Original question

半年付息的三个同步调整：期数、票息、收益率

For semiannual bonds, align periods, coupons, and yields

来源：题库 PDF 第 90 页（固收第 22 页）；答案 PDF 第 106 页。

Source: question bank p. 90 (Fixed Income p. 22); solution PDF p. 106.

某债券具有下列特征：

A bond has the following characteristics:

到期年限 Time to maturity	3年 3 years
票面利率 Coupon rate	12%
付息周期 Payment period	半年 Semiannual

如果到期收益率为4%，该债券每100面值的价格最接近：

For a yield to maturity of 4%, the price of the bond per 100 of par value is closest to:

A. 122.20.

B. 122.41.

C. 122.53.

正确答案 / Answer: B

半年付息需要同步改三项：3年对应6期；每100面值的年票息12分成每期6；年化YTM4%分成每半年2%。折现六笔票息及末期本金后，价格122.405724，最接近122.41。票息12%高于收益率4%，溢价方向合理。

Semiannual payment requires three aligned inputs: three years become six periods, an annual coupon of 12 per 100 par becomes 6 per period, and the 4% annualized YTM becomes 2% per half-year. Discount the six coupons and final principal to obtain 122.405724, closest to 122.41. Coupon above yield correctly implies a premium.

1. 先把所有输入转换为半年口径。

Convert every input to a half-year basis.

```
N = 3 × 2 = 6; PMT = 100 × 12%/2 = 6; i = 4%/2 = 2%
```

2. 使用年金现值加本金现值。

Add the coupon-annuity and principal present values.

```
P = 6[1-(1.02)^(-6)]/0.02 + 100(1.02)^(-6) = 122.405724
```

错误选项为什么不成立 / Why the other options fail

A. 把半年付息误当成年付息，按3期、每期12、收益率4%计算。

A. It treats the bond as annual-pay, using three periods, a coupon of 12, and a 4% periodic yield.

C. 122.53并非按题目的半年期末付息现金流折现所得。所附答案对该干扰项的解释还有算术问题，见提示。

C. 122.53 is not the value of the stated semiannual end-of-period cash flows. The source's explanation of this distractor also has an arithmetic issue, noted below.

> 原资料提示 / 使用条件：原答案B已复算确认。其对C的文字称"年付息、年折现、期初付息"得到122.53，但按该文字将12在第0、1、2年支付、100在第3年偿还，计算为123.5328，不能复现122.53。保留原选项数值，不把该干扰项解释当作正确公式。
>
> Source note / condition: Answer B has been independently verified. The source says C uses annual discounting and coupons at the beginning of each year to obtain 122.53. But paying 12 at times 0, 1, and 2 and repaying 100 at time 3 gives 123.5328, not 122.53. The original option is retained; its supplied rationale should not be learned as a valid formula.

Q087 | 原题 / Original question

票息率决定现金流，市场收益率决定折现

Coupon rate determines cash flows; market yield determines discounting

来源：题库 PDF 第 86 页（固收第 18 页）；答案 PDF 第 87 页。

Source: question bank p. 86 (Fixed Income p. 18); solution PDF p. 87.

分析师收集到某只每年付息债券的下列信息：

An analyst gathers the following information about a bond that pays interest annually:

票面利率 Coupon rate	4%
到期年限 Time-to-maturity	3年 3 years
面值 Par value	$100,000

如果市场折现率为5%，该债券的市场价值最接近：

If the market discount rate is 5%, the market value of this bond is closest to:

A. $89,839.

B. $97,277.

C. $102,775.

正确答案 / Answer: B

每年票息是100,000×4%=4,000；市场要求收益率5%用于折现。前两年各收4,000，第3年收104,000，现值合计97,276.75。票息率低于市场收益率，结果低于面值，方向合理。

Annual coupon cash flow is 100,000×4%=4,000; the 5% market yield is the discount rate. The first two payments are 4,000 each and the final payment is 104,000. Their present values sum to 97,276.75. Coupon below market yield correctly implies a discount to par.

1. 先由票面利率算出每年票息。

Calculate the annual coupon using the coupon rate.

```
C = 100000 × 0.04 = 4000
```

2. 折现全部现金流。

Discount every cash flow.

```
P = 4000/1.05 + 4000/(1.05)^2 + 104000/(1.05)^3 = 97276.751971
```

错误选项为什么不成立 / Why the other options fail

A. 只折现末年104,000，遗漏第1年和第2年的票息。

A. It discounts only the final 104,000 payment and omits coupons in years 1 and 2.

C. 将票息率4%与折现率5%互换了。

C. It swaps the 4% coupon rate and the 5% discount rate.

Q035 | 原题 / Original question

每笔现金流用对应期限的即期利率折现

Discount each cash flow at its matching spot rate

来源：题库 PDF 第 75 页（固收第 7 页）；答案 PDF 第 35 页。

Source: question bank p. 75 (Fixed Income p. 7); solution PDF p. 35.

分析师收集到以下即期收益率曲线资料：

An analyst gathers the following information about a spot curve:

剩余期限 Term to Maturity	即期利率 Spot Rate
1 年 1year	6%
2 年 2years	5%
3 年 3yaers	4%

如果一只三年期、每年付息一次的债券票息率为 4%，其价格最接近：

If the coupon rate of a 3-year annual-pay bond is 4%, the price of the bond is closest to:

A. 97.28.

B. 99.86.

C. 100.00.

正确答案 / Answer: B

按每 100 面值，年度票息为 4；第 3 年收到本金加票息 104。分别用 1 年 6%、2 年 5%、3 年 4% 折现，得到 99.8573，选 B。不能因为 3 年即期利率刚好等于票息率，就直接断定这只付息债券按面值交易。

Per 100 of par, the annual coupon is 4 and the final cash flow is 104. Discount the first, second, and third cash flows at the corresponding 6%, 5%, and 4% spot rates to obtain 99.8573. Equality between the coupon rate and the three-year spot rate alone does not make this coupon bond trade at par.

1. 匹配现金流日期与即期利率期限。

Match each cash-flow date with its spot-rate maturity.

```
P = 4/1.06 + 4/(1.05)^2 + 104/(1.04)^3 = 99.8573
```

错误选项为什么不成立 / Why the other options fail

A. 错误地把三个即期利率取算术平均为 5%，再以 5% 折现全部现金流，会得到约 97.28。

A. Taking the arithmetic average of the three spot rates as 5% and discounting every cash flow at that rate gives approximately 97.28.

C. 把所有现金流都用 4% 折现才会得到 100；题目给的是不同期限的即期利率。

C. Discounting every cash flow at 4% would give 100, but the question provides different spot rates for different maturities.

> 原资料提示 / 使用条件：原表英文将 “3 years” 误拼为 “3yaers”，此处英文表格保留原文，中文按明确含义译为“3 年”。按原选项与所附解答，价格使用每 100 面值的口径。
>
> Source note / condition: The source table misspells “3 years” as “3yaers”; the English table preserves it and the Chinese translation gives its clear meaning. Consistent with the original options and supplied solution, the price is quoted per 100 of par.

Q014 | 原题 / Original question

即期利率定价与 YTM 定价的一致性

Consistency between spot-rate pricing and YTM pricing

来源：题库 PDF 第 71 页（固收第 3 页）；答案 PDF 第 14 页。

Source: question bank p. 71 (Fixed Income p. 3); solution PDF p. 14.

一只以无套利价值交易的债券处于溢价。用各期即期利率折现债券现金流得到的现值之和：

A bond trading at its no-arbitrage value is priced at a premium. The sum of the present value of the bond's cash flows discounted at spot rates is:

A. 小于用债券到期收益率折现各项现金流得到的现值之和。
less than the sum of the present values of the bond's cash flows discounted at its yield to maturity.

B. 等于用债券到期收益率折现各项现金流得到的现值之和。
equal to the sum of the present values of the bond's cash flows discounted at its yield to maturity.

C. 大于用债券到期收益率折现各项现金流得到的现值之和。
greater than the sum of the present values of the bond's cash flows discounted at its yield to maturity.

正确答案 / Answer: B

无套利价格可以用每期匹配的即期利率折现求得。YTM 则是使同一组现金流的现值恰好等于该市场价格的单一内部收益率。因此两种计算都等于同一个价格，债券溢价并不会改变这个恒等关系。

The no-arbitrage price is obtained by discounting each cash flow at its corresponding spot rate. The YTM is the single internal rate of return that makes those same cash flows equal the observed price. Both therefore produce the same price; trading at a premium does not alter this identity.

错误选项为什么不成立 / Why the other options fail

A. 不能因为债券溢价就判定即期利率定价更低；YTM 是由该价格反推的。

A. A premium does not make spot-rate pricing lower; the YTM is solved from that same price.

C. 逐期折现和单一 YTM 折现不同，不代表现值总和不同。

C. Different discounting methods do not imply different total present values when the YTM is calibrated to the price.

收益率口径与复利频率 / Yield conventions and compounding

Q023 | 原题 / Original question

当期收益率只比较年度票息与当前价格

Current yield compares annual coupon cash flow with price

来源：题库 PDF 第 72 页（固收第 4 页）；答案 PDF 第 23 页。

Source: question bank p. 72 (Fixed Income p. 4); solution PDF p. 23.

一只票息率 4.5%、期限 10 年的债券，到期面值为 100 美元，目前价格为 85.70 美元。其当期收益率最接近：

The current yield for a 4.5% coupon,10-year bond, with a maturity par value of $100 and currently priced at $85.70 is closest to:

A. 4.50%.

B. 5.25%.

C. 5.93%.

正确答案 / Answer: B

先用票息率乘面值得到每年票息 4.50 美元，再除以当前价格 85.70，得到 5.250875%。当期收益率不考虑到期回归面值所产生的资本损益，也不考虑票息再投资；不能把它直接当作 YTM。

The annual coupon is 4.5% of the $100 par value, or $4.50. Divide by the current price of $85.70 to obtain 5.250875%. Current yield ignores the capital gain or loss from repayment at par and coupon reinvestment, so it is not the same as YTM.

1. 先算票息金额，再除以现价。

Calculate coupon cash flow before dividing by price.

```
Current yield = (0.045*100)/85.70 = 5.250875%
```

错误选项为什么不成立 / Why the other options fail

A. 4.50% 是票息率，分母为面值，不是市场价格。

A. 4.50% is the coupon rate, whose denominator is par value rather than market price.

C. 把折价 14.30 平均摊到 10 年得到 1.43，再直接加到 4.50 得到 5.93；这既混入本金增值，也未以当前价格为分母，不是当期收益率。

C. Dividing the 14.30 discount by 10 years gives 1.43 and adding it directly to 4.50 gives 5.93. This mixes in principal appreciation and omits division by the current price, so it is not current yield.

Q002 | 原题 / Original question

不同复利频率之间的名义年化收益率换算

Converting nominal annual yields across compounding frequencies

来源：题库 PDF 第 69 页（固收第 1 页）；答案 PDF 第 2 页。

Source: question bank p. 69 (Fixed Income p. 1); solution PDF p. 2.

一只五年期、每半年付息一次的债券，到期收益率为 8%。换算为每季度复利口径后，到期收益率最接近：

A five-year semiannual bond has a yield to maturity of 8%. Converted to a quarterly periodicity, the yield to maturity is closest to:

A. 1.98%.

B. 3.92%.

C. 7.92%.

正确答案 / Answer: C

必须保持有效年收益率不变。8% 是半年复利口径的名义年化收益率，半年利率为 4%。换算后的季度利率约为 1.9804%，再乘 4 才得到题目要求的名义年化收益率 7.9216%，选 C。五年期限不参与这一频率换算。

Preserve the effective annual return. The 8% quote is a nominal annual yield with semiannual compounding, so its half-year rate is 4%. The equivalent quarterly rate is approximately 1.9804%; multiply by four to obtain a nominal annual yield of 7.9216%. The five-year maturity is irrelevant to this frequency conversion.

1. 先统一有效年收益率。

First equate effective annual returns.

```
(1+0.08/2)^2 = (1+y4/4)^4
```

2. 求季度利率，再换成年化报价。

Solve for the quarterly rate and annualize the quotation.

```
y4 = 4[(1.04)^(1/2)-1] = 7.9216%
```

错误选项为什么不成立 / Why the other options fail

A. 1.98% 是单季度利率，尚未乘以 4 年化。

A. 1.98% is the quarterly periodic rate, before multiplying by four.

B. 3.92% 错误地把 8% 当作另一种年化口径开平方，未保持原收益率的复利等价关系。

B. 3.92% results from treating the 8% quotation as a different annual rate and applying an inappropriate square-root conversion.

Q010 | 原题 / Original question

半年复利名义年利率换算为有效年利率

Converting a semiannual nominal yield to an effective annual yield

来源：题库 PDF 第 70 页（固收第 2 页）；答案 PDF 第 10 页。

Source: question bank p. 70 (Fixed Income p. 2); solution PDF p. 10.

按半年付息口径报价的名义年化到期收益率为 3.66%。有效年收益率最接近：

The stated annual yield to maturity on a semiannual bond basis is 3.66%. The effective annual yield is closest to:

A. 3.63%.

B. 3.69%.

C. 7.45%.

正确答案 / Answer: B

先除以 2 得到半年的周期利率 1.83%，再将半年的增长因子连续复利两次。有效年收益率为 3.693489%，略高于名义报价 3.66%，选 B。

Divide by two to obtain the half-year periodic rate of 1.83%, then compound that growth factor twice. The effective annual yield is 3.693489%, slightly above the nominal quotation of 3.66%.

1. 周期利率必须与复利次数配套。

Match the periodic rate to the compounding frequency.

```
EAR = (1+0.0366/2)^2-1 = 0.03693489 = 3.693489%
```

错误选项为什么不成立 / Why the other options fail

A. 该结果接近把有效年收益率反向换成半年复利名义报价，方向错了。

A. This resembles the reverse conversion from an effective annual yield to a semiannual nominal quotation.

C. 它把 3.66% 误当作每半年的利率，没有先除以 2。

C. This incorrectly treats 3.66% as the half-year rate without dividing by two.

Q029 | 原题 / Original question

负收益率的复利口径换算

Compounding-frequency conversion for a negative yield

来源：题库 PDF 第 74 页（固收第 6 页）；答案 PDF 第 29 页。

Source: question bank p. 74 (Fixed Income p. 6); solution PDF p. 29.

对于一只收益率为负的付息债券，如果一年内的复利频率提高，换算后的到期收益率将：

For a coupon bond with a negative yield, compounding more frequently within the year results in a yield-to-maturity that is:

A. 变得更负。
more negative.

B. 保持不变。
the same.

C. 变得较不负。
less negative.

正确答案 / Answer: A

这里是在相同债券价格、相同有效年回报下，换算不同复利频率的名义年化报价。若有效年收益率 E 为负，名义报价 y_m = m[(1+E)^(1/m)-1]；m 增加时，报价变得更负。不要把“名义报价换算”与“固定名义报价后改变复利次数”混为一谈。

This question converts nominal annual quotations while keeping the bond price and effective annual return unchanged. For a negative effective annual yield E, y_m = m[(1+E)^(1/m)-1]. Increasing m makes the equivalent nominal quotation more negative. This differs from holding a nominal quotation fixed while changing its compounding frequency.

错误选项为什么不成立 / Why the other options fail

B. 周期收益率需要调整，等价名义年化报价不会保持完全相同。

B. The periodic rate must adjust, so the equivalent nominal annual quotation does not remain unchanged.

C. 在本题保持有效回报不变的换算下，方向相反。

C. This reverses the direction of the equivalent-yield conversion used in this question.

Q090 | 原题 / Original question

货币市场报价一般按单利年化

Money-market quotations generally use simple annualization

来源：题库 PDF 第 87 页（固收第 19 页）；答案 PDF 第 90 页。

Source: question bank p. 87 (Fixed Income p. 19); solution PDF p. 90.

货币市场收益率：

Money market yields are:

A. 经年化并采用复利。
annualized and compounded.

B. 采用统一的每年期数。
stated for a common periodicity.

C. 以单利方式报价。
stated on a simple interest basis.

正确答案 / Answer: C

本题所用货币市场报价是将短期收益按天数比例作单利年化。年化不等于复利：365/D或360/D只是比例缩放。不同工具的期限、计息日数基准以及分母可能不同，比较前应统一口径。

The money-market convention tested annualizes a short-period return using a day-count proportion on a simple-interest basis. Annualization is not compounding: 365/D or 360/D is a scaling factor. Maturity, day-count basis, and the quotation denominator can differ across instruments.

错误选项为什么不成立 / Why the other options fail

A. 常规货币市场报价虽年化，但不按债券YTM方式复利。

A. These money-market quotes are annualized but not compounded like conventional bond YTMs.

B. 不同剩余期限对应不同的年化期数，并非统一periodicity。

B. Different maturities imply different annualization periodicities; there is no single common periodicity.

Q104 | 原题 / Original question

贴现率用到期面值作分母

Discount-rate quotations use redemption value as the denominator

来源：题库 PDF 第 89 页（固收第 21 页）；答案 PDF 第 104 页。

Source: question bank p. 89 (Fixed Income p. 21); solution PDF p. 104.

某商业票据具有下列特征：

A commercial paper has the following characteristics:

到期偿还金额 Redemption value	$5,000,000
赚取的利息 Interest earned	$140,500
剩余到期天数 Days to maturity	160

按一年365天计算，贴现率最接近：

For a 365-day year, the discount rate is closest to:

A. 6.2%.

B. 6.4%.

C. 6.6%.

正确答案 / Answer: B

题目明确要求discount rate，分母应为到期偿还金额5,000,000，不是购买价格。利息140,500除以到期金额，再乘365/160按单利年化，得到6.4103125%，最接近6.4%。

The question explicitly asks for the discount rate, whose denominator is redemption value of 5,000,000 rather than purchase price. Divide interest of 140,500 by redemption value and simple-annualize using 365/160. The result is 6.4103125%, closest to 6.4%.

1. 贴现率分母用面值/到期金额。

Use face or redemption value as the discount-rate denominator.

```
DR = (FV-PV)/FV × 365/Days
```

2. 代入原题数据。

Substitute the original figures.

```
DR = (140500/5000000) × (365/160) = 6.4103125%
```

错误选项为什么不成立 / Why the other options fail

A. 将已是到期金额的5,000,000再加上利息，错误扩大分母。

A. It incorrectly adds interest to 5,000,000 even though that amount is already the redemption value.

C. 以购买价格5,000,000-140,500作分母，算出的是附加利率约6.60%。

C. It uses purchase price, 5,000,000-140,500, as the denominator and obtains the add-on rate of about 6.60%.

矩阵估值、最差收益率与时间变化 / **Matrix pricing, yield to worst and time passage**

Q031 | 原题 / Original question

矩阵定价使用可比债券估值

Matrix pricing uses comparable bonds

来源：题库 PDF 第 74 页（固收第 6 页）；答案 PDF 第 31 页。

Source: question bank p. 74 (Fixed Income p. 6); solution PDF p. 31.

矩阵定价最可能用于估计下列哪一种债券的价格？

Matrix pricing is most likely used to estimate the price of a bond that:

A. 流动性很高的债券。
is highly liquid.

B. 尚未发行的债券。
is not yet issued.

C. 信用质量未知的债券。
has unknown credit quality.

正确答案 / Answer: B

矩阵定价从信用质量、期限及结构相似的债券推导所需收益率，再估计目标债券价格，适用于缺少可靠交易价格的债券，包括新发行定价。本题给出的选项中，“尚未发行”最符合这一用途。

Matrix pricing uses yields on bonds with comparable credit quality, maturity, and structure to estimate a target bond’s required yield and price. It is useful when a reliable market quotation is unavailable, including pricing a new issue. “Not yet issued” is the best of the listed choices.

错误选项为什么不成立 / Why the other options fail

A. 流动性很高的债券通常已有可观察的市场价格，不是矩阵定价最典型的对象。

A. A highly liquid bond generally has observable market prices, making it a less typical target.

C. 信用质量是选择可比债券的重要依据，未知信用质量反而妨碍可靠比较。

C. Credit quality helps identify appropriate comparables; unknown credit quality impairs the estimation.

Q060 | 原题 / Original question

矩阵定价中的收益率线性插值

Linear yield interpolation in matrix pricing

来源：题库 PDF 第 80 页（固收第 12 页）；答案 PDF 第 60 页。

Source: question bank p. 80 (Fixed Income p. 12); solution PDF p. 60.

分析师收集了以下信息：

An analyst gathers the following information:

	到期期限 Time to Maturity	到期收益率 Yield to Maturity
债券 1 Bond 1	3 年 Three years	3.4%
债券 2 Bond 2	8 年 Eight years	5.4%

仅根据这些信息，一只信用质量相近的 5 年期债券的估计市场贴现率是多少？

Based only on this information, the estimated market discount rate for a 5-year bond with similar credit quality is:

A. 4.2%.

B. 4.4%.

C. 4.6%.

正确答案 / Answer: A

只给出相似信用质量的 3 年与 8 年债券，应按期限距离线性插值。5 年位于区间内的比例为 (5 - 3)/(8 - 3) = 2/5。估计收益率 = 3.4% + (2/5) × (5.4% - 3.4%) = 4.2%。这是一种基于可比债券的估计，不是完整的无套利即期曲线定价。

With only comparable three- and eight-year yields, use linear interpolation by maturity. Five years lies (5 - 3)/(8 - 3) = 2/5 of the way across the interval. The estimated yield is 3.4% + (2/5) × (5.4% - 3.4%) = 4.2%. This is a comparable-bond estimate, not full no-arbitrage valuation from a spot curve.

1. 按目标期限在区间内的位置插值。

Interpolate by the target maturity's position.

```
y5 = 3.4% + [(5 - 3) / (8 - 3)] * (5.4% - 3.4%) = 4.2%
```

错误选项为什么不成立 / Why the other options fail

B. 简单平均得到 4.4%，对应 5.5 年的中点，非题目要求的 5 年。

B. A simple average of 4.4% corresponds to the 5.5-year midpoint, not five years.

C. 把到两端的距离权重倒置，给较长端过高权重。

C. This reverses the distance weights and overweights the longer maturity.

Q082 | 原题 / Original question

最差收益率取所有适用赎回收益率与到期收益率中的最低值

Yield to worst is the minimum of applicable call yields and YTM

来源：题库 PDF 第 85 页（固收第 17 页）；答案 PDF 第 82 页。

Source: question bank p. 85 (Fixed Income p. 17); solution PDF p. 82.

分析师收集到某只每年付息的可赎回债券的下列信息：

An analyst gathers the following information about a callable bond that pays interest annually:

当前价格 Current price	每100面值为105 105 per 100 of par value
到期年限 Time-to-maturity	4年 4 years
票面利率 Coupon rate	4%
第2年赎回价格 Call price at year 2	每100面值为103 103 per 100 of par value
第3年赎回价格 Call price at year 3	每100面值为101 101 per 100 of par value

这只债券的最差收益率是：

This bond's yield to worst is the:

A. 到期收益率。
yield to maturity.

B. 第一次可赎回日收益率。
yield to first call.

C. 第二次可赎回日收益率。
yield to second call.

正确答案 / Answer: C

每一种终止日期都是一组独立现金流，须分别求内部收益率。到期收益率约2.6656%；第2年按103赎回对应2.8706%；第3年按101赎回对应2.5718%。最低的是第二次可赎回日收益率，所以选C。不要直接假定最早赎回就最差。

Each termination date defines a separate cash-flow stream and requires its own internal rate of return. YTM is about 2.6656%; redemption at 103 in year 2 gives 2.8706%; redemption at 101 in year 3 gives 2.5718%. The second-call yield is lowest. The earliest call is not automatically the worst outcome.

1. 到期：最后一期现金流为104。

At maturity, the final payment is 104.

```
105 = 4/(1+y) + 4/(1+y)^2 + 4/(1+y)^3 + 104/(1+y)^4; y = 2.6656%
```

2. 第2年赎回：最后一期现金流为107。

At the first call, the final payment is 107.

```
105 = 4/(1+y) + 107/(1+y)^2; y = 2.8706%
```

3. 第3年赎回：最后一期现金流为105。

At the second call, the final payment is 105.

```
105 = 4/(1+y) + 4/(1+y)^2 + 105/(1+y)^3; y = 2.5718%
```

错误选项为什么不成立 / Why the other options fail

A. 2.6656%高于第二次赎回的2.5718%，因此不是最低收益率。

A. Its 2.6656% yield exceeds the second-call yield of 2.5718%.

B. 2.8706%是三者中最高的收益率。

B. At 2.8706%, the first-call yield is the highest of the three.

Q092 | 原题 / Original question

经过一年后重新定价：期限减少，收益率口径统一

Reprice after one year: shorten maturity and align yield conventions

来源：题库 PDF 第 87 页（固收第 19 页）；答案 PDF 第 92 页。

Source: question bank p. 87 (Fixed Income p. 19); solution PDF p. 92.

一只三年期、半年付息、面值$100、票面利率5%的债券以$108购入。一年后，如果到期收益率下降100个基点，该债券价值的变动最接近：

A three-year, semiannual-pay bond with a $100 par value and a 5% coupon rate is purchased for $108. One year later, if the yield to maturity has decreased by 100 basis points, the change in the value of this bond is closest to:

A. $0.57.

B. $1.52.

C. $3.08.

正确答案 / Answer: A

先用初始6个半年期、每期票息2.5及价格108解出每半年收益率1.114191%，对应年化YTM约2.228382%。一年后还剩4个半年期；年化YTM先减1个百分点，再除以2，得到每半年0.614191%。新价格107.428819，所以价格下降0.571181。A对应变动的绝对金额。

First solve for the initial semiannual yield using six periods, a 2.5 coupon per period, and a price of 108. The semiannual yield is 1.114191%, or an annualized YTM of 2.228382%. One year later, only four semiannual periods remain. Reduce the annual YTM by one percentage point and divide by two: 0.614191% per half-year. The new price is 107.428819, a decline of 0.571181. A matches the absolute amount of the change.

1. 初始收益率：注意计算器求出的是每半年收益率。

The initial calculator yield is per half-year.

```
N = 6; PMT = 2.5; FV = 100; PV = -108; i_half = 1.114191%; y_annual = 2.228382%
```

2. 剩余两年，共4期，重新定价。

Reprice using four periods over the remaining two years.

```
i_new = (2.228382% - 1.00%)/2 = 0.614191%; P_new = 107.428819
```

3. 区分带符号的价格变化与题目选项的绝对金额。

Distinguish the signed price change from the absolute amount in the options.

```
ΔP = 107.428819 - 108 = -0.571181; |ΔP| ≈ 0.57
```

错误选项为什么不成立 / Why the other options fail

B. 把100个基点直接从半年收益率中扣除，等于年化收益率下降200个基点。

B. It subtracts 100 basis points from the semiannual rate, implying a 200-basis-point decline in annualized YTM.

C. 仍用6个半年期，忽略已经过去一年而剩余期限缩短。

C. It retains six half-year periods and ignores the year that has already elapsed.

> 原资料提示 / 使用条件：原答案选择A，并用108-107.43表示变动金额。按通常"新价格减旧价格"的定义，带符号变动是-$0.57，表示下跌；原选项未写负号，按变动绝对金额理解。题目只问债券价格变化，不能把它当作包含已收票息的总回报。
>
> Source note / condition: The source selects A and computes 108-107.43 as the amount changed. Under the usual new-price-minus-old-price convention, the signed change is -$0.57, a decrease. The unsigned option must be read as the absolute amount. This is a price change, not total return including coupons received.

CHAPTER 03 / 8 ORIGINAL QUESTIONS

利率曲线、远期利率与利差

Yield curves, forward rates and spreads

> 把远期利率读成一段时间，而不是只看一个数字。
> Read a forward rate as a time interval rather than an isolated number.

先做这些原题检验理解：Q59, Q73, Q79, Q93。
Start with these original checkpoints: Q59, Q73, Q79, Q93.

3.1 三种利率的区别

3.1 Distinguish three rate concepts

利率 / Rate	回答的问题 / What it describes
即期利率 / Spot rate	今天到某一未来时点的零息投资利率。 A zero-coupon investment rate from today to a future date.
远期利率 / Forward rate	今天隐含的未来某段投资期间的利率。 A rate implied today for a future investment interval.
平价利率 / Par rate	使某期限附息债券按面值定价的票息率。 The coupon rate that prices a coupon bond at par.

2y1y 表示“两年后开始、持续一年”，覆盖第 2 年末至第 3 年末。4y2y 表示“四年后开始、持续两年”，覆盖第 4 年末至第 6 年末。先在时间轴上圈出起止日期。
2y1y starts two years from today and lasts one year, covering years 2 to 3. 4y2y starts in four years and lasts two years, covering years 4 to 6. Mark the start and end dates first.

$$(1+z_{a+b})^{a+b} = (1+z_a)^a(1+f_{a,b})^b$$

左边是一次投到终点；右边是先投到 a 年，再按远期利率投资 b 年。无套利要求两条路径得到相同的终值。
The left side invests directly to the final date. The right side invests to year a and then reinvests for b years at the forward rate. No arbitrage equates the two terminal values.

> Q59：f(2,1) = (1.035)^3/(1.025)^2 − 1 = 5.5294%，选 B。不能简单用 3.5% − 2.5%，也不能将期限权重忽略。
> Q59: f(2,1) = (1.035)^3/(1.025)^2 − 1 = 5.5294%, answer B. Neither a simple yield difference nor ignoring maturity weights gives the correct compounded rate.

Q73 | Match two investment paths at year 6

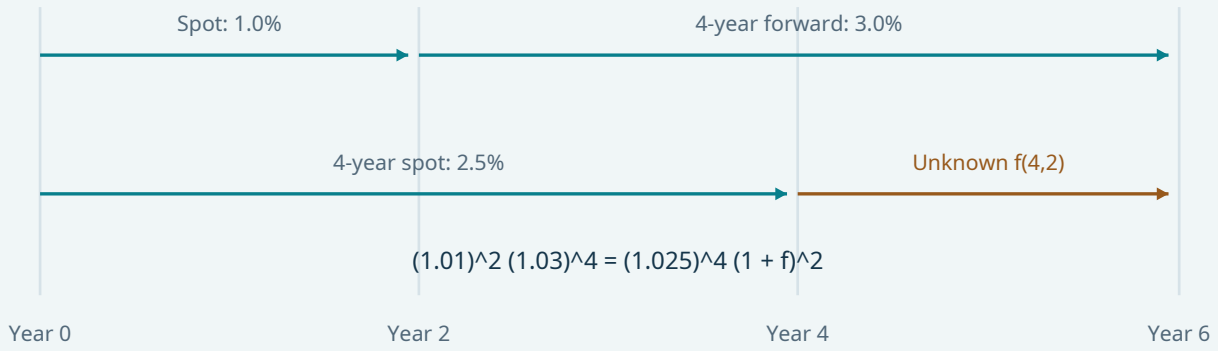


图 2：原题 Q73 的两条投资路径在第 6 年末必须相等。

Figure 2. The two investment paths in original Q73 must agree at the end of year 6.

```
(1.01)^2 x (1.03)^4 = (1.025)^4 x (1 + f_(4,2))^2
```

Q73 的 3% 是从第 2 年至第 6 年的四年期远期利率；求的是第 4 年至第 6 年的两年期利率。解得约 1.9878%，选 A。

The 3% in Q73 is a four-year forward rate from year 2 to year 6. The unknown is the two-year rate from year 4 to year 6. Solving gives about 1.9878%, answer A.

3.2 逐年远期利率也能直接折现

3.2 Discount directly with consecutive forward rates

$$DF_t = \frac{1}{(1+f_{0,1})(1+f_{1,1})\cdots(1+f_{t-1,1})}$$

Q79 的一年期远期利率依次为 1%、2%、4%。第三年的现金流必须依次除以 1.01、1.02、1.04，而不是只除以 (1.04)^3。

Q79 supplies successive one-year forward rates of 1%, 2% and 4%. The year-three cash flow is discounted by 1.01 x 1.02 x 1.04, not by (1.04)^3.

```
P = 1/1.01 + 1/(1.01 x 1.02) + 101/(1.01 x 1.02 x 1.04)
```

原题 Q79 的价格为 96.2293，选 C。第一年只经过第一段利率；第二年经过两段；第三年经过三段。

The original Q79 price is 96.2293, answer C. The first cash flow crosses one rate interval, the second crosses two and the third crosses three.

Q13 与 Q39 的共同概念是“盈亏平衡再投资利率”：远期利率使不同零息投资路径产生相同终值。它由当前曲线隐含，并不保证未来实际短期利率一定等于该值。

Q13 and Q39 describe a breakeven reinvestment rate: the forward rate equates terminal values across zero-coupon investment paths. It is implied by current prices and is not a guarantee of the future short rate.

3.3 记利差时先记基准

3.3 Identify the benchmark behind each spread

利差 / Spread	比较基准 / Benchmark
G-spread	同币种、相应期限政府债券收益率，可插值。Q96 A corresponding government-bond yield, with interpolation if needed. Q96
I-spread	同币种、相应期限标准掉期利率。Q93 A same-currency standard swap rate of corresponding tenor. Q93
Z-spread	加到整条即期曲线上，使现金流现值等于价格的恒定利差。 A constant spread added to spot rates to reproduce price.
OAS	在期权定价模型中剔除嵌入期权影响后的利差。 A model-based spread after allowing for embedded options.

Q42：企业债需求旺盛通常支持价格并压低收益率利差。信用恶化或做市流动性下降，通常要求更高而不是更低的风险补偿。

In Q42, strong corporate-bond demand generally supports prices and narrows spreads. Deteriorating credit quality or weaker market liquidity generally requires more risk compensation, not less.

> 做题顺序：画时间段 → 确认年化和复利口径 → 写“终值相等” → 代数求解。利差题则先圈出 government 或 swap。
>
> Workflow: draw the interval, check annualization and compounding, equate terminal values and solve. For spread questions, first identify government versus swap.

即期与远期的复利关系 / Compounding spot and forward rates

Q013 | 原题 / Original question

隐含远期利率是盈亏平衡再投资利率

An implied forward rate is a breakeven reinvestment rate

来源：题库 PDF 第 71 页（固收第 3 页）；答案 PDF 第 13 页。

Source: question bank p. 71 (Fixed Income p. 3); solution PDF p. 13.

盈亏平衡再投资利率最可能等同于：

A break-even reinvestment rate is most likely equivalent to a(n):

A. 平价利率。
par rate.

B. 即期利率。
spot rate.

C. 隐含远期利率。
implied forward rate.

正确答案 / Answer: C

在相同终点比较两条无套利投资路径：直接买较长期零息债，或先买短期零息债、到期后再投资。使两条路径的累计回报相等的后段再投资利率，就是由当前即期利率推导的隐含远期利率；它不等于对未来实际利率的保证。

Compare two no-arbitrage paths ending on the same date: buy a longer zero-coupon bond, or buy a shorter one and reinvest after it matures. The later-period rate that equalizes their accumulated returns is the implied forward rate. It is not a guarantee of the future realized interest rate.

错误选项为什么不成立 / Why the other options fail

A. 平价利率使一只付息债券按面值定价。

A. A par rate makes a coupon bond trade at par.

B. 即期利率从现在起覆盖到某个到期日；远期利率才连接两个未来时间点。

B. A spot rate runs from today to a maturity date; a forward rate links two future dates.

Q039 | 原题 / Original question

即期利率与隐含远期利率

Spot rates and implied forward rates

来源：题库 PDF 第 76 页（固收第 8 页）；答案 PDF 第 39 页。

Source: question bank p. 76 (Fixed Income p. 8); solution PDF p. 39.

隐含远期利率的最佳定义是：

Implied forward rates are best defined as the:

A. 即期利率的几何平均数。
geometric average of spot rates.

B. 不同期限零息债券之间的盈亏平衡再投资利率。
breakeven reinvestment rates between zero-coupon bonds.

C. 不同期限零息债券目前的到期收益率。
current yield to maturity on zero-coupon bonds of different maturities.

正确答案 / Answer: B

即期利率用于从今天到某一到期日的单笔投资。隐含远期利率则连接两个未来时点，使“直接投资较长期零息债券”与“先投资较短期零息债券、再按远期利率再投资”得到相同终值。它是无套利下的盈亏平衡再投资利率，并不保证未来实际利率等于这个数。

A spot rate prices an investment from today to one maturity date. An implied forward rate connects future dates so that investing directly in a longer zero-coupon bond matches investing first in a shorter zero-coupon bond and then reinvesting at that forward rate. It is a no-arbitrage breakeven reinvestment rate, not a guarantee of the future realized rate.

错误选项为什么不成立 / Why the other options fail

A. 即期利率可由分段远期增长因子组合得出；不能把远期利率直接定义为即期利率的几何平均。

A. Spot rates can be built from successive forward growth factors; a forward rate is not simply the geometric average of spot rates.

C. 这描述的是即期利率。

C. This describes spot rates.

Q059 | 原题 / Original question

2y1y：两年后开始、期限一年的远期利率

2y1y: a one-year forward rate beginning in two years

来源：题库 PDF 第 80 页（固收第 12 页）；答案 PDF 第 59 页。

Source: question bank p. 80 (Fixed Income p. 12); solution PDF p. 59.

分析师观察到以下零息政府债券的到期收益率：

An analyst observes the following yields to maturity on zero-coupon government bonds:

到期期限 Maturity	到期收益率 Yield to Maturity
1 年 1 year	1.5%
2 年 2 years	2.5%
3 年 3 years	3.5%

2y1y 隐含远期利率最接近多少？

The 2y1y implied forward rate is closest to:

A. 4.5%.

B. 5.5%.

C. 6.6%.

正确答案 / Answer: B

零息债券的 YTM 就是相应期限的即期利率。2y1y 表示第 2 年末到第 3 年末的一年利率，不是两年期利率。无套利关系为 (1.035)^3 = (1.025)^2 × (1 + f)，求得 f = 5.5294%，最接近 5.5%。1 年期 1.5% 在本题不需要使用。

For zero-coupon bonds, YTM is the spot rate for that maturity. 2y1y means the one-year rate from year 2 to year 3, not a two-year rate. No arbitrage gives (1.035)^3 = (1.025)^2 × (1 + f), yielding f = 5.5294%, closest to 5.5%. The one-year spot rate of 1.5% is not needed.

1. 把投资到第 3 年与先到第 2 年再投资 1 年连接。

Equate a three-year investment with two years followed by one year.

```
(1 + 0.035)^3 = (1 + 0.025)^2 * (1 + f)
```

2. 解出远期利率。

Solve for the forward rate.

```
f = 1.035^3 / 1.025^2 - 1 = 5.5294% ≈ 5.5%
```

错误选项为什么不成立 / Why the other options fail

A. 4.5% 对应 1y2y：一年后开始、期限两年的远期利率，时间段不同。

A. About 4.5% corresponds to 1y2y, a two-year forward rate beginning in one year, which covers a different interval.

C. 没有使用第 2 年和第 3 年累计增长因子的正确比值。

C. This does not use the correct ratio of the year-3 and year-2 accumulation factors.

> 原资料提示 / 使用条件：答案原页的推导有排版笔误，把部分增长因子写成“(1 + 1.035)^3”。正确应为“(1 + 0.035)^3”或“1.035^3”；原答案 B 及约 5.53% 的结果正确。
>
> Source note / condition: The source solution contains formula typos, including “(1 + 1.035)^3”. The correct factor is “(1 + 0.035)^3” or “1.035^3”. Its answer B and approximately 5.53% result remain correct.

Q073 | 原题 / Original question

用相同终点配对即期与远期增长因子

Match spot and forward growth factors at the same end date

来源：题库 PDF 第 83 页（固收第 15 页）；答案 PDF 第 73 页。

Source: question bank p. 83 (Fixed Income p. 15); solution PDF p. 73.

分析师收集到下列即期利率及远期利率：

An analyst gathers the following spot and forward rates:

两年期即期利率 2-year spot rate	1.0%
四年期即期利率 4-year spot rate	2.5%
两年后开始的四年期远期利率 4-year forward rate, two years from today	3.0%

从今天起四年后开始的两年期远期利率最接近：

The 2-year forward rate, four years from today is closest to:

A. 2%.

B. 3%.

C. 4%.

正确答案 / Answer: A

先画时间线：已知从第2年到第6年的四年期远期利率；要求从第4年到第6年的两年期远期利率。两条投资路径都在第6年结束，因此累计增长因子必须相等。解得年化两年期远期利率约1.9878%，选择A。

Draw the time line first: the known four-year forward rate runs from year 2 to year 6; the requested two-year forward rate runs from year 4 to year 6. Equate the accumulated growth factors of two paths ending at year 6. The annualized two-year forward rate is about 1.9878%, so A is closest.

1. 让两条路径都到达第6年。

Set both paths to end at year 6.

```
(1.01)^2(1.03)^4 = (1.025)^4(1 + f_4,2)^2
```

2. 求两年累计因子，再转换为年利率。

Solve for the two-year factor and annualize it.

```
f_4,2 = [(1.01)^2(1.03)^4/(1.025)^4]^(1/2) - 1 = 1.9878%
```

错误选项为什么不成立 / Why the other options fail

B. 3%覆盖第2年至第6年，并非第4年至第6年；远期期间不同不能直接照搬。

B. The 3% rate covers years 2 to 6, not years 4 to 6. Forward rates for different intervals are not interchangeable.

C. 约4.01%是两年的累计增长率；还要开平方，转换为每年的利率。

C. About 4.01% is the cumulative two-year return. Take its growth factor to the power 1/2 to annualize it.

Q079 | 原题 / Original question

用逐年远期增长因子折现现金流

Discount cash flows with successive forward growth factors

来源：题库 PDF 第 84 页（固收第 16 页）；答案 PDF 第 79 页。

Source: question bank p. 84 (Fixed Income p. 16); solution PDF p. 79.

分析师收集到下列远期利率：

An analyst gathers the following information about forward rates:

时间区间 Time Period	远期利率 Forward Rate
0y1y	1%
1y1y	2%
2y1y	4%

仅使用这些信息，一只三年期、每年付息、票面利率1%的债券，每100面值的价格最接近：

Using only this information, the price per 100 of par value of a 3-year, 1% annual coupon bond is closest to:

A. 84.05.

B. 91.74.

C. 96.23.

正确答案 / Answer: C

0y1y是现在开始的一年期利率；1y1y覆盖第1年至第2年；2y1y覆盖第2年至第3年。第t年的现金流应除以截至该年经历的全部增长因子之积。每年票息为1，末年收到101，现值合计96.2293。

The rate 0y1y applies from now to year 1, 1y1y from year 1 to year 2, and 2y1y from year 2 to year 3. Discount a year-t cash flow by the product of all annual growth factors up to that year. Annual coupons are 1 and the final payment is 101, giving a total value of 96.2293.

1. 分别折现三笔现金流。

Discount the three cash flows separately.

```
P = 1/1.01 + 1/(1.01 × 1.02) + 101/(1.01 × 1.02 × 1.04)
```

2. 相加得到债券价格。

Add the present values.

```
P = 0.990099 + 0.970685 + 94.268477 = 96.229261
```

错误选项为什么不成立 / Why the other options fail

A. 已将各年的增长因子连乘后，又把乘积重复乘方，导致过度折现。

A. It compounds the already-multiplied annual growth factors again, causing excessive discounting.

B. 把逐年远期利率误当成对应期限即期利率，直接分别按1%、2%、4%乘方折现。

B. It incorrectly treats the one-year forward rates as one-, two-, and three-year spot rates.

利差口径和变化 / **Spread measures and spread changes**

Q042 | 原题 / Original question

利差收窄：信用或流动性补偿下降

Spread tightening: lower required credit or liquidity compensation

来源：题库 PDF 第 76 页（固收第 8 页）；答案 PDF 第 42 页。

Source: question bank p. 76 (Fixed Income p. 8); solution PDF p. 42.

公司债收益率利差收窄最可能与以下哪种情况相关？

Tightening corporate bond yield spreads are most likely associated with:

A. 发行人信用状况恶化。
issuers' deteriorating creditworthiness.

B. 公司债需求旺盛的时期。
periods of high demand for corporate bonds.

C. 经纪交易商做市能力与意愿下降。
broker-dealers' reduced ability and willingness to make markets.

正确答案 / Answer: B

收益率利差是公司债收益率超过相应基准收益率的部分。需求旺盛时，买盘提高债券价格、压低相对收益率，利差往往收窄。信用恶化或市场流动性下降则通常使投资者要求更高补偿，利差扩大。

A yield spread is the corporate bond yield above an appropriate benchmark. Strong demand raises bond prices and reduces relative yields, so spreads tend to tighten. Deteriorating credit quality or weaker market liquidity generally increases the compensation investors demand and widens spreads.

错误选项为什么不成立 / Why the other options fail

A. 信用恶化通常提高信用风险补偿，使利差扩大。

A. Deteriorating creditworthiness normally increases the required spread.

C. 做市能力下降会削弱流动性，通常扩大流动性风险溢价。

C. Reduced market making usually worsens liquidity and increases the liquidity premium.

Q093 | 原题 / Original question

I-spread以同币种、同期限互换利率为基准

I-spread uses a same-currency, same-tenor swap benchmark

来源：题库 PDF 第 87 页（固收第 19 页）；答案 PDF 第 93 页。

Source: question bank p. 87 (Fixed Income p. 19); solution PDF p. 93.

某债券收益率相对于同币种、同期限标准互换利率的利差，最恰当的名称是：

The yield spread of a specific bond over the standard swap rate in that currency of the same tenor best describes the:

A. I利差。
I-spread.

B. Z利差。
Z-spread.

C. 期权调整利差。
option-adjusted spread.

正确答案 / Answer: A

看到swap rate即可锁定I-spread：将债券YTM与相同币种、相同期限的互换利率相减。期限不完全匹配时可以插值，但字母I不是“只要插值都叫I-spread”。

The swap-rate benchmark identifies the I-spread: bond YTM minus the swap rate in the same currency and tenor. Interpolation may be needed to match tenor, but not every spread involving interpolation is an I-spread.

错误选项为什么不成立 / Why the other options fail

B. Z-spread是加到整条基准即期曲线上、使现金流现值等于价格的固定利差。

B. A Z-spread is a constant spread added to a benchmark spot curve so that discounted cash flows match price.

C. OAS还要借助含权定价模型剔除嵌入期权影响，题目没有这种处理。

C. OAS uses an option-pricing model to adjust for embedded-option effects; that procedure is not described here.

Q096 | 原题 / Original question

G-spread取政府债收益率作为基准

G-spread uses a government-bond yield benchmark

来源：题库 PDF 第 88 页（固收第 20 页）；答案 PDF 第 96 页。

Source: question bank p. 88 (Fixed Income p. 20); solution PDF p. 96.

相对于经插值处理的主权债券收益率计算的利差，最恰当的名称是：

The yield spread over an interpolated sovereign bond is best described as a(n):

A. I利差。
I-spread.

B. G利差。
G-spread.

C. Z利差。
Z-spread.

正确答案 / Answer: B

关键是基准为government/sovereign bond，因此选G-spread。即使政府债期限要插值才能与目标债匹配，其基准仍然是政府债；不能因出现interpolated一词就选I-spread。

The benchmark is a government or sovereign bond, so the answer is G-spread. Interpolating government yields to match the target maturity does not turn the benchmark into a swap rate. The word “interpolated” alone does not imply I-spread.

错误选项为什么不成立 / Why the other options fail

A. I-spread使用互换利率，而非政府债收益率。

A. I-spread uses a swap-rate benchmark, not a government-bond yield.

C. Z-spread对基准即期曲线的每个现金流折现率加上相同利差，不是简单对一个政府债收益率相减。

C. Z-spread adds a constant spread to the spot rates used for all cash-flow discounting, rather than subtracting one government-bond yield.

CHAPTER 04 / 27 ORIGINAL QUESTIONS

久期：选对指标，再算敏感度

Duration: choose the measure, then calculate sensitivity

> 先判断变动对象是 YTM 还是基准曲线；再判断需要百分比、金额还是组合风险。
> Determine whether the shock affects YTM or the benchmark curve, then whether the answer requires percentage, money or portfolio risk.

先做这些原题检验理解：Q48, Q25, Q55, Q56, Q78。
Start with these original checkpoints: Q48, Q25, Q55, Q56, Q78.

4.1 四种久期放在一张表里

4.1 Four duration measures in one table

指标 / Measure	核心意思与使用场景 / Meaning and use
麦考利久期 / Macaulay duration	现金流现值加权的平均收款时间；以年计。 PV-weighted average time to receive cash flows; measured in years.
修正久期 / Modified duration	价格对 YTM 微小变动的百分比敏感度；通常用于现金流固定的债券。 Percentage price sensitivity to a small YTM change, generally for fixed cash flows.
有效久期 / Effective duration	基准曲线变化并重新估值后的价格敏感度；允许现金流随利率变化。 Sensitivity to a benchmark-curve shift with repricing, allowing rate-dependent cash flows.
关键利率久期 / Key rate duration	只改变曲线某一期限点；分析陡峭化、平坦化等形状变化。 Sensitivity to a selected curve maturity; useful for steepening, flattening and other shape changes.

含权债券的赎回或提前还款会改变现金流路径，因此 Q33、Q34、Q40 指向有效久期。无期权的普通美国国债在 Q46 中可用修正久期。Q3、Q52 看到曲线形状变化，优先想到关键利率久期。
Options can alter cash flows through calls or prepayments, so Q33, Q34 and Q40 point to effective duration. The plain US Treasury in Q46 can use modified duration. Curve-shape changes in Q3 and Q52 call for key rate duration.

$$D_{\text{Mac}} = \frac{\sum_t t\,PV(CF_t)}{P_{\text{full}}}$$

零息债券只在到期收到一笔钱，所以 MacDur = 到期期限（Q76）。附息债券较早收回部分现金，通常 MacDur 小于到期期限。
A zero-coupon bond has only one payment, so its Macaulay duration equals maturity (Q76). Coupon bonds return some cash earlier and generally have a shorter Macaulay duration than maturity.

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1 + y/m}$$

换算用收益率，不是票息率。Q48 是年度付息：ModDur = 10/1.045 = 9.5694。
Use the yield, not the coupon rate, in the conversion. Q48 pays annually: ModDur = 10/1.045 = 9.5694.

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\Delta y$$

Q48：利率下降 100 bp = −0.01；估计涨幅 = −9.5694 × (−0.01) = 9.5694%，选 B。负号体现价格与收益率反向。

In Q48, a 100 bp fall means a yield change of −0.01. The estimated price gain is −9.5694 x (−0.01) = 9.5694%, answer B. The minus sign captures the inverse price-yield relationship.

Q83 的一般比较：其他条件相同，较高 YTM 会使较远期现金流现值衰减得更多，权重偏向较早收款时点，因此降低 Macaulay 久期。原题声称三者均溢价与表内数据矛盾，详见该题提示。

For Q83, all else equal, a higher YTM discounts distant cash flows more heavily and shifts present-value weights toward earlier receipts, reducing Macaulay duration. Its claim that all three bonds trade at a premium contradicts the table; see the source note.

4.2 由上下变动后的价格估算

4.2 Estimate duration from up and down scenarios

$$D \approx \frac{P_- - P_+}{2P_0\Delta y}$$

P_minus 是利率下降后的价格；P_plus 是利率上升后的价格。delta_y 是单边冲击幅度，不是上下情景之间的总差。输入必须转为小数。

P_minus is the price after a yield decrease and P_plus after an increase. delta_y is the one-sided shock, not the entire gap between scenarios. Express it as a decimal.

若扰动对象为 YTM，这个差分近似给出修正久期；若扰动基准曲线并重新计算含权现金流，得到有效久期。公式外形类似，经济含义取决于被扰动的变量。

If the shock changes YTM, the difference quotient approximates modified duration. If it shifts the benchmark curve and reprices option-dependent cash flows, it estimates effective duration. Similar algebra does not imply the same risk measure.

> 原题 Q25：delta_y = 0.1% = 0.001；D_eff = (100.75 − 99.26)/(2 × 100 × 0.001) = 7.45，选 B。原题 Q26：delta_y = 0.5% = 0.005，不能把 0.5 直接代入。
>
> Original Q25: delta_y = 0.1% = 0.001 and D_eff = (100.75 − 99.26)/(2 x 100 x 0.001) = 7.45, answer B. In Q26, delta_y = 0.5% = 0.005, not 0.5.

Q50 给上下 YTM 情景，先算近似修正久期 4.45，再乘 1.04，得到 MacDur = 4.628。题目问哪种久期，要在最后再检查一次。

Q50 gives YTM scenarios: first estimate modified duration as 4.45, then multiply by 1.04 to get MacDur = 4.628. Recheck which duration measure the question actually requests.

4.3 从百分比敏感度到金额敏感度

4.3 Convert percentage risk to money risk

$$\text{Money duration} = D_{\text{mod}}\,V_{\text{full}}$$

用全价或全市场价值，不是面值，也不是净价。Q84 先将 114.75 + 1.625 = 116.375，再乘 4.825，得到 561.5094。

Use the full price or full market value, not par or clean price. Q84 first adds 114.75 + 1.625 = 116.375, then multiplies by 4.825 to obtain 561.5094.

$$PVBP \approx D_{\text{mod}} V_{\text{full}} (0.0001)$$

1 bp = 0.01% = 0.0001。Q55：103.50 × 6.2 × 0.0001 = 0.06417，每 100 面值的价格单位。不要误读成整笔持仓的美元金额。

1 bp = 0.01% = 0.0001. Q55: 103.50 x 6.2 x 0.0001 = 0.06417 in price units per 100 par. It is not automatically the dollar value for an entire holding.

$$\Delta V \approx -D_{\text{mod}} V_{\text{full}} \Delta y$$

Q54 用市场价值 912,575，不用面值 1,000,000。2.4 × 912,575 × 0.005 = 10,950.90，因此约为 +11,000。

Q54 uses the market value 912,575 rather than par 1,000,000. 2.4 x 912,575 x 0.005 = 10,950.90, giving an approximate gain of 11,000.

4.4 组合久期一定先看市值权重

4.4 Portfolio duration uses market-value weights

$$D_{\text{portfolio}} = \sum_i w_i D_i$$

Q86：权重是 120,000/300,000 与 180,000/300,000，组合久期 = 0.4 × 5 + 0.6 × 4 = 4.40。不是用面值 100,000 与 200,000 加权。

Q86 weights are 120,000/300,000 and 180,000/300,000. Portfolio duration is 0.4 x 5 + 0.6 x 4 = 4.40. Do not weight by the par amounts 100,000 and 200,000.

Q78 给的是每只债券的 Macaulay 久期，而题目要组合修正久期。先逐只转换，再按市值加权；不要先平均 YTM 后统一转换。Q56 给 money duration，则可以先加总金额敏感度，再除总市值。

Q78 gives individual Macaulay durations but asks for portfolio modified duration. Convert each bond first, then market-value weight the results. Do not average yields first. In Q56, aggregate money duration and divide by total market value.

4.5 久期的边界条件

4.5 Conditions and limitations

Q19 的“付息日接近”是指两次付息之间：在其他条件不变时，Macaulay 久期随时间推移下降。付息发生后，最早现金流从剩余现金流集合消失，可出现跳升，不能把一个方向延伸到整个付息过程。

Q19 concerns approaching a coupon date between payments: holding other factors constant, Macaulay duration falls as time passes. Immediately after payment, removal of the earliest cash flow can cause an upward jump; the between-date trend does not describe the entire coupon cycle.

Q44 的准确读法是：在一致模型与曲线划分下，各关键利率久期之和近似对应平行移动的有效久期，而不是每一个关键利率久期都等于有效久期。Q85 的原答案为 C：在题目采用的标准附息债框架中，完全平坦的收益率曲线使有效久期与修正久期一致；不要反过来认为二者在所有无期权债券上永远相等。严格数学意义上的单笔现金流等例外，在该题提示中另作说明。

Q44 is best read as the sum of key rate durations approximating effective duration for a parallel curve shift under a consistent model, not each key rate duration being identical. The source answer to Q85 is C: a perfectly flat curve aligns effective and modified duration in its standard coupon-bond framework. Do not assume they are always equal for every option-free bond. Mathematical exceptions, including single-cash-flow cases, are noted with the question.

> 本章四步检查：冲击变量 → 久期类别 → bp 转小数 → 百分比/金额/组合单位。每次错题标出错在哪一步，重复题就会明显减少。
>
> Four checks: identify the shocked variable, select the duration measure, convert bp to decimals and verify percentage, money or portfolio units. Tag the failed step when reviewing an error.

四种久期如何选择 / Choosing among duration measures

Q027 | 原题 / Original question

曲线久期与收益率久期

Curve duration versus yield duration

来源：题库 PDF 第 73 页（固收第 5 页）；答案 PDF 第 27 页。

Source: question bank p. 73 (Fixed Income p. 5); solution PDF p. 27.

下列哪项属于曲线久期指标？

Which of the following is a curve duration measure?

A. 修正久期。
Modified duration

B. 有效久期。
Effective duration

C. 麦考利久期。
Macaulay duration

正确答案 / Answer: B

有效久期通过移动基准收益率曲线并重新估值来衡量价格敏感度，因此属于曲线久期。修正久期围绕债券自身 YTM 的变化；麦考利久期是现金流回收时间的现值加权平均。

Effective duration reprices the bond after shifting a benchmark yield curve, so it is a curve-duration measure. Modified duration relates price to the bond's own YTM, while Macaulay duration is the present-value-weighted average timing of cash flows.

错误选项为什么不成立 / Why the other options fail

A. 修正久期针对债券自身的到期收益率变化。

A. Modified duration is a yield-duration measure based on the bond's own YTM.

C. 麦考利久期首先描述现金流的加权回收时间，不是基准曲线位移敏感度。

C. Macaulay duration describes weighted cash-flow timing, not sensitivity to a benchmark-curve shift.

Q033 | 原题 / Original question

有内嵌期权时使用有效久期

Use effective duration for embedded options

来源：题库 PDF 第 74 页（固收第 6 页）；答案 PDF 第 33 页。

Source: question bank p. 74 (Fixed Income p. 6); solution PDF p. 33.

对于含有内嵌期权的债券，衡量其价格对利率变化的敏感度，最合适的指标是：

For bonds with embedded options, the most appropriate measure of price sensitivity to interest rate changes is:

A. 有效久期。
effective duration.

B. 修正久期。
modified duration.

C. 麦考利久期。
Macaulay duration.

正确答案 / Answer: A

利率变化可能改变赎回、回售或提前还款行为，进而改变现金流。有效久期在基准曲线上下移动后重新估值，并允许模型中的现金流随之调整，因此最适合此类债券。

Rate changes can alter calls, puts, or prepayments and therefore change expected cash flows. Effective duration reprices the instrument after benchmark-curve shifts while allowing the model's cash flows to adjust, making it the appropriate measure.

错误选项为什么不成立 / Why the other options fail

B. 修正久期通常假设现金流固定，无法充分反映期权引起的现金流变化。

B. Modified duration ordinarily assumes fixed cash flows and does not fully capture option-driven changes.

C. 麦考利久期以既定现金流的现值权重计算，不能充分处理这种不确定性。

C. Macaulay duration weights a specified set of cash flows and does not adequately handle this uncertainty.

Q034 | 原题 / Original question

有效久期的原因是现金流会变化

Changing cash flows motivate effective duration

来源：题库 PDF 第 74 页（固收第 6 页）；答案 PDF 第 34 页。

Source: question bank p. 74 (Fixed Income p. 6); solution PDF p. 34.

对于含有内嵌期权的债券，有效久期是最适合的利率风险指标，因为该债券的：

For a bond with an embedded option, effective duration is the most appropriate measure of interest rate risk because the bond's:

A. 未来现金流不确定。
future cash flows are uncertain.

B. 内部收益率有明确定义。
internal rate of return is well-defined.

C. 定价对信用利差变化敏感。
pricing is sensitive to changes in credit spreads.

正确答案 / Answer: A

内嵌期权使未来现金流取决于利率路径和行权行为。例如利率降低时，发行人可能提前赎回，原本假定的未来票息便不再发生。有效久期重新评估各利率情景下的价格，可把这些现金流变化纳入。

Embedded options make future cash flows contingent on interest-rate paths and exercise decisions. A call after rates fall, for example, can eliminate coupons previously expected in later years. Effective duration incorporates such changes through scenario repricing.

错误选项为什么不成立 / Why the other options fail

B. 恰恰因为实际现金流不确定，以固定现金流定义的单一收益率未必能可靠表达风险。

B. The uncertainty of realized cash flows limits the usefulness of a single fixed-cash-flow yield for measuring this risk.

C. 信用利差敏感性不是选择有效久期处理内嵌期权的核心原因。

C. Credit-spread sensitivity is not the central reason for using effective duration for embedded options.

Q040 | 原题 / Original question

现金流不确定时使用有效久期

Use effective duration when cash flows depend on rates

来源：题库 PDF 第 76 页（固收第 8 页）；答案 PDF 第 40 页。

Source: question bank p. 76 (Fixed Income p. 8); solution PDF p. 40.

有效久期是含嵌入期权债券最适合的利率风险指标，因为该债券没有明确定义的：

Effective duration is the most appropriate measure of interest rate risk for a bond with an embedded option because the bond does not have a well-defined:

A. 有效凸性。
effective convexity.

B. 内部收益率。
internal rate of return.

C. 收益率曲线久期指标。
curve duration statistic.

正确答案 / Answer: B

赎回、回售或提前偿还会使现金流随利率路径而改变，因此不能把一套确定现金流的单一内部收益率当作完整的利率风险基础。有效久期对基准曲线上下移动后重新估值，并允许现金流随之变化。这里的内部收益率不确定，是针对含权债券真实的、依赖行权的现金流。

Calls, puts or prepayments can change cash flows as rates change. A single internal rate of return based on fixed cash flows is therefore an inadequate basis for measuring the full rate risk. Effective duration reprices the bond after benchmark-curve shifts while allowing option-dependent cash flows to change. The issue is the contingent cash-flow stream.

错误选项为什么不成立 / Why the other options fail

A. 有效凸性同样可以通过曲线移动后的重新估值计算。

A. Effective convexity can also be estimated by repricing after curve shifts.

C. 有效久期本身就是收益率曲线久期指标。

C. Effective duration is itself a curve-duration measure.

> 原资料提示 / 使用条件：题目沿用“没有明确定义的内部收益率”的教材表述。可以针对假定现金流计算 YTM 或 YTC，但它们不能消除实际行权及现金流的不确定性。
>
> Source note / condition: The source uses the textbook phrase “no well-defined internal rate of return.” A YTM or YTC can still be calculated for assumed cash flows, but it does not remove uncertainty about exercise and realized cash flows.

Q046 | 原题 / Original question

固定现金流用修正久期，含权现金流用有效久期

Modified duration for fixed cash flows; effective duration for contingent cash flows

来源：题库 PDF 第 77 页（固收第 9 页）；答案 PDF 第 46 页。

Source: question bank p. 77 (Fixed Income p. 9); solution PDF p. 46.

修正久期最适合衡量以下哪种证券的利率风险？

Modified duration is the most appropriate measure of interest rate risk for which of the following securities?

A. 可赎回债券。
Callable bond

B. 美国国债。
US Treasury bond

C. 抵押贷款支持债券。
Mortgage-backed bond

正确答案 / Answer: B

本题的美国国债按标准无嵌入期权、固定现金流债券理解，因此可用修正久期衡量价格对其 YTM 的敏感度。可赎回债券会受发行人行权影响，抵押贷款支持债券会受提前还款影响，两者现金流随利率变化，应优先使用有效久期。

The Treasury bond in this question is understood as a standard option-free bond with fixed cash flows. Modified duration therefore measures its price sensitivity to YTM. A callable bond can be redeemed early, while a mortgage-backed bond is exposed to prepayments; both have rate-dependent cash flows and are better assessed with effective duration.

错误选项为什么不成立 / Why the other options fail

A. 赎回权会改变现金流，优先用有效久期。

A. The call option can change cash flows, favoring effective duration.

C. 提前偿还会改变现金流，优先用有效久期。

C. Mortgage prepayments can change cash flows, favoring effective duration.

Q047 | 原题 / Original question

先看冲击变量：自身 YTM 还是基准曲线

Identify the shock: own YTM or benchmark curve

来源：题库 PDF 第 77 页（固收第 9 页）；答案 PDF 第 47 页。

Source: question bank p. 77 (Fixed Income p. 9); solution PDF p. 47.

已知债券到期收益率的变动，最适合估计债券价格百分比变化的指标是：

The percentage price change for a bond, given a change in its yield to maturity, is best estimated by:

A. 有效久期。
effective duration.

B. 修正久期。
modified duration.

C. 麦考利久期。
Macaulay duration.

正确答案 / Answer: B

关键词是“其到期收益率的变动”。一阶估计为价格百分比变动约等于负的修正久期乘收益率变化，收益率必须用小数。麦考利久期要先除以每期增长因子才能转为修正久期；有效久期则由基准曲线移动后的重新估值计算。

The wording specifies a change in the bond’s yield to maturity. The first-order percentage price estimate is minus modified duration times the yield change, expressed as a decimal. Macaulay duration must first be converted to modified duration; effective duration instead uses repricing after benchmark-curve shifts.

1. 一阶价格近似。

First-order price approximation.

```
DeltaP / P ≈ -ModDur * DeltaYTM
```

错误选项为什么不成立 / Why the other options fail

A. 适用于基准收益率曲线冲击，尤其是现金流会变化的含权债券。

A. It addresses benchmark-curve shocks, especially for bonds whose cash flows can change.

C. 它是时间加权指标；不能直接作为对 YTM 的价格敏感度系数。

C. It is a time-weighted measure and is not directly the price-sensitivity coefficient to YTM.

Q003 | 原题 / Original question

关键利率久期识别收益率曲线形状变化

Key rate duration identifies changes in yield-curve shape

来源：题库 PDF 第 69 页（固收第 1 页）；答案 PDF 第 3 页。

Source: question bank p. 69 (Fixed Income p. 1); solution PDF p. 3.

关键利率久期最适合衡量债券对下列哪种变化的敏感程度？

The key rate duration best measures a bond's sensitivity to a change in the:

A. 到期收益率水平的变化。
level of the yield-to-maturity.

B. 最差收益率曲线斜率的变化。
slope of the yield-to-worst curve.

C. 基准收益率曲线形状的变化。
shape of the benchmark yield curve.

正确答案 / Answer: C

关键利率久期只改变某一特定期限点或期限区间的基准利率，并衡量债券价格反应。把多个期限点的敏感度放在一起，可以分析曲线变陡、变平等非平行变动。

Key rate duration changes the benchmark rate at a selected maturity point or segment and measures the price response. The resulting set of sensitivities helps analyze nonparallel curve shifts, including steepening and flattening.

错误选项为什么不成立 / Why the other options fail

A. 对债券自身到期收益率变化的敏感度主要由修正久期衡量。

A. Modified duration measures sensitivity to changes in the bond’s own yield to maturity.

B. 最差收益率是多个可能偿还情景中最低的收益率，不是关键利率久期所使用的曲线。

B. Yield to worst is the lowest yield across relevant redemption scenarios; it is not the curve underlying key rate duration.

Q052 | 原题 / Original question

非平行曲线变动使用关键利率久期

Key rate duration for nonparallel curve shifts

来源：题库 PDF 第 78 页（固收第 10 页）；答案 PDF 第 52 页。

Source: question bank p. 78 (Fixed Income p. 10); solution PDF p. 52.

以下哪一种久期指标最适合衡量债券价格对收益率曲线变平的敏感度？

Which of the following duration statistics best measures the sensitivity of a bond's price to a flattening of the yield curve?

A. 关键利率久期。
Key rate duration

B. 有效久期。
Effective duration

C. 麦考利久期。
Macaulay duration

正确答案 / Answer: A

收益率曲线变平表示长短端利率的变化幅度不同，是非平行移动。关键利率久期把风险拆到不同期限节点，能够分别估计短端、长端变化的价格影响。单一有效久期主要概括平行移动，不能完整描述曲线形状变化。

A flattening curve means short- and long-maturity yields move by different amounts: a nonparallel shift. Key rate durations separate exposure by maturity point and can estimate the price effects of these different moves. A single effective duration principally summarizes a parallel shift and cannot fully describe a change in curve shape.

错误选项为什么不成立 / Why the other options fail

B. 主要概括整条基准曲线平行移动的敏感度。

B. It primarily summarizes sensitivity to a parallel benchmark-curve shift.

C. 是现金流现值加权的回收时间，无法独立刻画不同期限节点的风险。

C. It is a cash-flow-weighted timing measure and does not isolate maturity-point exposures.

Q044 | 原题 / Original question

平行移动下：各关键利率久期加总

Parallel shifts: sum the key rate durations

来源：题库 PDF 第 76 页（固收第 8 页）；答案 PDF 第 44 页。

Source: question bank p. 76 (Fixed Income p. 8); solution PDF p. 44.

对于含嵌入期权的债券，当基准收益率曲线平行移动时，有效久期最可能与以下哪项反映相同的利率敏感度？

With respect to a bond with an embedded option, for parallel shifts in the benchmark yield curve, effective duration most likely indicates the same interest rate sensitivity as:

A. 关键利率久期。
key rate durations.

B. 修正久期。
modified duration.

C. 麦考利久期。
Macaulay duration.

正确答案 / Answer: A

关键利率久期分别衡量曲线上不同期限节点的局部变动；有效久期衡量整条基准曲线的平行移动。采用一致模型和微小冲击时，所有节点同时等幅变动的总效应，约等于各关键利率久期之和。因此本题选择 A。不能把某一个节点的久期直接等同于整只债券的有效久期。

Each key rate duration measures sensitivity to a local maturity point on the benchmark curve; effective duration measures a parallel curve shift. Under consistent modeling and small shocks, shifting all points equally gives an aggregate sensitivity approximately equal to the sum of the key rate durations. Thus A is intended. A single key rate duration is not generally equal to effective duration.

错误选项为什么不成立 / Why the other options fail

B. 修正久期针对债券自身 YTM 的变动，不能完整处理含权现金流变化。

B. Modified duration measures sensitivity to the bond's own YTM and does not fully accommodate option-driven cash-flow changes.

C. 麦考利久期是现金流现值的时间加权指标，不是含权债券平移曲线后的完整价格敏感度。

C. Macaulay duration is a cash-flow-weighted time measure, not the full curve-shift sensitivity of an option-bearing bond.

> 原资料提示 / 使用条件：原选项 A 只写 key rate durations。准确理解应为“一组关键利率久期所反映的合计敏感度”，通常表述为其总和约等于有效久期。
>
> Source note / condition: Option A says only “key rate durations.” Read it as their aggregate sensitivity: the sum is approximately equal to effective duration under consistent assumptions.

Q085 | 原题 / Original question

有效久期与修正久期的定价变量不同

Effective and modified duration use different pricing variables

来源：题库 PDF 第 86 页（固收第 18 页）；答案 PDF 第 85 页。

Source: question bank p. 86 (Fixed Income p. 18); solution PDF p. 85.

有效久期：

Effective duration is:

A. 只对含嵌入期权的债券才是有用的利率风险指标。
a useful interest rate risk measure only for bonds with embedded options.

B. 只有假设的基准利率变动很小时，才可以准确估计利率风险。
an accurate estimate of interest rate risk only when the assumed benchmark rate change is small.

C. 只有收益率曲线完全平坦时，才与无期权债券的修正久期相同。
the same as the modified duration of an option-free bond only when the yield curve is perfectly flat.

正确答案 / Answer: C

按所附答案的标准附息债口径选择C。修正久期改变单一YTM；有效久期则平移基准收益率曲线，并允许模型重估现金流。完全平坦的曲线配合一致的复利及平移假设，使两者能够对应。有效久期也可用于无期权债。

C is the supplied answer under the source's standard coupon-bond convention. Modified duration changes one YTM, while effective duration shifts the benchmark curve and allows the model to re-estimate cash flows. An entirely flat curve with consistent compounding and shift assumptions aligns the measures. Effective duration is also useful for option-free bonds.

错误选项为什么不成立 / Why the other options fail

A. 有效久期并不限于含权债券；普通债券也可用它衡量基准曲线风险。

A. Effective duration is not restricted to bonds with embedded options; it also measures benchmark-curve risk for ordinary bonds.

B. 含权模型中，赎回或再融资可能需要足够的利率变动才触发。减小冲击幅度不一定改善风险估计。

B. In option models, calling or refinancing may require a sufficiently large rate move to overcome transaction costs. A smaller assumed shift does not necessarily improve the risk estimate.

> 原资料提示 / 使用条件：“只有完全平坦才相等”是答案材料在标准附息债框架下的简化表述，不应当推广为所有数学情形的必要条件；单笔现金流等特殊情形可以例外。学习重点是两者的风险变量不同，不能仅凭“无期权”就默认相等。
>
> Source note / condition: The source's “only when flat” statement is a simplification for its standard coupon-bond framework, not a universal mathematical necessity; single-cash-flow cases can be exceptions. The key lesson is that the measures use different risk variables, so “option-free” alone does not justify treating them as equal.

现金流时间与久期换算 / Cash-flow timing and duration conversion

Q019 | 原题 / Original question

票息支付日之间的麦考利久期变化

Macaulay duration between coupon dates

来源：题库 PDF 第 72 页（固收第 4 页）；答案 PDF 第 19 页。

Source: question bank p. 72 (Fixed Income p. 4); solution PDF p. 19.

对于一只以溢价交易、没有内嵌期权的固定利率债券，随着下一次付息日临近，麦考利久期最可能：

For an option-free fixed-rate bond trading at a premium, as the coupon payment date approaches, Macaulay duration most likely.

A. 下降。
decreases

B. 保持不变。
remains constant.

C. 上升。
increases.

正确答案 / Answer: A

麦考利久期是按现金流现值加权的平均等待时间。在下一次票息尚未付出前，时间推移使每笔尚未收到的现金流都更接近，因此久期下降。注意区分“付息日临近”与“刚付完票息”：刚付完时，最短的一笔现金流被移除，久期可能向上跳动。

Macaulay duration is the present-value-weighted average waiting time for cash flows. Before the next coupon is paid, each remaining cash flow moves closer as time passes, so duration declines. Distinguish approaching the payment date from immediately after payment, when removing the nearest cash flow can cause an upward jump.

错误选项为什么不成立 / Why the other options fail

B. 剩余现金流的等待时间正在缩短，久期不会仅因票息固定就保持不变。

B. Waiting times shorten; a fixed coupon does not make duration constant.

C. 上升通常对应付息后的离散跳动，而非题目所问的付息日前临近过程。

C. An upward jump can occur just after payment, not during the approach to that payment.

Q076 | 原题 / Original question

麦考利久期是现金流收回时间的现值加权平均

Macaulay duration as the present-value-weighted time to cash flows

来源：题库 PDF 第 84 页（固收第 16 页）；答案 PDF 第 76 页。

Source: question bank p. 84 (Fixed Income p. 16); solution PDF p. 76.

零息债券的麦考利久期最可能：

The Macaulay duration of a zero-coupon bond is most likely:

A. 短于到期年限。
less than the time to maturity.

B. 等于到期年限。
equal to the time to maturity.

C. 长于到期年限。
greater than the time to maturity.

正确答案 / Answer: B

零息债券在到期前没有票息，全部现金流都发生在到期日，因此100%的现值权重都落在到期日，麦考利久期正好等于到期年限。不要把这个结论套用到修正久期。

A zero-coupon bond pays no interim coupons. All cash flow occurs at maturity, so 100% of its present-value weight is at that date. Its Macaulay duration therefore equals maturity. This equality does not apply to modified duration.

错误选项为什么不成立 / Why the other options fail

A. 有提前票息的普通附息债才通常具有短于到期年限的麦考利久期。

A. A conventional coupon bond usually has Macaulay duration below maturity because some cash arrives earlier.

C. 本题所有现金流均不晚于到期日，加权平均收款时间不可能超过它。

C. No cash flow occurs after maturity, so the weighted-average receipt time cannot exceed it.

Q048 | 原题 / Original question

从麦考利久期转换到修正久期

Convert Macaulay duration to modified duration

来源：题库 PDF 第 77 页（固收第 9 页）；答案 PDF 第 48 页。

Source: question bank p. 77 (Fixed Income p. 9); solution PDF p. 48.

分析师收集了以下年付息债券的信息：

An analyst gathers the following information about an annual-pay bond:

票息率 Coupon rate	6.0%
到期收益率 Yield to maturity	4.5%
麦考利久期 Macaulay duration	10.0

如果到期收益率下降 100 个基点，债券价格的预计百分比变动最接近多少？

If the yield to maturity decreases by 100 basis points, the expected percentage change in the bond's price is closest to:

A. 9.43%

B. 9.57%

C. 10.00%.

正确答案 / Answer: B

债券一年付息一次，故用 YTM 4.5% 而非票息 6% 进行转换。修正久期为 10/1.045 = 9.5694。100 个基点等于 0.01；收益率下降意味着变化量为 -0.01，价格近似上涨 9.5694%。题目未提供凸性，因此使用久期的一阶近似。

With annual coupons, convert using the 4.5% YTM, not the 6% coupon: modified duration is 10/1.045 = 9.5694. A 100-basis-point decline is a yield change of -0.01, giving an estimated 9.5694% price increase. Convexity is not supplied, so this is a first-order duration estimate.

1. 先把麦考利久期转为修正久期。

Convert to modified duration.

```
ModDur = 10 / (1 + 0.045) = 9.5694
```

2. 代入收益率下降的负号。

Apply the negative yield change.

```
DeltaP / P ≈ -9.5694 * (-0.01) = 9.57%
```

错误选项为什么不成立 / Why the other options fail

A. 用票息 6% 作分母，10/1.06 = 9.4340，误把票息当作贴现率。

A. Using 10/1.06 = 9.4340 incorrectly substitutes the coupon rate for the yield.

C. 直接用麦考利久期乘收益率变动，漏掉了转为修正久期的一步。

C. This uses Macaulay duration directly without converting to modified duration.

Q050 | 原题 / Original question

先用价格差估算修正久期，再还原麦考利久期

Estimate modified duration from prices, then recover Macaulay duration

来源：题库 PDF 第 78 页（固收第 10 页）；答案 PDF 第 50 页。

Source: question bank p. 78 (Fixed Income p. 10); solution PDF p. 50.

分析师收集了一只票息率 4%、一年付息一次、当前到期收益率 4.0% 的债券信息：

An analyst gathers the following information about a 4% annual-payment bond with a current yield-to-maturity of 4.0%:

年化到期收益率 Annualized Yield to Maturity	债券价格 Bond Price
3.9%	100.45
4.0%	100.00
4.1%	99.56

该债券的年化麦考利久期最接近多少？

The bond's annualized Macaulay duration is closest to:

A. 4.28.

B. 4.45.

C. 4.63.

正确答案 / Answer: C

上下各移动 10 个基点，因此分母的单边收益率变化为 0.001。先算修正久期：(100.45 - 99.56)/(2 × 100 × 0.001) = 4.45。题目要的是年化麦考利久期，年付息时应再乘以 1.04，得到 4.628，选 4.63。

The yield moves 10 basis points in either direction, so the one-sided change is 0.001. Approximate modified duration is (100.45 - 99.56)/(2 × 100 × 0.001) = 4.45. The question asks for annualized Macaulay duration, so multiply by 1.04 for this annual-pay bond: 4.628, or 4.63.

1. 求近似修正久期。

Find approximate modified duration.

```
ModDur ≈ (100.45 - 99.56) / (2 * 100 * 0.001) = 4.45
```

2. 乘以每期增长因子。

Multiply by the per-period growth factor.

```
MacDur = 4.45 * 1.04 = 4.628 ≈ 4.63
```

错误选项为什么不成立 / Why the other options fail

A. 把修正久期再次除以 1.04，转换方向反了。

A. It divides modified duration by 1.04, reversing the required conversion.

B. 4.45 是修正久期，尚未转为麦考利久期。

B. 4.45 is modified duration, before conversion to Macaulay duration.

Q083 | 原题 / Original question

比较票息、收益率及期限对久期的共同影响

Compare the effects of coupon, yield, and maturity on duration

来源：题库 PDF 第 85 页（固收第 17 页）；答案 PDF 第 83 页。

Source: question bank p. 85 (Fixed Income p. 17); solution PDF p. 83.

分析师收集到三只无嵌入期权债券的下列信息，题干称它们均以溢价交易：

An analyst gathers the following information about three option-free bonds, each trading at a premium:

	票面利率 Coupon	到期收益率 Yield To Maturity	到期年限 Time to Maturity
债券1 Bond 1	4%	4%	10年 10 years
债券2 Bond 2	4%	5%	11年 11 years
债券3 Bond 3	5%	5%	9年 9 years

其他条件相同时，麦考利久期最低的债券最可能是：

All else being equal, the bond with the lowest Macaulay duration is most likely:

A. 债券1。
Bond 1.

B. 债券2。
Bond 2.

C. 债券3。
Bond 3.

正确答案 / Answer: C

按表中数据，债券3期限最短、票息最高，且收益率不低于其余两只。较早收款和较高票息都使现金流现值权重偏向较早时点，支持其麦考利久期最低。原答案选择C。

Using the table, Bond 3 has the shortest maturity, the highest coupon, and a yield no lower than either other bond. Earlier repayment and higher coupons shift present-value weights toward earlier dates. The supplied answer is C, the lowest Macaulay duration.

错误选项为什么不成立 / Why the other options fail

A. 债券1相对债券3期限更长、票息更低、收益率更低，不支持其久期最低。

A. Relative to Bond 3, Bond 1 has a longer maturity, a lower coupon, and a lower yield.

B. 债券2相对债券3收益率相同，但期限更长、票息更低。

B. Relative to Bond 3, Bond 2 has the same yield but a longer maturity and a lower coupon.

> 原资料提示 / 使用条件：原题存在内部矛盾：在票息与YTM采用相同频率口径的通常定价下，债券1和3的票息等于YTM，应为平价；债券2票息低于YTM，应为折价，不是三者均为溢价。此处保留原题与原答案C，并按表中数据讲解。
>
> Source note / condition: The source is internally inconsistent. Under standard pricing with consistent coupon and YTM frequency conventions, Bonds 1 and 3 are at par because coupon equals YTM; Bond 2 is at a discount because coupon is below YTM. They are not all at a premium. The original wording and answer C are retained, and the explanation follows the table.

由上下价格估算久期 / **Estimating duration from repriced bonds**

Q025 | 原题 / Original question

基准利率变化下的有效久期

Effective duration from benchmark-rate changes

来源：题库 PDF 第 73 页（固收第 5 页）；答案 PDF 第 25 页。

Source: question bank p. 73 (Fixed Income p. 5); solution PDF p. 25.

分析师收集到以下一只票息率 6%、目前按面值交易的债券资料：

An analyst gathers the following information about a 6% coupon bond currently trading at par:

基准利率 Benchmark Rate	每 100 面值的价格 Price per 100 of Par Value
5.9%	100.75
6.0%	100.00
6.1%	99.26

该债券的有效久期最接近：

The bond's effective duration is closest to:

A. 3.75.

B. 7.45.

C. 7.50.

正确答案 / Answer: B

基准利率从 6.0% 向上和向下各移动 0.1 个百分点，所以单边变化量是 0.001。用降息后的高价减去加息后的低价，再除以 2 倍原价乘以单边利率变化，得到 7.45。

The benchmark moves up and down by 0.1 percentage point from 6.0%, so the one-sided rate change is 0.001. Subtract the up-rate price from the down-rate price, then divide by twice the initial price times that one-sided change. The result is 7.45.

1. 利率变化用小数，且只取单边变化。

Use a decimal rate change measured in one direction.

```
EffDur = (100.75-99.26)/(2*100.00*0.001) = 7.45
```

错误选项为什么不成立 / Why the other options fail

A. 误用当前价格 100.00 代替加息后的价格 99.26，算成 (100.75-100.00)/(2×100×0.001)=3.75。

A. This substitutes the current price of 100.00 for the up-rate price of 99.26: (100.75-100.00)/(2*100*0.001)=3.75.

C. 只使用降息一侧的价格变化得到约 7.50；题目提供两侧价格，应使用中心差分。

C. Using only the price increase from a rate decline gives about 7.50; both repriced values should be used.

Q026 | 原题 / Original question

负债现值也可计算有效久期

Effective duration also applies to liabilities

来源：题库 PDF 第 73 页（固收第 5 页）；答案 PDF 第 26 页。

Source: question bank p. 73 (Fixed Income p. 5); solution PDF p. 26.

分析师收集到以下养老金计划负债资料：

An analyst gathers the following information about a pension plan's liabilities:

利率假设 Interest Rate Assumption	负债现值（百万美元） Present Value of Liabilities (in $Millions)
0.5%	198
1.0%	186
1.5%	174

如果当前利率为 1.0%，负债的有效久期最接近：

If interest rates are currently 1.0%, the effective duration of the liabilities is closest to:

A. 6.5.

B. 12.9.

C. 25.8.

正确答案 / Answer: B

久期可衡量任何利率敏感现金流现值的变化，包括养老金负债。当前负债现值为 186，单边利率变化 0.5 个百分点 = 0.005。代入中心差分得 12.9032，选 B。所有现值使用同一“百万美元”单位，会在比率中约掉。

Duration can measure the rate sensitivity of liability present values as well as bond prices. The current liability value is 186 and the one-sided rate shift is 0.5 percentage point, or 0.005. Central differencing gives 12.9032. The common million-dollar units cancel in the ratio.

1. 把负债现值代入与债券相同的中心差分公式。

Apply the central-difference formula to liability present values.

```
EffDur = (198-174)/(2*186*0.005) = 12.9032
```

错误选项为什么不成立 / Why the other options fail

A. 把上下两端合计 1 个百分点当作单边变化，重复计算了公式中的 2。

A. Using the full 1-percentage-point endpoint difference double-counts the factor of two.

C. 遗漏分母中的 2，会把结果放大一倍。

C. Omitting the factor of two in the denominator doubles the result.

Q064 | 原题 / Original question

双边价格差估计修正久期

Estimate modified duration from two-sided price changes

来源：题库 PDF 第 81 页（固收第 13 页）；答案 PDF 第 64 页。

Source: question bank p. 81 (Fixed Income p. 13); solution PDF p. 64.

分析师观察到一只无嵌入期权债券的以下价格与收益率关系：

An analyst observes the following price-yield relationship for an option-free bond:

价格（每 100 面值） Price per 100 of Par Value	到期收益率 Yield to Maturity
100.95	7.45%
101.80	6.75%
103.40	6.05%

如果该债券每 100 面值的交易价格为 101.80，其近似修正久期最接近多少？

If the bond trades at 101.80 per 100 of par value, its approximate modified duration is closest to:

A. 1.72.

B. 2.25.

C. 3.44.

正确答案 / Answer: A

当前收益率为 6.75%，上下各变化 0.70 个百分点，即 0.007。降息价格为 103.40，升息价格为 100.95，当前价格为 101.80。近似修正久期 = (103.40 - 100.95)/(2 × 101.80 × 0.007) = 1.7191，约 1.72。

The current yield is 6.75%, with a 0.70-percentage-point move in either direction, or 0.007. The lower-yield price is 103.40, the higher-yield price is 100.95, and the current price is 101.80. Approximate modified duration is (103.40 - 100.95)/(2 × 101.80 × 0.007) = 1.7191, or 1.72.

1. 取当前收益率到其中一侧的变化。

Use the one-sided change from the current yield.

```
DeltaYTM = 0.0745 - 0.0675 = 0.007
```

2. 代入双边价格差。

Use the two-sided price difference.

```
ModDur ≈ (103.40 - 100.95) / (2 * 101.80 * 0.007) = 1.7191
```

错误选项为什么不成立 / Why the other options fail

B. 约等于只用降息一侧的差价估计，没有按题目给定的双边价格差计算。

B. This is close to a one-sided estimate using only the yield-decrease price, rather than the supplied two-sided approximation.

C. 分母漏掉 2，导致久期扩大一倍。

C. This omits the factor of two from the denominator and doubles duration.

Q065 | 原题 / Original question

有效久期：移动的是基准收益率

Effective duration: the shock is to the benchmark yield

来源：题库 PDF 第 81 页（固收第 13 页）；答案 PDF 第 65 页。

Source: question bank p. 81 (Fixed Income p. 13); solution PDF p. 65.

分析师收集了一只当前每 100 面值交易价格为 95.35 的债券信息：

An analyst gathers the following information about a bond which is currently trading at 95.35 per 100 par:

基准收益率 Benchmark Yield	债券价格 Bond Price
3.50%	99.50
4.00%	95.35
4.50%	92.25

该债券的有效久期最接近多少？

The effective duration of the bond is closest to:

A. 6.5.

B. 7.6.

C. 8.7.

正确答案 / Answer: B

基准收益率从 4.00% 上下各变动 0.50 个百分点，所以单边变化为 0.005。有效久期 = (99.50 - 92.25)/(2 × 95.35 × 0.005) = 7.6036，约 7.6。这里分母使用基准利率的变化，因此属于有效久期。

The benchmark yield moves 0.50 percentage point in either direction from 4.00%, giving a one-sided change of 0.005. Effective duration is (99.50 - 92.25)/(2 × 95.35 × 0.005) = 7.6036, or 7.6. Because the denominator uses a benchmark-rate change, this is effective duration.

1. 使用当前全价和单边基准利率变化。

Use the current full price and one-sided benchmark-rate change.

```
EffDur = (99.50 - 92.25) / (2 * 95.35 * 0.005) = 7.6036 ≈ 7.6
```

错误选项为什么不成立 / Why the other options fail

A. 约为只采用升息后的价格下降所算的一侧敏感度。

A. This is approximately the one-sided sensitivity using only the price decrease when yields rise.

C. 约为只采用降息后的价格上升所算的一侧敏感度。

C. This is approximately the one-sided sensitivity using only the price increase when yields fall.

Q069 | 原题 / Original question

久期分母使用单边收益率变化与当前价格

Duration uses a one-sided yield shock and the current price

来源：题库 PDF 第 82 页（固收第 14 页）；答案 PDF 第 69 页。

Source: question bank p. 82 (Fixed Income p. 14); solution PDF p. 69.

投资者得到以下有关一只无嵌入期权债券价格敏感度的信息：

An investor determines the following information about the price sensitivity of an option-free bond:

利率变化 Rate Change	价格 Price
上升 75 个基点 75 basis point increase	104.0
下降 75 个基点 75 basis point decrease	108.5

如果当前价格为 106，该债券的久期最接近多少？

If the current price is 106, the duration of this bond is closest to:

A. 2.1.

B. 2.8.

C. 3.0.

正确答案 / Answer: B

降息价格 108.5，升息价格 104.0，当前价格 106。75 个基点为 0.0075。用双边公式：(108.5 - 104.0)/(2 × 106 × 0.0075) = 2.8302，最接近 2.8。不要把两侧总相差 150 个基点再放入已经带 2 的分母。

The lower-yield price is 108.5, the higher-yield price is 104.0, and the current price is 106. Seventy-five basis points is 0.0075. The two-sided duration formula gives (108.5 - 104.0)/(2 × 106 × 0.0075) = 2.8302, closest to 2.8. Do not use the full 150-basis-point range in a denominator that already includes a factor of two.

1. 双边价格差除以两倍当前价格和单边利率变化。

Divide the two-sided price difference by twice current price and the one-sided rate shock.

```
Duration ≈ (108.5 - 104.0) / (2 * 106 * 0.0075) = 2.8302
```

错误选项为什么不成立 / Why the other options fail

A. 误把 75 个基点写为 100 个基点，分母使用 0.01，得到约 2.12。

A. This uses a 100-basis-point shock of 0.01 instead of 75 basis points and gives about 2.12.

C. 约使用面值 100 代替当前价格 106；久期需要当前价格作分母。

C. This is close to substituting par of 100 for the current price of 106; the denominator requires current price.

Q072 | 原题 / Original question

有效久期的基点换算

Basis-point conversion in effective duration

来源：题库 PDF 第 83 页（固收第 15 页）；答案 PDF 第 72 页。

Source: question bank p. 83 (Fixed Income p. 15); solution PDF p. 72.

分析师收集了一只当前按面值交易的债券信息：

An analyst gathers the following information about a bond currently trading at par:

基准曲线变化 Change in Benchmark Curve	价格（每 100 面值） Price per 100 of Par Value
上升 25 个基点 +25 bps	98
下降 25 个基点 -25 bps	103

该债券的有效久期最接近多少？

The effective duration of this bond is closest to:

A. 2.5.

B. 5.0.

C. 10.0.

正确答案 / Answer: C

按面值交易，所以当前价格为 100。基准曲线向下移动 25 个基点时价格为 103，向上移动时价格为 98；单边利率变化为 0.0025。有效久期 = (103 - 98)/(2 × 100 × 0.0025) = 10。记住：25 个基点 = 0.25% = 0.0025。

At par, the current price is 100. The price is 103 after a 25-basis-point downward curve shift and 98 after an upward shift, with a one-sided rate change of 0.0025. Effective duration is (103 - 98)/(2 × 100 × 0.0025) = 10. Remember: 25 basis points = 0.25% = 0.0025.

1. 先识别面值交易对应的当前价格。

Identify the current price from the par quotation.

```
P0 = 100; DeltaCurve = 25 / 10000 = 0.0025
```

2. 使用有效久期公式。

Apply the effective-duration formula.

```
EffDur = (103 - 98) / (2 * 100 * 0.0025) = 10.0
```

错误选项为什么不成立 / Why the other options fail

A. 只把价格差除以 2，(103 - 98)/2 = 2.5，漏掉当前价格与基准利率变化。

A. This only halves the price difference: (103 - 98)/2 = 2.5, omitting current price and the benchmark-rate change.

B. 只算两侧价格差 103 - 98 = 5，没有除以完整分母。

B. This takes only the price difference of 103 - 98 = 5 and omits the full denominator.

金额久期、基点价值与组合 / **Money duration, PVBP and portfolios**

Q054 | 原题 / Original question

金额价格变动以当前市值为基数

Dollar price changes use current market value

来源：题库 PDF 第 79 页（固收第 11 页）；答案 PDF 第 54 页。

Source: question bank p. 79 (Fixed Income p. 11); solution PDF p. 54.

分析师收集了一只无嵌入期权债券的以下信息：

An analyst gathers the following information about an option-free bond:

面值 Par value	1,000,000 美元 $1,000,000
当前市值 Current market value	912,575 美元 $912,575
久期 Duration	2.4

如果预期收益率下降 50 个基点，该债券的预计价格变动最接近多少？

If yields are expected to decrease by 50 basis points, the expected price change for the bond is closest to:

A. 11,000 美元。
$11,000.

B. 22,000 美元。
$22,000.

C. 24.000 美元〔按原文保留〕。
$24.000.

正确答案 / Answer: A

以当前市值计算，不用面值。收益率下降 50 个基点，即变化量 -0.005。价格百分比近似上涨 2.4 × 0.005 = 1.2%，金额上涨 912,575 × 1.2% = 10,950.90 美元，最接近 11,000 美元。

Use current market value, not par value. A 50-basis-point decline is a yield change of -0.005. The estimated price increase is 2.4 × 0.005 = 1.2%, or 912,575 × 1.2% = $10,950.90, closest to $11,000.

1. 先转为收益率的小数变化。

Convert the yield change to decimal form.

```
DeltaYTM = -50 / 10000 = -0.005
```

2. 按当前市值计算金额变化。

Apply duration to current market value.

```
DeltaP ≈ -2.4 * (-0.005) * 912575 = 10950.90
```

错误选项为什么不成立 / Why the other options fail

B. 相当于把 50 个基点误当成 100 个基点，约把结果翻倍。

B. This is approximately double the correct result, as if 50 basis points were treated as 100.

C. 即使按 24,000 美元理解，也同时偏离了 50 个基点和当前市值的正确使用。

C. Even read as $24,000, it does not use the stated 50-basis-point change and current market value correctly.

> 原资料提示 / 使用条件：原题仅写 Duration，未明确久期种类；按所附答案及选项，2.4 被作为修正久期使用。原英文选项 C 为“$24.000.”，疑似千位分隔符笔误，原文保留，不影响正确答案 A。
>
> Source note / condition: The stem states only “Duration.” The supplied answer and choices treat 2.4 as modified duration. Original option C reads “$24.000.”, apparently a separator typo; its wording is preserved and does not affect answer A.

Q055 | 原题 / Original question

PVBP：一个基点对应的价格变动绝对值

PVBP: the absolute price change for one basis point

来源：题库 PDF 第 79 页（固收第 11 页）；答案 PDF 第 55 页。

Source: question bank p. 79 (Fixed Income p. 11); solution PDF p. 55.

一只债券的全价为 103.50，修正久期为 6.2，其一个基点的价格价值（PVBP）最接近多少？

The price value of a basis point (PVBP) for a bond with a full price of 103.50 and a modified duration of 6.2 is closest to:

A. 0.0642.

B. 0.6420.

C. 6.4200.

正确答案 / Answer: A

一个基点是 0.01 个百分点，即小数 0.0001。PVBP 近似等于修正久期 × 全价 × 0.0001 = 6.2 × 103.50 × 0.0001 = 0.06417，约 0.0642。结果使用与输入价格相同的计价单位；通常表示每 100 面值的价格变化。

One basis point is 0.01 percentage point, or 0.0001 in decimal form. PVBP is approximately modified duration × full price × 0.0001 = 6.2 × 103.50 × 0.0001 = 0.06417, or 0.0642. Its unit matches the quoted price, typically the price change per 100 of par.

1. 用全价和基点的小数单位。

Use full price and the decimal basis-point unit.

```
PVBP ≈ 6.2 * 103.50 * 0.0001 = 0.06417 ≈ 0.0642
```

错误选项为什么不成立 / Why the other options fail

B. 把一个基点误写成 0.001，结果扩大 10 倍。

B. This uses 0.001 for one basis point and overstates the result tenfold.

C. 把一个基点误写成 0.01，结果扩大 100 倍。

C. This uses 0.01 for one basis point and overstates the result one hundredfold.

Q084 | 原题 / Original question

货币久期必须使用全价

Money duration uses the full price

来源：题库 PDF 第 85 页（固收第 17 页）；答案 PDF 第 84 页。

Source: question bank p. 85 (Fixed Income p. 17); solution PDF p. 84.

分析师收集到某债券的下列信息：

An analyst gathers the following information about a bond:

净价（每100面值） Clean price (per 100 of par value)	114.75
年化修正久期 Annual modified duration	4.8250
麦考利久期（年） Macaulay duration (years)	4.9469
应计利息（每100面值） Accrued interest (per 100 of par value)	1.6250

该债券的货币久期（每100面值）最接近：

The bond's money duration (per 100 of par value) is closest to:

A. 553.67.

B. 561.51.

C. 575.70.

正确答案 / Answer: B

货币久期等于修正久期乘全价。先将净价114.75加上应计利息1.625，得到全价116.375；再乘修正久期4.8250，结果561.509375。这里的计量基准仍是每100面值。

Money duration equals modified duration times full price. Add accrued interest of 1.625 to the clean price of 114.75 to obtain 116.375, then multiply by 4.8250. The result is 561.509375 per 100 of par value.

1. 先求全价。

Find the full price.

```
P_full = 114.75 + 1.6250 = 116.375
```

2. 再乘修正久期。

Multiply by modified duration.

```
MoneyDur = 116.375 × 4.8250 = 561.509375
```

错误选项为什么不成立 / Why the other options fail

A. 使用净价114.75乘修正久期，漏加应计利息。

A. It multiplies modified duration by the clean price and omits accrued interest.

C. 使用全价乘麦考利久期，选错久期口径。

C. It multiplies full price by Macaulay duration instead of modified duration.

Q056 | 原题 / Original question

组合修正久期 = 合计金额久期 / 合计市值

Portfolio modified duration = total money duration / total market value

来源：题库 PDF 第 79 页（固收第 11 页）；答案 PDF 第 56 页。

Source: question bank p. 79 (Fixed Income p. 11); solution PDF p. 56.

分析师收集了以下债券组合资料：

An analyst gathers the following details about a bond portfolio:

债券 Bond	市场价格（每 100 面值） Market Price (per 100 of par value)	金额久期（每 100 面值） Money Duration (per 100 of par value)
1	95	730
2	120	515

如果每只债券的面值均为 2,500 万英镑，该债券组合的修正久期最接近多少？

If each bond has a par value of £25 million, the modified duration of this bond portfolio is closest to:

A. 5.8.

B. 6.1.

C. 6.2.

正确答案 / Answer: A

两只债券的面值相同，因此每 100 面值的价格和金额久期可直接相加，公共倍数抵消。组合修正久期 = (730 + 515)/(95 + 120) = 5.7907，约 5.8。也可先分别求 730/95 和 515/120，再按市值 95:120 加权。不能因面值相同就给两只债券相同权重。

The bonds have equal par values, so the common scaling factor cancels when adding money duration and market value per 100 of par. Portfolio modified duration is (730 + 515)/(95 + 120) = 5.7907, or 5.8. Equivalently, compute each modified duration as money duration divided by price, then weight by market values 95:120. Equal par does not imply equal market-value weights.

1. 合计金额久期除以合计市值。

Divide aggregate money duration by aggregate market value.

```
Portfolio ModDur = (730 + 515) / (95 + 120) = 5.7907 ≈ 5.8
```

错误选项为什么不成立 / Why the other options fail

B. 把每 100 面值的金额久期先除以 100，再按市值加权，会得到约 6.1；单券金额久期应除以该券的市价，不能统一除以 100。

B. Dividing each money-duration quotation by 100 and then market-value weighting gives about 6.1. Each quotation must instead be divided by that bond's own market price.

C. 把金额久期按总面值 200 而非总市值 215 相除，会得到约 6.2。

C. Dividing total money duration by total par of 200 instead of total market value of 215 gives about 6.2.

Q078 | 原题 / Original question

组合修正久期：先转换，再用市值加权

Portfolio modified duration: convert first, then market-value weight

来源：题库 PDF 第 84 页（固收第 16 页）；答案 PDF 第 78 页。

Source: question bank p. 84 (Fixed Income p. 16); solution PDF p. 78.

某债券组合由下列无嵌入期权、每年付息的债券组成：

A bond portfolio consists of the following option-free annual-pay coupon bonds:

	债券1 Bond 1	债券2 Bond 2
面值 Par value	$300,000	$450,000
市值 Market value	$200,000	$400,000
到期收益率 Yield to maturity	4%	3%
麦考利久期 Macaulay duration	7.5	5.4

该组合的修正久期最接近：

The modified duration of this portfolio is closest to:

A. 5.9.

B. 6.0.

C. 6.1.

正确答案 / Answer: A

这道题有两道关：表中给的是麦考利久期，而问题要修正久期；组合权重要使用市值而非面值。分别算出7.211538和5.242718，再按1/3与2/3加权，结果5.898992。

There are two checks: the table gives Macaulay durations, while the question asks for modified duration; portfolio weights use market values, not par values. Convert the individual durations to 7.211538 and 5.242718, then weight them by 1/3 and 2/3 to obtain 5.898992.

1. 年付息：分别除以1加各自YTM。

For annual payments, divide by one plus each bond's YTM.

```
D_1 = 7.5/1.04 = 7.211538; D_2 = 5.4/1.03 = 5.242718
```

2. 用各自市值占总市值的比例加权。

Weight by each bond's share of total market value.

```
D_port = (200000/600000)D_1 + (400000/600000)D_2 = 5.898992
```

错误选项为什么不成立 / Why the other options fail

B. 使用面值权重40%及60%，会得到约6.0。

B. Using par-value weights of 40% and 60% gives about 6.0.

C. 使用市值权重但没有转换久期，得到的是组合麦考利久期6.1。

C. Market-value weighting the unconverted Macaulay durations gives the portfolio Macaulay duration of 6.1.

Q086 | 原题 / Original question

组合久期以市值为权重

Portfolio duration uses market-value weights

来源：题库 PDF 第 86 页（固收第 18 页）；答案 PDF 第 86 页。

Source: question bank p. 86 (Fixed Income p. 18); solution PDF p. 86.

投资者收集到由两只无期权债券组成的组合的下列信息：

An investor gathers the following information about a bond portfolio comprised of two option-free bonds:

债券 Bond	面值 Par Value	市值 Market Value	久期 Duration
1	$100,000	$120,000	5
2	$200,000	$180,000	4

该组合的久期最接近：

The duration of the portfolio is closest to:

A. 4.33.

B. 4.40.

C. 4.55.

正确答案 / Answer: B

总市值为300,000。债券1和债券2的市值权重分别为40%和60%；组合久期为0.4×5+0.6×4=4.40。面值只表示偿还本金，不代表当前组合风险权重。

Total market value is 300,000. Bonds 1 and 2 have market-value weights of 40% and 60%, so portfolio duration is 0.4×5+0.6×4=4.40. Par value is the promised principal repayment, not the current risk weight in the portfolio.

1. 按市值计算权重并求加权平均。

Compute market-value weights and take the weighted average.

```
D_port = (120000/300000) × 5 + (180000/300000) × 4 = 4.40
```

错误选项为什么不成立 / Why the other options fail

A. 4.33来自以100,000和200,000面值加权。

A. A value of 4.33 comes from weighting by par values of 100,000 and 200,000.

C. 4.55来自以久期本身作为权重，不是市值加权。

C. A value of 4.55 comes from using durations themselves as weights rather than market values.

CHAPTER 05 / 10 ORIGINAL QUESTIONS

凸性：为什么涨跌不对称

Convexity and asymmetric price changes

> 久期是切线，凸性是曲线修正；计算时保留凸性的正负号。
> Duration gives the tangent-line effect; convexity corrects for curvature. Preserve the sign of convexity.

先做这些原题检验理解：Q4, Q12, Q75, Q100。
Start with these original checkpoints: Q4, Q12, Q75, Q100.

5.1 先理解图形，再记公式

5.1 Understand the shape before memorizing the formula

普通无期权债券的价格与收益率反向，且通常有正凸性：收益率下降同样幅度时，价格上涨的幅度大于收益率上升时价格下跌的幅度。Q37 就是在问这个不对称。
An option-free bond has an inverse price-yield relationship and generally positive convexity: an equal yield decrease produces a larger price gain than the loss from an equal yield increase. Q37 tests this asymmetry.

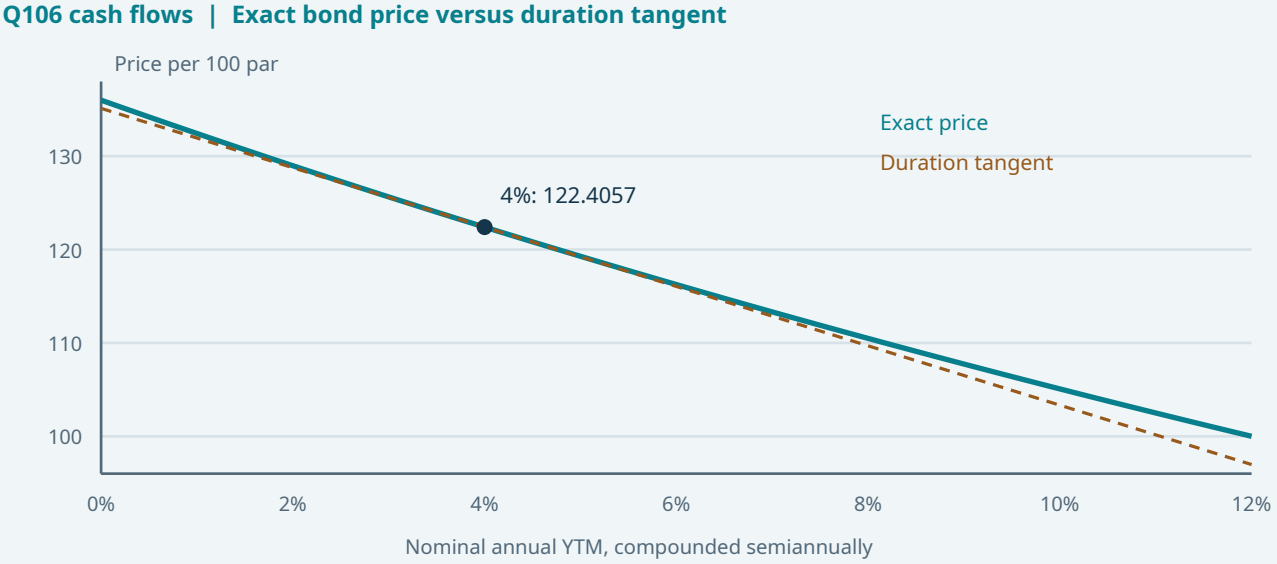


图 3：用原题 Q106 的现金流绘制价格曲线；切线只适用于当前收益率附近。
Figure 3. Price-yield curve from the original Q106 cash flows; the tangent is a local approximation.

Q36 给出“降息 100 bp，涨 7%”。反方向若要跌同样的 7%，通常需要加息超过 100 bp，因为正凸性减缓了加息造成的下跌。
Q36 states that a 100 bp yield fall produces a 7% price rise. A 7% fall typically requires a yield increase greater than 100 bp because positive convexity cushions the price decline.

5.2 用久期加凸性估计价格

5.2 Estimate price using duration and convexity

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}\,\text{Conv}\,(\Delta y)^2$$

第一项有利率变动的方向；第二项中利率变动被平方，但凸性本身的符号仍然保留。负凸性产生负的修正项。
The duration term includes the direction of the yield change. The yield change is squared in the convexity term, but convexity retains its own sign. Negative convexity gives a negative correction.

> Q4：delta_y = +0.0075，D = 6.9，凸性 = −212。百分比变化 = −6.9×0.0075 + 0.5×(−212)×0.0075² = −5.77125%；新价格 = 99.4×(1−0.0577125) = 93.6634，选 A。
>
> Q4: delta_y = +0.0075, D = 6.9 and convexity = −212. Percentage change = −6.9x0.0075 + 0.5x(−212)x0.0075² = −5.77125%; the new price is 99.4x(1−0.0577125) = 93.6634, answer A.

$$\text{Conv} \approx \frac{P_- + P_+ - 2P_0}{P_0(\Delta y)^2}$$

这里没有久期公式中的分母 2。Q12：delta_y = 10 bp = 0.001；(96.30+95.80−2×96)/(96×0.001²) = 1,041.6667，选 B。

There is no factor of 2 in the denominator as there is for duration. In Q12, delta_y = 10 bp = 0.001; (96.30+95.80−2x96)/(96x0.001²) = 1,041.6667, answer B.

Q75 是零息债券，MacDur = 5；按年度复利口径 ModDur = 5/1.02。再加凸性项：−(5/1.02)×(−0.01) + 0.5×28.835×0.01² = 5.0461%，选 B。不要把 Macaulay 久期直接放入价格变动公式。

Q75 is a zero-coupon bond, so MacDur = 5 and annual-basis ModDur = 5/1.02. Adding convexity gives −(5/1.02)x(−0.01) + 0.5x28.835x0.01² = 5.0461%, answer B. Do not insert Macaulay duration directly into the price-change formula.

5.3 风险大小与可赎回债

5.3 Relative sensitivity and callable bonds

Q81：固定现金流企业债的总收益率可以拆为基准收益率与信用利差。两者任何一项变化都可能改变总收益率，因此久期和凸性均可用于分析其价格影响；此处假设其他现金流条件不变。

In Q81, a fixed-cash-flow corporate bond's total yield reflects the benchmark yield plus a credit spread. A change in either component can change total yield, so duration and convexity are relevant to both, holding other cash-flow assumptions fixed.

其他条件相同，期限更长、票息更低的无期权债券一般更敏感。Q38 中 Bond 3 同时具有低票息与长到期期限，降息时涨幅最大。对高折价或异常收益率情景，不要把一般规律当成无条件定理。

All else equal, an option-free bond generally becomes more rate-sensitive with longer maturity and lower coupon. Bond 3 in Q38 combines the lower coupon with the longer maturity. Do not treat this general relationship as unconditional in unusual discount or yield cases.

可赎回债券在利率低、赎回更可能发生时，价格上涨受限制，可能呈负凸性。在利率较高、赎回不太可能时，它仍可呈正凸性。因此 Q101 选“可能正也可能负”，不是“永远负”。

A callable bond may have negative convexity when low rates make exercise more likely and limit price appreciation. At higher rates, when a call is unlikely, it can retain positive convexity. Q101 therefore allows either positive or negative convexity.

Q105 在相同久期、相同起始价格等条件下比较凸性：收益率上升时，更高正凸性一般使跌幅较小，表现相对更好。这里的“表现更好”可以只是“跌得更少”。

Q105 compares bonds with equal duration and initial price. A higher convexity generally cushions the decline after a yield increase. Outperformance can simply mean losing less.

> 计算题只记三个检查点：bp 先转小数；久期用 ModDur；凸性符号保留。概念题再问：是否含有发行人的赎回权？
>
> For calculations, convert bp to decimals, use modified duration and preserve the convexity sign. For conceptual questions, check whether the issuer holds a call option.

正凸性与价格敏感度 / **Positive convexity and price sensitivity**

Q036 | 原题 / Original question

正凸性导致涨跌幅不对称

Positive convexity creates asymmetric price changes

来源：题库 PDF 第 75 页（固收第 7 页）；答案 PDF 第 36 页。

Source: question bank p. 75 (Fixed Income p. 7); solution PDF p. 36.

一只无内嵌期权债券在到期收益率下降 100 个基点时，价格上涨 7%。如果该债券价格下跌 7%，到期收益率最可能上升：

The price of an option-free bond increases by 7% when the yield to maturity decreases by 100 basis points. If the price of this bond decreases by 7%,the yield to maturity most likely increases by:

A. 少于 100 个基点。
less than 100 basis points.

B. 100 个基点。
100 basis points.

C. 多于 100 个基点。
more than 100 basis points.

正确答案 / Answer: C

以同一初始价格和收益率为起点，正凸性意味着相同幅度的降息带来的价格上涨，大于加息带来的价格下跌。因此加息 100 bp 的跌幅小于 7%；要让价格下跌完整的 7%，利率需要上升超过 100 bp。

Starting from the same initial price and yield, positive convexity means that an equal-sized yield decline produces a larger price gain than the loss produced by a yield increase. A 100 bp rise would therefore cause a decline of less than 7%; achieving a 7% decline requires a rise of more than 100 bp.

错误选项为什么不成立 / Why the other options fail

A. 它把正凸性造成的价格变化不对称方向反过来了。

A. This reverses the asymmetry created by positive convexity.

B. 这相当于把价格与收益率关系当成直线，忽略凸性。

B. This treats the price-yield relationship as linear and ignores convexity.

Q037 | 原题 / Original question

正凸性：降息涨得多，等幅加息跌得少

Positive convexity: an equal yield fall produces a larger gain than the loss from a yield rise

来源：题库 PDF 第 75 页（固收第 7 页）；答案 PDF 第 37 页。

Source: question bank p. 75 (Fixed Income p. 7); solution PDF p. 37.

其他条件相同，无嵌入期权债券的价格百分比变动的绝对值最可能：

All else being equal, the absolute value of the percentage price change for an option-free bond is most likely:

A. 市场贴现率下降时，小于贴现率等幅上升时的绝对变动。
less when the market discount rate decreases than when it increases by the same amount.

B. 无论市场贴现率等幅下降或上升，绝对变动均相同。
the same whether the market discount rate decreases or increases by the same amount.

C. 市场贴现率下降时，大于贴现率等幅上升时的绝对变动。
greater when the market discount rate decreases than when it increases by the same amount.

正确答案 / Answer: C

普通无期权债券的价格与收益率呈向上弯曲的反向关系。久期给出近似对称的一阶变动，正凸性则为升息和降息两种情形都增加正的修正项。因此，等幅降息带来的涨幅，大于升息带来的跌幅。比较的是百分比变动的绝对值，不是带正负号的数值。

An option-free bond has a convex price-yield curve. Duration supplies a roughly symmetric first-order effect, while positive convexity adds a positive second-order adjustment for either direction of the yield change. The gain from an equal yield decrease therefore exceeds the absolute loss from a yield increase.

错误选项为什么不成立 / Why the other options fail

A. 把正凸性的方向写反了。

A. This reverses the effect of positive convexity.

B. 这是忽略凸性的线性久期近似，不是实际价格曲线的特征。

B. Equality is a linear duration approximation that ignores convexity.

Q038 | 原题 / Original question

票息、期限与利率敏感度

Coupon, maturity and interest-rate sensitivity

来源：题库 PDF 第 75 页（固收第 7 页）；答案 PDF 第 38 页。

Source: question bank p. 75 (Fixed Income p. 7); solution PDF p. 38.

分析师收集了以下三只无嵌入期权债券的信息：

An analyst gathers the following information about three option-free bonds:

	债券 1 Bond 1	债券 2 Bond 2	债券 3 Bond 3
票息率 Coupon rate	5%	3%	3%
到期期限（年） Maturity (years)	15	10	15

其他条件相同，如果市场贴现率下降 50 个基点，哪只债券的价格百分比变动最可能最大？

All else being equal, if the market discount rate decreases by 50 basis points, the bond most likely to experience the greatest percentage price change is:

A. 债券 1。
Bond 1.

B. 债券 2。
Bond 2.

C. 债券 3。
Bond 3.

正确答案 / Answer: C

其他条件相同，较低票息使更多价值集中于较远期的本金，较长期限也会提高一般债券的利率敏感度。债券 3 与债券 1 同为 15 年，但票息更低；与债券 2 同为 3% 票息，但期限更长。因此，本题中债券 3 对 50 个基点的利率下降最敏感，价格百分比涨幅最大。

All else equal, a lower coupon places more value in the distant principal payment, and a longer maturity generally increases rate sensitivity. Bond 3 has a lower coupon than Bond 1 at the same 15-year maturity, and a longer maturity than Bond 2 at the same 3% coupon. It therefore has the greatest percentage price increase for the stated 50-basis-point yield decline.

错误选项为什么不成立 / Why the other options fail

A. 与债券 3 的期限相同，但票息更高，较早收回现金，敏感度较低。

A. It has the same maturity as Bond 3 but a higher coupon, bringing cash flows forward and reducing sensitivity.

B. 与债券 3 的票息相同，但期限更短，敏感度较低。

B. It has the same coupon as Bond 3 but a shorter maturity and lower sensitivity.

Q081 | 原题 / Original question

公司债总收益率由基准利率与利差构成

Corporate bond yield combines a benchmark yield and a spread

来源：题库 PDF 第 85 页（固收第 17 页）；答案 PDF 第 81 页。

Source: question bank p. 85 (Fixed Income p. 17); solution PDF p. 81.

对无嵌入期权的固定利率公司债，久期与凸性指标最可能适用于下列哪类变动：

For an option-free fixed-rate corporate bond, the duration and convexity statistics are most likely relevant for a change in:

A. 只有信用利差的变动。
the credit spread only.

B. 只有基准收益率的变动。
the benchmark yield only.

C. 信用利差和基准收益率的变动两者。
both the credit spread and the benchmark yield.

正确答案 / Answer: C

保持约定现金流不变时，公司债价格取决于总折现收益率。总收益率的变动可以来自基准收益率，也可以来自信用利差，或同时来自两者。因此同一组以YTM为变量的久期与凸性可以估计这些变动引起的价格效应。

With contractual cash flows unchanged, the price of an option-free corporate bond depends on its total discount yield. That yield can change because of the benchmark yield, the credit spread, or both. The same YTM-based duration and convexity can estimate the resulting price effect.

错误选项为什么不成立 / Why the other options fail

A. 基准收益率改变也会改变总折现收益率。

A. A benchmark-yield change also changes the total discount yield.

B. 即使基准利率不动，信用利差扩大也会提高总收益率并压低价格。

B. Even with an unchanged benchmark, a wider credit spread raises total yield and reduces price.

Q105 | 原题 / Original question

同久期下，更高凸性减少升息造成的价格跌幅

With equal duration, higher convexity reduces the loss from rising yields

来源：题库 PDF 第 89 页（固收第 21 页）；答案 PDF 第 105 页。

Source: question bank p. 89 (Fixed Income p. 21); solution PDF p. 105.

两只平价债券具有相同久期但不同凸性。其他条件相同时，如果到期收益率上升10个基点，最可能：

Two par bonds have the same duration but different convexity. All else being equal, if yields to maturity increase by 10 basis points, it is most likely that.

A. 凸性较高的债券表现落后于凸性较低的债券。
the more convex bond underperforms the less convex bond.

B. 两只债券价格下降相同金额。
both bond prices decrease by the same amount.

C. 凸性较高的债券表现优于凸性较低的债券。
the more convex bond outperforms the less convex bond.

正确答案 / Answer: C

两只债券的初始价格和久期相同，所以一阶久期效应相同。凸性修正为0.5×Convexity×(Δy)^2；在同样的收益率变动下，凸性较高的一只拥有更大的正修正，价格跌得较少。这里的outperforms表示相对少跌，并不意味着升息时一定上涨。

Equal starting prices and durations give the two bonds the same first-order duration effect. The convexity adjustment is 0.5×Convexity×(Δy)^2. For the same yield change, the bond with greater convexity has the larger positive adjustment and loses less. Outperformance here means a smaller decline, not necessarily a price increase.

1. 10个基点写成0.001；相同的一阶效应抵消后，只比较凸性项。

Write 10 basis points as 0.001; the common first-order effect cancels when comparing the bonds.

```
Return_high - Return_low ≈ 0.5(C_high - C_low)(0.001)^2 > 0
```

错误选项为什么不成立 / Why the other options fail

A. 把更高凸性的有利方向颠倒了。

A. It reverses the benefit of higher convexity.

B. 只考虑久期，忽略了凸性差异带来的二阶影响。

B. It uses duration alone and ignores the second-order effect of different convexities.

凸性计算与价格估计 / **Convexity calculations and price estimates**

Q012 | 原题 / Original question

用上下移收益率后的价格计算近似凸性

Approximating convexity from up-and-down repricing

来源：题库 PDF 第 70 页（固收第 2 页）；答案 PDF 第 12 页。

Source: question bank p. 70 (Fixed Income p. 2); solution PDF p. 12.

分析师收集到以下债券资料：

An analyst gathers the following information about a bond:

当前价格 Price	96.00
到期收益率上升 10 个基点后的价格 Price with yield to maturity 10 basis points higher	95.80
到期收益率下降 10 个基点后的价格 Price with yield to maturity 10 basis points lower	96.30

这只债券的近似凸性最接近：

The approximate convexity of this bond is closest to:

A. 521.

B. 1,042.

C. 2,604.

正确答案 / Answer: B

凸性利用上下两侧价格之和相对于 2 倍原价的差额。10 bp = 0.001，分母中的收益率变化必须平方。代入后得到 1,041.6667，选 B。凸性公式的分母没有久期公式里的额外系数 2。

Convexity uses the sum of the up-shifted and down-shifted prices relative to twice the initial price. Convert 10 bp to 0.001 and square the yield change in the denominator. The result is 1,041.6667. Unlike the duration formula, this convexity denominator has no extra factor of two.

1. 价格相加，不是相减。

Add the two repriced bond prices.

```
Conv = (96.30+95.80-2*96.00)/(96.00*0.001^2) = 1041.6667
```

错误选项为什么不成立 / Why the other options fail

A. 在分母多乘一个 2，会得到约 521。

A. Adding an extra factor of two to the denominator gives approximately 521.

C. 它混用了久期的价格差分子与凸性的平方收益率变化分母。

C. This mixes the duration-style price-difference numerator with a squared yield change in the denominator.

Q004 | 原题 / Original question

久期与负凸性的价格变化近似

Approximating a price change with duration and negative convexity

来源：题库 PDF 第 69 页（固收第 1 页）；答案 PDF 第 4 页。

Source: question bank p. 69 (Fixed Income p. 1); solution PDF p. 4.

一只债券价格为 99.4，修正久期为 6.9，年化凸性为 -212。如果市场收益率上升 75 个基点，该债券价格最接近：

A bond priced at 99.4 has a modified duration of 6.9 and an annual convexity statistic of-212.If the market yield increases by 75 basis points, the price of this bond is closest to:

A. 93.7.

B. 94.3.

C. 94.9.

正确答案 / Answer: A

先把 75 bp 写成 0.0075。久期项为 -5.175%，负凸性项为 -0.59625%，合计价格变化约为 -5.77125%。用原价 99.4 乘以 1 加上这个变化率，得到 93.6634，选 A。负凸性在本题中会进一步扩大价格跌幅。

Convert 75 bp to 0.0075. The duration term is -5.175% and the negative-convexity term is -0.59625%, giving an estimated change of -5.77125%. Apply this return to the original price of 99.4 to obtain 93.6634. Negative convexity increases the estimated loss in this example.

1. 计算价格变化率。

Estimate the percentage price change.

```
dP/P = -6.9(0.0075) + 0.5(-212)(0.0075)^2 = -0.0577125
```

2. 把变化率应用到原价。

Apply the percentage change to the original price.

```
P1 = 99.4(1-0.0577125) = 93.6634
```

错误选项为什么不成立 / Why the other options fail

B. 应以实际价格 99.4 为起点，不能用面值 100 减去按 99.4 算出的金额损失。

B. Start with the actual price, 99.4; do not subtract a loss calculated on 99.4 from a par price of 100.

C. 把 -212 错当成正凸性会抵消一部分跌幅，导致价格估得过高。

C. Treating -212 as positive convexity offsets part of the loss and overstates the resulting price.

> 原资料提示 / 使用条件：所附解答在中间一行将负数之和写成了正的 0.0577125，存在漏写负号；正确的价格变化率是 -0.0577125。原答案 A 及最终价格方向正确。
>
> Source note / condition: The supplied solution drops a minus sign in an intermediate equality and writes the sum as positive 0.0577125. The correct signed price change is -0.0577125. Its answer A and final price direction remain correct.

Q075 | 原题 / Original question

零息债的久期与凸性修正

Zero-coupon duration and the convexity adjustment

来源：题库 PDF 第 83 页（固收第 15 页）；答案 PDF 第 75 页。

Source: question bank p. 83 (Fixed Income p. 15); solution PDF p. 75.

分析师收集到某零息债券的下列信息：

An analyst gathers the following information about a zero-coupon bond:

到期年限 Time to maturity	5年 5 years
年度凸性 Annual convexity	28.835
年到期收益率 Annual yield to maturity	2%

如果收益率下降1个百分点，该债券价格的百分比变动最接近：

If the yield decreases by 1%, the bond's percentage change in price is closest to:

A. 4.76%.

B. 5.05%.

C. 5.14%.

正确答案 / Answer: B

零息债的麦考利久期等于到期年限5年；先除以1.02得到修正久期。收益率下降1个百分点要写作Δy=-0.01。久期项和正凸性项在本题都增加价格，合计约5.0461%。

A zero-coupon bond has a Macaulay duration equal to its five-year maturity. Divide by 1.02 to obtain modified duration. A one-percentage-point yield decline means Δy=-0.01. Both the duration term and the positive convexity term increase price here, giving about 5.0461%.

1. 先转换久期口径。

Convert the duration measure.

```
D_mod = 5/(1 + 0.02) = 4.901961
```

2. 代入久期与凸性近似式。

Apply the duration-convexity approximation.

```
ΔP/P ≈ -4.901961(-0.01) + 0.5(28.835)(-0.01)^2 = 5.04614%
```

错误选项为什么不成立 / Why the other options fail

A. 把正凸性修正项减掉了。凸性项含收益率变动的平方，不能因降息而改成负数。

A. It subtracts the positive convexity adjustment. Squaring the yield change makes this term positive for positive convexity, even when yields fall.

C. 直接使用5年的麦考利久期，未转换为修正久期。

C. It uses the five-year Macaulay duration without converting it to modified duration.

可赎回债的价格上限 / **Callable bonds and limited price appreciation**

Q100 | 原题 / Original question

可赎回条款限制降息时的价格上涨

Call provisions limit price appreciation when rates fall

来源：题库 PDF 第 88 页（固收第 20 页）；答案 PDF 第 100 页。

Source: question bank p. 88 (Fixed Income p. 20); solution PDF p. 100.

其他条件相同时，当市场利率低于债券票面利率时，哪种债券的潜在价格上涨最受限制？

All else being equal, when the market interest rate falls below a bond's coupon rate, potential price appreciation is most limited for a:

A. 可回售债券。
putable bond.

B. 可赎回债券。
callable bond.

C. 无期权债券。
option-free bond.

正确答案 / Answer: B

降息使高票息旧债对投资者更有价值，但也提高发行人提前赎回并低息再融资的动力。可赎回债的投资者可能提前被偿还，难以保留全部高票息收益，因此其价格上涨受到赎回条款限制。

Falling rates make an existing high-coupon bond more valuable to investors, but also encourage the issuer to call it and refinance more cheaply. Investors may be repaid early and lose future high coupons, so the call provision limits price appreciation.

错误选项为什么不成立 / Why the other options fail

A. 回售权属于投资者，主要在利率上升、价格下跌时提供保护。

A. The investor owns the put, which mainly protects against price declines when rates rise.

C. 无期权债没有发行人赎回条款造成的上涨限制。

C. An option-free bond has no issuer call provision imposing this limit on appreciation.

Q101 | 原题 / Original question

可赎回债的凸性随利率区域改变

Callable-bond convexity changes across yield regions

来源：题库 PDF 第 88 页（固收第 20 页）；答案 PDF 第 101 页。

Source: question bank p. 88 (Fixed Income p. 20); solution PDF p. 101.

可赎回债券会表现出：

Callable bonds exhibit:

A. 只有正凸性。
positive convexity only.

B. 只有负凸性。
negative convexity only.

C. 可能为正凸性，也可能为负凸性。
either positive or negative convexity.

正确答案 / Answer: C

当利率较高、赎回可能性很小时，可赎回债表现接近普通债，具有正凸性。利率下降到赎回更有吸引力的区域时，上涨空间受限，可能进入负凸性区间。因此不能给可赎回债贴上“永远负凸性”的标签。

At high yields, when calling is unlikely, a callable bond behaves much like an ordinary bond and has positive convexity. At lower yields, calling becomes more attractive and caps appreciation, so the bond may enter a negative-convexity region. Callable bonds are not always negatively convex.

错误选项为什么不成立 / Why the other options fail

A. 忽略了赎回选择权变得有价值时可能出现的负凸性。

A. It ignores possible negative convexity when the call option becomes valuable.

B. 忽略了赎回不太可能发生时的正凸性区域。

B. It ignores the positive-convexity region where calling is unlikely.

CHAPTER 06 / 9 ORIGINAL QUESTIONS

持有期回报、再投资风险与久期缺口

Holding-period return, reinvestment risk and duration gap

> 把到期收到的钱拆为卖价或本金、票息、票息再投资收益三部分。
> Separate terminal wealth into sale proceeds or principal, coupons and reinvestment income.

先做这些原题检验理解：Q66, Q103, Q99, Q91。
Start with these original checkpoints: Q66, Q103, Q99, Q91.

6.1 一个利率变化，同时产生两个方向的影响

6.1 A rate change creates two opposing effects

利率变动 / Rate move	价格效应 / Price effect	再投资效应 / Reinvestment effect
下降 / Fall	固定现金流现值上升。 Fixed cash flows become more valuable.	未来票息再投资所得下降。 Coupon reinvestment income falls.
上升 / Rise	固定现金流现值下降。 Fixed cash flows become less valuable.	未来票息再投资所得上升。 Coupon reinvestment income rises.

Q94 直接考上表。Q80 在第一笔票息收到后、到期前卖出：既要面对未来卖价不确定，也要面对已收到票息再投资利率不确定，因此两种风险都有。
Q94 tests the table directly. Selling after receiving the first coupon but before maturity in Q80 exposes the investor both to an uncertain sale price and to reinvestment uncertainty for received coupons.

若持有到期且按约还本，就没有提前卖出的终值价格不确定性；但附息债券仍有再投资风险。按 Q5 的通常假设，利率上升提高未来票息再投资收益，可使实现回报高于购买时 YTM。可赎回债的再投资风险更突出，因为发行人常在低利率时赎回（Q16）。
If a bond is held to maturity and pays as promised, there is no uncertain terminal sale price, but coupon reinvestment risk remains. Under Q5's standard assumptions, rising rates can raise realized return above the purchase YTM. Callable bonds add reinvestment risk because issuers tend to call when rates are low (Q16).

6.2 票息终值：越早收到，能再投资越久

6.2 Coupon future value: earlier coupons compound longer

$$FV_{\text{coupons}} = \sum_{t} C_t(1+r)^{H-t}$$

Q66：年票息 5.5，持有 3 年。第 1 年票息再投资 2 年；第 2 年再投资 1 年；第 3 年刚收到，不再复利。
Q66 pays an annual coupon of 5.5 over a three-year holding period. The first coupon compounds for two years, the second for one and the third is received at the horizon without additional compounding.

```
FV_coupons = 5.5 x [(1.0475)^2 + 1.0475 + 1] = 17.2962
```

选 B。问题只问票息终值，所以不把卖出债券的价格或到期本金加进去。
Answer B. The question asks only for the future value of coupons; do not add sale proceeds or principal.

6.3 持有期收益率：先算总终值，再年化

6.3 Horizon yield: calculate terminal wealth, then annualize

$$r_H = \left(\frac{P_{\text{sale}} + FV_{\text{coupons}}}{P_{\text{buy}}}\right)^{1/H} - 1$$

Q103：终值 = 102.06 + 24.28 = 126.34；H = 7 年，而不是债券原来 10 年的期限。年化收益 = (126.34/95.27)^(1/7)−1 = 4.1147%，选 C。

In Q103, terminal wealth is 102.06 + 24.28 = 126.34. H is the seven-year holding period, not the original ten-year tenor. Annualized return is (126.34/95.27)^(1/7)−1 = 4.1147%, answer C.

Q91 反向思考：已知实现年化收益率 4.2%，再投资率只有 4.1%。按本题的持有到期假设，较低的再投资率把回报拉低，因此购买时 YTM 应高于 4.2%，选 C。

Work backward in Q91: the realized horizon yield is 4.2% while coupons were reinvested at 4.1%. Under the held-to-maturity assumptions, the lower reinvestment rate drags return down, so the purchase YTM must exceed 4.2%, answer C.

6.4 久期缺口：用平均收款时间匹配持有期

6.4 Duration gap: match average cash-flow timing to the horizon

$$\text{Duration gap} = D_{\text{Mac}} - H$$

缺口为零时，持有期等于 Macaulay 久期，而不是修正久期。它描述小幅、平行利率变化下价格与再投资效应的近似抵消；不是任何情景下都保证回报不变。

A zero gap matches the investment horizon to Macaulay duration, not modified duration. It approximately offsets price and reinvestment effects for small parallel rate shifts, rather than guaranteeing an unchanged return in every scenario.

```
Q99: H = MacDur = 7.4 x (1 + 0.09) = 8.066 years
```

选 C。原题未给付息频率；提供的答案采用年度口径。若有半年付息条件，换算应使用 1 + 0.09/2，因此本题需注明假设。

Answer C. The question omits payment frequency and the supplied solution uses an annual convention. Semiannual payments would require 1 + 0.09/2, so the annual assumption must be stated.

6.5 资本利得不一定等于卖价减购买价

6.5 Capital gain need not mean sale price minus purchase price

Q102 使用债券持有期分析中的账面价值口径：将卖价与出售时的 carrying value 比较，分开“折价摊增或溢价摊销”与额外资本利得。购买价用于计算整个投资的总收益，账面价值用于这道题的资本利得定义。

Q102 uses carrying value at the sale date to distinguish discount accretion or premium amortization from additional capital gain. Purchase price matters for total investment return; carrying value is the reference for the capital-gain definition tested here.

> 回报题先写三行：购买价格；持有期末收到的全部财富；实际持有年数。再看题目要“票息终值”“资本利得”还是“年化总回报”。
>
> For a return question, write down the purchase price, total terminal wealth and actual years held. Then identify whether it asks for coupon future value, capital gain or annualized total return.

价格风险与再投资风险 / **Price risk and reinvestment risk**

Q005 | 原题 / Original question

持有至到期时，再投资收益主导利率变化的影响

Reinvestment effects when a coupon bond is held to maturity

来源：题库 PDF 第 69 页（固收第 1 页）；答案 PDF 第 5 页。

Source: question bank p. 69 (Fixed Income p. 1); solution PDF p. 5.

如果持有期间利率上升，一只持有至到期的付息债券的总回报最可能：

If interest rates rise over the holding period, the total return of a coupon bond held until maturity is most likely to be:

A. 低于买入时的到期收益率。
less than the yield to maturity at purchase.

B. 等于买入时的到期收益率。
equal to the yield to maturity at purchase.

C. 高于买入时的到期收益率。
greater than the yield to maturity at purchase.

正确答案 / Answer: C

在题目默认无违约且持有至到期的情境下，最终收到的本金不受中途市价下跌影响；收到的票息却可以按更高利率再投资，因此实现收益率较买入时的 YTM 更高。YTM 等于实际复合回报需要票息按该 YTM 再投资等条件。

Under the question’s no-default, hold-to-maturity setting, the final principal repayment is not reduced by interim market-price declines. Coupons can instead be reinvested at higher rates, raising the realized compound return above the purchase YTM. Realizing the initial YTM requires, among other conditions, reinvestment at that yield.

错误选项为什么不成立 / Why the other options fail

A. 它把中途价格下跌当成了持有至到期后的已实现卖出损失。

A. This incorrectly treats an interim price decline as a realized sale loss for a bond held to maturity.

B. 只有满足相应再投资假设等条件，实现收益率才等于买入时 YTM。

B. Equality requires the relevant reinvestment assumptions and other conditions to hold.

Q016 | 原题 / Original question

赎回风险加剧再投资风险

Call risk increases reinvestment risk

来源：题库 PDF 第 71 页（固收第 3 页）；答案 PDF 第 16 页。

Source: question bank p. 71 (Fixed Income p. 3); solution PDF p. 16.

其他条件相同，下列哪一种债券的再投资风险最高？

All else being equal, reinvestment risk is greatest for:

A. 投资者可回售债券。
putable bonds.

B. 发行人可赎回债券。
callable bonds.

C. 不可赎回的可转换债券。
non-callable convertible bonds.

正确答案 / Answer: B

利率下降时，发行人更有动力赎回高票息债券，以更低成本再融资。投资者因此提前收回大额本金，却只能按较低市场利率再投资，所以可赎回债券的再投资风险最高。

When rates fall, the issuer has an incentive to call a high-coupon bond and refinance more cheaply. The investor then receives principal early and must reinvest it at lower market rates. This makes callable bonds particularly exposed to reinvestment risk.

错误选项为什么不成立 / Why the other options fail

A. 回售权由投资者控制，并非发行人在低利率时强迫投资者提前收回本金。

A. A put is controlled by the investor; it is not an issuer-driven repayment when rates are low.

C. 本选项明确不可赎回，缺少发行人在低利率时提前赎回本金的渠道。

C. The instrument is explicitly non-callable, so it lacks the issuer's low-rate early-redemption mechanism.

Q080 | 原题 / Original question

出售前既有再投资风险，也有价格风险

Reinvestment risk and sale-price risk before maturity

来源：题库 PDF 第 84 页（固收第 16 页）；答案 PDF 第 80 页。

Source: question bank p. 84 (Fixed Income p. 16); solution PDF p. 80.

就利率风险而言，投资者在收到第一笔票息之后、到期之前卖出固定利率债券，其面临：

With respect to interest rate risk, an investor who sells a fixed-rate bond after the first coupon is received and before its maturity is exposed to:

A. 只有市场价格风险。
market price risk, only.

B. 只有票息再投资风险。
coupon reinvestment risk, only.

C. 市场价格风险和票息再投资风险两者。
both market price risk and coupon reinvestment risk.

正确答案 / Answer: C

已收到的票息在出售前需要再投资，未来利率会影响其累积金额；债券又在到期前出售，出售价格受当时市场收益率影响。因此两种风险都存在。

Coupons received before sale must be reinvested, so future rates affect their accumulated value. Because the bond is sold before maturity, its sale price also depends on the market yield at that time. Both risks are present.

错误选项为什么不成立 / Why the other options fail

A. 忽略了已收到票息的再投资利率可能变化。

A. It ignores the changing reinvestment rate on coupons already received.

B. 忽略了到期前出售时，市场价格并非固定为面值。

B. It ignores the fact that the market sale price before maturity is not fixed at par.

Q091 | 原题 / Original question

持有至到期时，低再投资利率拖低实际收益

Low reinvestment rates reduce realized returns when held to maturity

来源：题库 PDF 第 87 页（固收第 19 页）；答案 PDF 第 91 页。

Source: question bank p. 87 (Fixed Income p. 19); solution PDF p. 91.

某投资组合经理收集到一只已持有至到期的无期权债券的下列信息：

A portfolio manager gathers the following information about an option-free bond that was held to maturity:

实际持有期年化收益率 Realized horizon yield	4.2%
再投资利率 Reinvestment rate	4.1%

购买时的到期收益率最可能：

The yield to maturity at purchase was most likely:

A. 低于4.2%。
less than 4.2%.

B. 等于4.2%。
equal to 4.2%.

C. 高于4.2%。
greater than 4.2%.

正确答案 / Answer: C

债券持有至到期，终点收到约定本金，无须按当时市场价格出售。本题再投资利率4.1%低于实际持有期年化收益4.2%；在正常付息、按约偿付的框架下，原始YTM必须高于4.2%，低再投资利率才会把实际结果拉低到4.2%。

The bond is held to maturity, so the terminal principal is contractually repaid rather than sold at a market price. Reinvestment earns 4.1%, below the 4.2% realized horizon yield. With normal positive coupons and promised repayment, the original YTM must exceed 4.2%; the lower reinvestment rate pulls the realized outcome down to 4.2%.

错误选项为什么不成立 / Why the other options fail

A. 若原YTM低于4.2%，再投资利率又只有4.1%，无法得到题目给出的4.2%结果。

A. An original YTM below 4.2%, combined with reinvestment at only 4.1%, cannot produce the stated 4.2% outcome in this framework.

B. 持有至到期要实现原YTM，票息须按原YTM再投资；4.1%与4.2%不一致。

B. Realizing the original YTM when held to maturity requires reinvestment at that YTM; 4.1% differs from 4.2%.

Q094 | 原题 / Original question

降息有利于当前价格，却降低未来票息再投资收入

Lower rates raise current price but reduce coupon reinvestment income

来源：题库 PDF 第 87 页（固收第 19 页）；答案 PDF 第 94 页。

Source: question bank p. 87 (Fixed Income p. 19); solution PDF p. 94.

对于固定利率债券，当利率下降时，再投资票息的未来价值最可能：

For a fixed-rate bond, when interest rates decrease, the future value of reinvested coupon payments most likely:

A. 下降，而债券市场价格上升。
decreases and the market price of the bond increases.

B. 上升，而债券市场价格下降。
increases and the market price of the bond decreases.

C. 上升，且债券市场价格上升。
increases and the market price of the bond increases.

正确答案 / Answer: A

同一利率变动有两个方向相反的影响：折现率下降，使尚未收取的约定现金流现值提高；票息收到后能赚到的再投资利率下降，使持有期末累计金额降低。先把“债券出售价格”和“已收票息的终值”分开判断。

The same rate decline has two opposite effects: lower discount rates raise the present value of remaining contractual cash flows, while lower rates earned after coupons are received reduce their accumulated value at the horizon. Analyze the sale price and the future value of received coupons separately.

错误选项为什么不成立 / Why the other options fail

B. 把两个方向都颠倒了：降息时再投资收入降低、债券价格提高。

B. It reverses both effects: lower rates reduce reinvestment income and raise bond price.

C. 正确判断了债券价格，但错误认为票息再投资终值也会提高。

C. It gets the bond-price direction right but incorrectly increases the reinvestment future value.

票息终值与年化回报 / Coupon future value and annualized return

Q066 | 原题 / Original question

票息再投资终值：每笔票息复利期数不同

Future value of coupons: each payment compounds for a different time

来源：题库 PDF 第 82 页（固收第 14 页）；答案 PDF 第 66 页。

Source: question bank p. 82 (Fixed Income p. 14); solution PDF p. 66.

分析师收集了一只债券的以下信息：

An analyst gathers the following information about a bond:

面值 Par value	100
票息率 Coupon rate	5.50%
付息频率 Coupon frequency	每年一次 Annual
到期期限 Time to maturity	5 年 5 years
持有期 Holding period	3 年 3 years

如果持有期内市场贴现率为 4.75%，持有期末再投资票息的终值最接近多少？

If the market discount rate is 4.75% for the holding period, the future value of reinvested coupons at the end of the holding period is closest to:

A. 15.05.

B. 17.30.

C. 18.12.

正确答案 / Answer: B

每年票息为 100 × 5.5% = 5.5。持有期为 3 年，因此第 1 年票息再投资 2 年，第 2 年票息再投资 1 年，第 3 年票息刚收到，复利 0 年。三笔终值之和为 5.5[(1.0475)^2 + 1.0475 + 1] = 17.2962，最接近 17.30。本题只问票息的终值，不包括卖债价格或本金。

The annual coupon is 100 × 5.5% = 5.5. Over the three-year holding period, the first coupon compounds for two years, the second for one year, and the third for zero years. Their future values sum to 5.5[(1.0475)^2 + 1.0475 + 1] = 17.2962, closest to 17.30. Only coupon future value is requested, excluding sale proceeds or principal.

1. 确定每年收到的票息。

Determine the annual coupon.

```
Coupon = 100 * 0.055 = 5.5
```

2. 按各自剩余再投资期数滚至第 3 年末。

Compound each coupon to the end of year 3.

```
FV(coupons) = 5.5 * 1.0475^2 + 5.5 * 1.0475 + 5.5 = 17.2962
```

错误选项为什么不成立 / Why the other options fail

A. 把票息率与再投资利率颠倒：用 4.75 作票息、用 5.50% 复利，得到约 15.05。

A. This swaps the coupon and reinvestment rates, using a coupon of 4.75 and compounding at 5.50% to obtain about 15.05.

C. 多复利了一年，相当于把每笔票息都提前一个期末收到。

C. This compounds each coupon for one extra year, as if all payments arrived one period earlier.

Q103 | 原题 / Original question

持有期年化收益率使用实际持有年数

Horizon yield uses the actual holding period

来源：题库 PDF 第 89 页（固收第 21 页）；答案 PDF 第 103 页。

Source: question bank p. 89 (Fixed Income p. 21); solution PDF p. 103.

投资者收集到某项十年期债券投资的下列信息：

An investor gathers the following information about an investment in a bond with a 10-year tenor:

购买价格 Purchase price	95.27
出售价格 Sale price	102.06
再投资票息的累积金额 Sum of reinvested coupon payments	24.28

如果持有期为七年，持有期年化收益率最接近：

If the holding period was seven years, the horizon yield is closest to:

A. 2.29%.

B. 2.86%.

C. 4.11%.

正确答案 / Answer: C

持有期末总财富=出售价格102.06+已再投资票息的累积金额24.28=126.34。将总财富除以初始投入95.27，再以实际持有7年进行年化。原债券期限10年不是投资者实际持有期。

Terminal wealth equals the sale price of 102.06 plus accumulated reinvested coupons of 24.28, or 126.34. Divide by the initial investment of 95.27 and annualize over the actual seven-year holding period. The bond's ten-year tenor is not the investor's holding period.

1. 计算期末总财富。

Calculate terminal wealth.

```
W_H = 102.06 + 24.28 = 126.34
```

2. 按实际持有七年复利年化。

Compound-annualize over the actual seven-year holding period.

```
r_H = (126.34/95.27)^(1/7) - 1 = 4.114712%
```

错误选项为什么不成立 / Why the other options fail

A. 颠倒了购买价格与出售价格。

A. It swaps purchase price and sale price.

B. 误用债券原期限10年进行年化，而非持有期7年。

B. It annualizes over the original ten-year tenor rather than the seven-year holding period.

久期缺口与资本利得口径 / **Duration gap and capital-gain convention**

Q099 | 原题 / Original question

久期缺口为零对应投资期限等于麦考利久期

Zero duration gap matches the horizon to Macaulay duration

来源：题库 PDF 第 88 页（固收第 20 页）；答案 PDF 第 99 页。

Source: question bank p. 88 (Fixed Income p. 20); solution PDF p. 99.

投资者持有一只具有下列特征的债券：

An investor holds a bond with the following characteristics:

到期收益率 Yield to maturity	9%
修正久期 Modified duration	7.4

如果久期缺口为零，投资期限最接近：

If the duration gap is zero, the investment horizon is closest to:

A. 6.8年。
6.8 years.

B. 7.4年。
7.4 years.

C. 8.1年。
8.1 years.

正确答案 / Answer: C

久期缺口定义为麦考利久期减投资期限。缺口为零时，投资期限必须等于麦考利久期。按原答案使用的年付息/年度收益率口径，麦考利久期=7.4×1.09=8.066年，最接近8.1年。

Duration gap is Macaulay duration minus the investment horizon. A zero gap therefore requires the horizon to equal Macaulay duration. Under the annual-payment/annual-yield convention used by the supplied answer, Macaulay duration is 7.4×1.09=8.066 years, closest to 8.1 years.

1. 缺口为零：投资期限等于麦考利久期。

A zero gap sets the horizon equal to Macaulay duration.

```
Gap = D_mac - H = 0
```

2. 按原答案年度口径进行转换。

Convert using the annual convention in the supplied answer.

```
H = D_mac = 7.4(1 + 0.09) = 8.066 years
```

错误选项为什么不成立 / Why the other options fail

A. 把修正久期转换为麦考利久期时应乘以1加每期收益率，不能再除一次。

A. Converting modified duration back to Macaulay duration requires multiplying by one plus the periodic yield, not dividing again.

B. 将修正久期直接当成麦考利久期。

B. It treats modified duration as though it were Macaulay duration.

> 原资料提示 / 使用条件：原题没有交代付息频率。原答案隐含每年付息（m=1）；通用关系是D_mac=D_mod×(1+y/m)。因此C依赖这一隐含口径，不能把7.4×1.09机械用于半年付息情形。
>
> Source note / condition: The question does not state coupon frequency. The supplied answer implicitly uses annual payments (m=1). The general relation is D_mac=D_mod×(1+y/m), so C relies on that convention; 7.4×1.09 should not be applied mechanically to a semiannual bond.

Q102 | 原题 / Original question

资本利得以出售时的恒定收益率账面价值为基准

Capital gain is measured against carrying value on the constant-yield path

来源：题库 PDF 第 89 页（固收第 21 页）；答案 PDF 第 102 页。

Source: question bank p. 89 (Fixed Income p. 21); solution PDF p. 102.

投资者出售一只最初以折价买入的固定利率债券。其资本利得或损失应将售价与下列哪项比较？

An investors sells a fixed-rate bond originally purchased at a discount. The resulting capital gain or loss should be measured by comparing the bond's selling price with its:

A. 面值。
par value.

B. 账面价值。
carrying value.

C. 买入价格。
purchase price.

正确答案 / Answer: B

本题采用债券收益分解口径：原始折价会随时间摊销，属于按购买时收益率逐步赚取的收益。出售日的账面价值体现这种恒定收益率价格轨迹；实际售价超出或低于该轨迹的部分，才是这里定义的资本利得或损失。

This question uses the bond-return decomposition convention. The original discount is accreted over time as part of the return earned at the purchase yield. Carrying value at the sale date reflects that constant-yield price path. The difference between sale price and that value is the capital gain or loss defined here.

错误选项为什么不成立 / Why the other options fail

A. 面值是到期偿还本金，不能代替出售日恒定收益率轨迹上的价值。

A. Par is the principal repaid at maturity, not the value on the constant-yield path at the sale date.

C. 售价减买价混合了折价摊销与额外的资本利得/损失。

C. Sale price minus purchase price mixes discount accretion with additional capital gain or loss.

> 原资料提示 / 使用条件：此处的carrying value是本题恒定收益率收益分解口径，不是在判断各种会计分类下财报列示的账面金额；直接"售价减买价"仍可描述价格差，但不是本题所问的资本利得分项。
>
> Source note / condition: Here, carrying value refers to the constant-yield return-decomposition convention, not every possible financial-reporting measurement category. Sale price minus purchase price describes a price difference, but not the capital-gain component requested here.

CHAPTER 07 / 13 ORIGINAL QUESTIONS

信用风险、偿债能力与清偿顺序

Credit risk, debt service and payment priority

> 区分“会不会违约”和“违约后能收回多少”，再看企业靠什么偿还。
> Separate the probability of default from recovery after default, then examine repayment capacity.

先做这些原题检验理解：Q30, Q51, Q18, Q89。
Start with these original checkpoints: Q30, Q51, Q18, Q89.

7.1 预期损失的三个组成部分

7.1 Three components of expected loss

$$EL = PD \times LGD \times EAD$$

PD 是违约概率；LGD 是违约损失比例；Exposure 是违约时风险敞口。在相同口径下 LGD = 1 − Recovery rate。
PD is default probability, LGD is the loss fraction given default and exposure is the amount at risk. Under matching conventions, LGD equals 1 minus the recovery rate.

Q30 只改变回收率：回收率提高，LGD 下降，所以预期损失下降。不能说它与回收率成正比，更不能说预期损失与回收率无关。
Q30 changes recovery rate alone. A higher recovery lowers LGD and therefore lowers expected loss. Expected loss is neither independent of recovery nor directly proportional to the recovery rate.

7.2 评级门槛和 notching

7.2 Rating thresholds and notching

要点 / Point	含义 / Meaning
投资级最低档 / Lowest investment grade	S&P/Fitch 为 BBB−；再降一级到 BB+ 即进入非投资级。Q17、Q20 BBB− on S&P/Fitch; one notch lower at BB+ is non-investment grade. Q17, Q20
发行人评级 / Issuer rating	用于评估发行人的整体信用状况。 Assesses the overall creditworthiness of the issuer.
债项评级 / Issue rating	考虑某一债项的担保、顺位等条款。 Reflects a specific instrument’s security and seniority.
评级调整 / Notching	按清偿顺位、回收预期等将债项评级相对基准上调或下调。 Adjusts issue ratings for seniority, expected recovery and related factors.

Q17 原文的 one category 在所附答案中指评级大类下降（BBB− 至 BB−），不同于下降一个 notch（BBB− 至 BB+）；两者都跨入非投资级。Q74 用 senior unsecured debt 作为通常的基准关系。Q57 与 Q89 强调：notching 的幅度和基准评级有关；高评级发行人通常需要较小调整。次级债较低评级主要反映违约后的损失严重程度，不等于同一发行人的每一类债务都有不同的违约发生概率。
In Q17, the supplied solution interprets one category as a broad-category move from BBB− to BB−, distinct from a single notch from BBB− to BB+; both cross into non-investment grade. Q74 uses senior unsecured debt as the usual rating reference. Q57 and Q89 emphasize that notching depends on the reference rating and is generally smaller for stronger credits. Lower subordinated ratings mainly reflect loss severity after default, not necessarily a different default-event probability for each debt class.

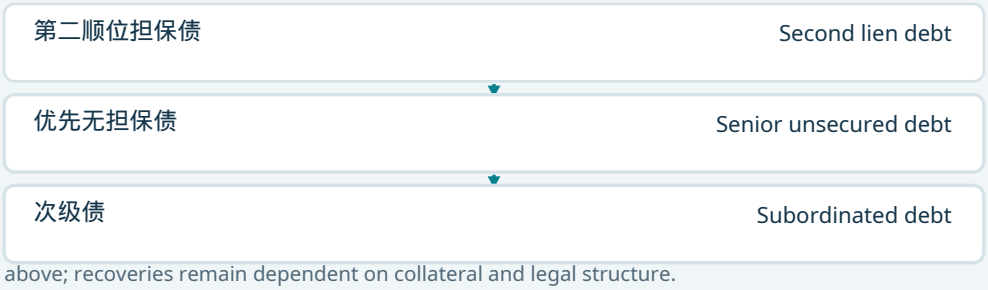
7.3 清偿顺序：看担保与法律层级

7.3 Payment priority: collateral and legal structure

Q15 两项债务都属于同一层级的 senior unsecured debt。期限较短不自动更优先，金额较大也不自动更优先；题目要求同级按应付金额比例分配。

Both claims in Q15 are senior unsecured debt of the same rank. Earlier maturity or a larger claim does not automatically create priority; equal-ranking creditors share proportionately under the question's assumptions.

Q95 | Standard relative priority among the listed claims



Higher priority above; recoveries remain dependent on collateral and legal structure.

图 4：原题 Q95 所比较的三类债务，在通常的教材顺位下排列；担保债的优先权首先针对相应担保资产。

Figure 4. Standard textbook priority among the three categories in Q95; secured priority first applies to the relevant collateral.

结构性次级来自公司组织层级。Q18 中经营子公司的债务先由子公司现金流偿还，剩余资金才能上划母公司；因此母公司债权人在经济上可能后于子公司债权人。这不只是把债券名称写成"次级"。

Structural subordination comes from the corporate organization. In Q18, operating-subsidiary debt is serviced before residual funds move up to the holding company, making holding-company creditors economically junior to subsidiary creditors.

7.4 偿债能力指标：先检查分子

7.4 Debt-service metrics: check the numerator

$$\text{EBIT coverage} = \frac{EBIT}{\text{Interest expense}}$$

Q51：operating income = 120；interest expense = 15；EBIT 利息覆盖倍数 = 8 倍。不要用净利润 85，也不要在题目明确要求 EBIT 时再加折旧摊销 15。

Q51: operating income is 120 and interest expense is 15, so EBIT interest coverage is 8 times. Do not use net income 85 or add depreciation and amortization 15 when EBIT coverage is explicitly requested.

$$\text{EBITDA coverage} = \frac{EBITDA}{\text{Interest expense}}$$

Q21 问指标类别：它是 coverage ratio（覆盖比率）。杠杆指标关心债务相对于资产、权益或盈利规模，盈利指标关心赚取利润的能力。

Q21 asks for the ratio category: it is a coverage ratio. Leverage metrics compare debt with assets, equity or earnings, whereas profitability metrics measure profit generation.

Q58 是项目收入支持的非主权政府债券，应优先看项目的债务服务覆盖能力，而不是只看居民人均收入或税基。Q53 则比较市场流动性：其他条件相同，投资级发行人通常面临较小的市场流动性风险。

Q58 concerns a revenue-backed non-sovereign government bond, so project debt-service coverage is central rather than general per-capita income or tax-base breadth. In Q53, an investment-grade issuer generally faces lower market-liquidity risk, all else equal.

评级、评级调整与流动性 / **Ratings, notching and liquidity**

Q017 | 原题 / Original question

投资级与非投资级的评级分界

The investment-grade boundary

来源：题库 PDF 第 71 页（固收第 3 页）；答案 PDF 第 17 页。

Source: question bank p. 71 (Fixed Income p. 3); solution PDF p. 17.

分析师担心某债券被标准普尔下调一个评级类别后成为非投资级。该债券目前的评级最可能是：

An analyst is concerned that a bond might be downgraded one category by Standard &Poor's and become non-investment grade. The current rating of this bond is most likely.

A. A-.

B. BB-.

C. BBB-.

正确答案 / Answer: C

标准普尔体系中，BBB- 是最低投资级。所附解答把 category 理解为评级大类，以 BBB- 下调至 BB- 为例，因此选 C。若只下调一个小档位（notch），BBB- 会降至 BB+，也会跨越投资级分界；不要把 category 与 notch 混用。

BBB- is the lowest investment-grade S&P rating. The supplied solution interprets category as a broad rating category and illustrates a move from BBB- to BB-, so C is correct. A one-notch downgrade from BBB- to BB+ would also cross the boundary, but a category and a notch are not the same unit.

错误选项为什么不成立 / Why the other options fail

A. 按所附解答，A- 下调一个评级大类至 BBB- 后仍为投资级。

A. In the supplied solution, a one-category move from A- to BBB- leaves the bond investment grade.

B. BB- 已经是非投资级，不是下调后才首次进入。

B. BB- is already non-investment grade.

Q020 | 原题 / Original question

S&P / Fitch 的最低投资级评级

Lowest investment-grade rating on the S&P / Fitch scales

来源：题库 PDF 第 72 页（固收第 4 页）；答案 PDF 第 20 页。

Source: question bank p. 72 (Fixed Income p. 4); solution PDF p. 20.

按标准普尔或惠誉的评级体系，投资级债券的最低评级是：

Using the rating scale from Standard &Poor's or Fitch, the lowest rating for an investment-grade bond is:

A. BBB-.

B. BBB.

C. BBB+.

正确答案 / Answer: A

从高到低看，BBB+、BBB、BBB- 都在投资级范围内；BBB- 下方的 BB+ 才进入非投资级。题目问“最低”，因此选择 BBB-。

BBB+, BBB, and BBB- are all investment grade. BB+, immediately below BBB-, is non-investment grade. Because the question asks for the lowest investment-grade rating, choose BBB-.

错误选项为什么不成立 / Why the other options fail

B. BBB 仍比最低投资级高一档。

B. BBB is one notch above the lowest investment-grade rating.

C. BBB+ 比 BBB- 高两档，不是最低投资级。

C. BBB+ is two notches above BBB-, so it is not the lowest investment-grade rating.

Q074 | 原题 / Original question

发行人评级与高级无担保债评级

Issuer ratings and senior unsecured debt

来源：题库 PDF 第 83 页（固收第 15 页）；答案 PDF 第 74 页。

Source: question bank p. 83 (Fixed Income p. 15); solution PDF p. 74.

某债券发行人的信用评级为BBB。仅根据此信息，该发行人的高级无担保债券评级最可能是：

A bond issuer has a credit rating of BBB. Based only on this information, the rating of a senior unsecured bond from this issuer is most likely to be:

A. 低于BBB。
lower than BBB.

B. BBB.

C. 高于BBB。
higher than BBB.

正确答案 / Answer: B

本题采用的评级基准是：发行人评级通常对应其高级无担保债。因此，没有担保、从属安排或其他额外信息时，最合理的判断是维持BBB。

In the convention used by this question, the issuer rating usually corresponds to senior unsecured debt. With no additional collateral, subordination, or other issue-specific information, BBB is the best estimate.

错误选项为什么不成立 / Why the other options fail

A. 题目未提供使该高级无担保债降级的结构或其他特征。

A. No issue-specific feature is supplied that would justify a lower senior unsecured rating.

C. 题目也未提供抵押担保等支持上调评级的特征。

C. No collateral or other feature is supplied that would justify a higher rating.

Q057 | 原题 / Original question

评级级差调整与受偿顺序

Notching and payment priority

来源：题库 PDF 第 79 页（固收第 11 页）；答案 PDF 第 57 页。

Source: question bank p. 79 (Fixed Income p. 11); solution PDF p. 57.

关于信用评级机构的评级级差调整，调整幅度：

With respect to the credit rating agencies' practice of notching, the size of a notching adjustment:

A. 对于评级更高的信用主体更大。
is larger for higher-rated credits.

B. 在不同评级机构之间标准化一致。
is standardized between rating agencies.

C. 取决于发行人优先无抵押债务的评级。
depends on the rating of the issuer's senior unsecured debt.

正确答案 / Answer: C

评级级差调整是在基准评级之上或之下，反映具体债项的受偿顺序、担保与违约后回收率。调整大小与发行人优先无抵押债务的信用等级有关。较高信用等级的发行人违约可能性低，受偿顺序差异对债项风险的影响通常较小，级差往往也较小。

Notching adjusts a reference rating upward or downward for an instrument’s seniority, security and expected recovery. The size of the adjustment depends in part on the rating of the issuer’s senior unsecured debt. For stronger issuers, default is less likely and differences in recovery priority usually have less impact, so notching is generally smaller.

错误选项为什么不成立 / Why the other options fail

A. 通常是评级越高，调整幅度越小，而非越大。

A. Adjustments are generally smaller, not larger, for higher-rated issuers.

B. 评级机构的方法和调整规则并非完全一致。

B. Rating agencies do not use identical standardized notching rules.

Q089 | 原题 / Original question

评级微调主要反映违约后的损失严重程度

Notching primarily reflects differences in loss severity

来源：题库 PDF 第 86 页（固收第 18 页）；答案 PDF 第 89 页。

Source: question bank p. 86 (Fixed Income p. 18); solution PDF p. 89.

关于信用评级机构采用的评级微调程序，某公司的次级债最可能：

With respect to the notching process adopted by credit rating agencies, a corporate's subordinated debt is most likely.

A. 以公司的更次级债务评级为基准向上调整。
notched up from the corporate's junior subordinated debt.

B. 公司发行人评级越高，评级微调幅度越小。
subject to a smaller notching adjustment the higher the corporate's issuer rating.

C. 因为违约概率更高，而从公司发行人评级向下调整。
notched down from the corporate's issuer rating due to higher probability of default.

正确答案 / Answer: B

本题中，具体债项评级以发行人评级为基准进行微调；不同清偿顺序主要影响违约后的回收率及损失严重程度。发行人本身信用越好，违约后的清偿差别对评级的影响通常越小，因此调整级数通常较少。

Here, issue ratings are notched from the issuer rating. Different payment priorities mainly affect recovery and loss severity after default. For stronger issuers, these differences generally have less effect on the rating, so fewer notches are typically applied.

错误选项为什么不成立 / Why the other options fail

A. 微调基准通常是发行人评级，而不是更次级债的评级。

A. The usual starting point is the issuer rating, not a more junior issue rating.

C. 本题考查的是同一发行人债项在损失严重程度上的差别，不能把次级身份直接解释为更高违约概率。

C. The distinction tested is loss severity among obligations of the same issuer, not automatically a higher probability of default for subordinated debt.

Q053 | 原题 / Original question

投资级与流动性风险的一般关系

Investment grade and market liquidity risk

来源：题库 PDF 第 78 页（固收第 10 页）；答案 PDF 第 53 页。

Source: question bank p. 78 (Fixed Income p. 10); solution PDF p. 53.

其他条件相同，与非投资级发行人相比，投资级债券发行人最可能具有：

All else being equal, an investment-grade bond issuer most likely has:

A. 较低的市场流动性风险。
less market liquidity risk than a below-investment-grade issuer.

B. 相同的市场流动性风险。
the same market liquidity risk as a below-investment-grade issuer.

C. 较高的市场流动性风险。
greater market liquidity risk than a below-investment-grade issuer.

正确答案 / Answer: A

市场流动性风险是无法以接近合理价值的价格迅速交易的风险。其他条件相同，投资级债券通常能吸引更广的投资者群体，交易市场相对更深，因而流动性风险较低。这是一般比较，不代表每只投资级债券都比每只高收益债券更活跃。

Market liquidity risk is the difficulty of trading promptly near fair value. All else equal, investment-grade bonds usually attract a broader investor base and have deeper markets, so liquidity risk is lower. This is a general comparison, not a claim that every investment-grade issue trades more actively than every high-yield issue.

错误选项为什么不成立 / Why the other options fail

B. 忽略了信用等级对投资者范围和交易流动性的一般影响。

B. This ignores the usual effect of credit quality on the investor base and trading liquidity.

C. 把一般关系写反了；较低等级债券通常面临较高流动性风险。

C. This reverses the usual relationship; lower-rated bonds generally have greater liquidity risk.

清偿顺序与结构性次级 / Claim priority and structural subordination

Q015 | 原题 / Original question

同一清偿顺位债权按金额比例分配

Claims of equal seniority share recovery proportionally

来源：题库 PDF 第 71 页（固收第 3 页）；答案 PDF 第 15 页。

Source: question bank p. 71 (Fixed Income p. 3); solution PDF p. 15.

分析师收集到一家仅有高级无担保债务的公司的以下资料：

An analyst gathers the following information about a company that only has senior unsecured debt:

类型 Type	未偿总额 Total Amount Outstanding	剩余期限 Time to Maturity
浮动利率票据（FRN） Floating-rate note (FRN)	5,000 万美元 $50 million	12 年 12 years
固定利率债券 Fixed-rate bond	1,000 万美元 $10 million	2 年 2 years

在破产情境下，如果严格执行债权清偿顺位，最可能出现哪种情况？

In a bankruptcy scenario, if the priority of claims is enforced, it is most likely that:

A. 固定利率债券先偿还，因为其到期日较早。
fixed-rate bond is repaid first because it matures earlier.

B. 两类债券持有人按各自被欠金额的比例获得偿还。
both bondholders are repaid proportionally to the amount owed.

C. 浮息票据先偿还，因为其未偿金额最大。
FRN is repaid first because it has the highest amount outstanding.

正确答案 / Answer: B

两类债务都是高级无担保债，清偿顺位相同。若严格执行相同顺位的分配规则，应按债权金额比例分配可回收资金，而不是按到期先后或谁金额较大决定优先权。本题两组债权金额之比为 50:10，即 5:1。

Both instruments are senior unsecured claims and therefore have the same payment priority. Under the stated enforcement assumption, recoveries are allocated in proportion to amounts owed, not by maturity or by which creditor has the larger claim. Their claim amounts are in a 50:10, or 5:1, ratio.

错误选项为什么不成立 / Why the other options fail

A. 到期日较早本身不会提高破产清偿顺位。

A. An earlier maturity does not by itself create a higher bankruptcy priority.

C. 债权金额较大意味着同等回收率下金额更多，不意味着优先获偿。

C. A larger claim receives a larger amount at an equal recovery rate, but does not gain priority.

Q018 | 原题 / Original question

控股公司债务的结构性次级

Structural subordination of holding-company debt

来源：题库 PDF 第 72 页（固收第 4 页）；答案 PDF 第 18 页。

Source: question bank p. 72 (Fixed Income p. 4); solution PDF p. 18.

一家集团的母控股公司和经营子公司都存在债务。经营层面的债务必须先得到偿付，资金才能上划用于偿还控股公司的债务。这种安排最符合：

A corporation with a holding company structure has debt at both its parent holding company and operating subsidiaries. Debt at the operating level must be serviced before funds can be upstreamed to pay debt at the holding company. This arrangement best describes:

A. 结构性次级。
structural subordination.

B. 交叉违约条款。
a cross-default provision.

C. 公司家族评级。
the corporate family rating.

正确答案 / Answer: A

母公司依赖子公司分配资金。子公司自己的债权人先对其资产和现金流享有债权，满足这些债务后才能把资金上划给母公司。因此母公司债权人因公司组织结构而处于较不利位置，这就是结构性次级。

The holding company relies on distributions from its operating subsidiaries. Subsidiary creditors first have claims on subsidiary assets and cash flows; funds can move upstream only after those obligations are met. The resulting disadvantage for parent creditors arises from the corporate structure.

错误选项为什么不成立 / Why the other options fail

B. 交叉违约条款规定一项债务违约可触发其他债务违约，不是现金先后分配关系。

B. A cross-default clause links defaults across debt obligations; it does not describe this upstream cash-flow hierarchy.

C. 公司家族评级是对集团整体信用质量的评估，不是这种清偿结构的名称。

C. A corporate family rating assesses group credit quality; it is not the name of this payment structure.

Q095 | 原题 / Original question

有担保债的清偿优先顺序

Priority of secured debt claims

来源：题库 PDF 第 87 页（固收第 19 页）；答案 PDF 第 95 页。

Source: question bank p. 87 (Fixed Income p. 19); solution PDF p. 95.

下列债务类别中，哪一类的偿付优先级最高？

Which of the following debt categories has the highest ranking in terms of priority of payment?

A. 第二留置权债务。
Second lien debt

B. 次级债务。
Subordinated debt

C. 高级无担保债务。
Senior unsecured debt

正确答案 / Answer: A

在本题标准债务优先顺序下，第二留置权债仍属有抵押担保债：对相关抵押物的权利排在第一留置权之后，但排在无担保债权之前。因此在给定三项中选A。“第二”并不表示低于所有“高级”债。

Under the standard ranking tested here, second-lien debt is still secured: its claim on the pledged collateral follows the first lien but precedes unsecured claims. It therefore ranks highest among these choices. “Second” does not mean below all debt labeled “senior.”

错误选项为什么不成立 / Why the other options fail

B. 次级债在无担保债内部的顺位也低于高级无担保债。

B. Subordinated debt also ranks below senior unsecured debt within the unsecured category.

C. “高级”是相对其他无担保债而言，并不使其对抵押物的权利优先于有担保债。

C. “Senior” distinguishes it from junior unsecured claims; it does not give priority over secured claims on pledged collateral.

预期损失与偿债指标 / **Expected loss and debt-service ratios**

Q030 | 原题 / Original question

预期损失与回收率、违约损失率

Expected loss, recovery rate, and loss given default

来源：题库 PDF 第 74 页（固收第 6 页）；答案 PDF 第 30 页。

Source: question bank p. 74 (Fixed Income p. 6); solution PDF p. 30.

其他条件相同，一项债务工具的预期损失：

All else being equal, expected loss for a debt instrument:

A. 与回收率无关。
is independent of the recovery rate.

B. 随回收率提高而下降。
decreases as the recovery rate increases.

C. 与回收率成正比变化。
changes proportionally to the recovery rate.

正确答案 / Answer: B

预期损失可写成 PD × LGD × EAD，其中 LGD 的比例等于 1 减回收率。在违约概率和违约风险敞口不变时，回收率越高，真正无法收回的金额越少，预期损失越低。

Expected loss can be written as PD × LGD × EAD, where the proportional LGD equals one minus the recovery rate. Holding default probability and exposure constant, a higher recovery rate reduces the amount lost and therefore lowers expected loss.

1. 区分违约发生概率与违约后的损失比例。

Separate the chance of default from the fraction lost after default.

```
Expected loss = PD*(1-Recovery rate)*EAD
```

错误选项为什么不成立 / Why the other options fail

A. 回收率通过 LGD 直接影响预期损失。

A. Recovery directly affects expected loss through LGD.

C. 预期损失与 1 减回收率成正比，而非与回收率本身成正比。

C. Expected loss is proportional to one minus the recovery rate, not to the recovery rate itself.

Q021 | 原题 / Original question

利息覆盖倍数：盈利能覆盖几倍利息

Interest coverage: earnings relative to interest obligations

来源：题库 PDF 第 72 页（固收第 4 页）；答案 PDF 第 21 页。

Source: question bank p. 72 (Fixed Income p. 4); solution PDF p. 21.

EBITDA / 利息费用这一比率最适合归为：

The EBITDA/interest expense ratio is best classified as a:

A. 杠杆比率。
leverage ratio.

B. 覆盖比率。
coverage ratio.

C. 盈利能力比率。
profitability ratio.

正确答案 / Answer: B

分母是需要支付的利息，分子是用于衡量偿债能力的一种盈利指标。这个比率回答“EBITDA 是利息费用的多少倍”，所以是利息覆盖比率。它不是实际自由现金流，也不直接等同于现金已经足够。

The denominator is interest owed and the numerator is an earnings measure used to assess debt-service capacity. The ratio asks how many times EBITDA covers interest expense, making it a coverage ratio. EBITDA is not itself free cash flow or proof that cash is available.

错误选项为什么不成立 / Why the other options fail

A. 杠杆比率通常比较债务与资产、权益或盈利规模，例如债务/EBITDA。

A. Leverage ratios compare debt with assets, equity, or earnings, such as debt/EBITDA.

C. 盈利能力比率衡量利润相对于收入、资产或权益的水平。

C. Profitability ratios measure earnings relative to revenue, assets, or equity.

Q051 | 原题 / Original question

利息保障倍数：EBIT 除以利息费用

Interest coverage: EBIT divided by interest expense

来源：题库 PDF 第 78 页（固收第 10 页）；答案 PDF 第 51 页。

Source: question bank p. 78 (Fixed Income p. 10); solution PDF p. 51.

分析师收集了一家公司的以下资料（单位：百万美元）：

An analyst gathers the following information about a company (in $ millions):

营业利润 Operating income	120
净利润 Net income	85
利息费用 Interest expense	15
折旧和摊销 Depreciation and amortization	15

使用 EBIT 计算的利息保障倍数最接近多少？

Interest coverage using EBIT is closest to:

A. 7 倍。
7x.

B. 8 倍。
8x.

C. 9 倍。
9x.

正确答案 / Answer: B

本题把营业利润 120 作为 EBIT，表示尚未扣除利息和所得税的利润。EBIT 利息保障倍数 = 120/15 = 8 倍。净利润 85 不是所需分子；折旧摊销只有在计算 EBITDA 利息保障倍数时才加回。

Operating income of 120 is used as EBIT here: earnings before interest and tax. EBIT interest coverage is 120/15 = 8 times. Net income of 85 is not the required numerator. Depreciation and amortization are added back only for an EBITDA-based coverage measure.

1. 选择 EBIT 口径。

Use the EBIT numerator.

```
Interest coverage = EBIT / Interest expense = 120 / 15 = 8x
```

错误选项为什么不成立 / Why the other options fail

A. 把净利润加折旧摊销后除以利息，(85 + 15)/15 = 6.67，约为 7 倍；这个分子不是 EBIT。

A. (85 + 15)/15 = 6.67, about 7 times, incorrectly uses net income plus depreciation and amortization rather than EBIT.

C. (120 + 15)/15 = 9 是 EBITDA 口径，本题要求 EBIT。

C. (120 + 15)/15 = 9 uses EBITDA, whereas the question asks for EBIT.

> 原资料提示 / 使用条件：原题表格前有一个孤立的“<”符号，无对应的比较关系；本讲义将其视为排版残留，不纳入题干，表格数据保持原样。
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Q058 | 原题 / Original question

项目收入债：看项目现金流能否覆盖债务服务

Revenue bonds: assess project cash flows against debt service

来源：题库 PDF 第 79 页（固收第 11 页）；答案 PDF 第 58 页。

Source: question bank p. 79 (Fixed Income p. 11); solution PDF p. 58.

分析以特定收入作偿债来源的非主权政府债券时，以下哪项最可能是关键因素？

Which of the following is most likely a key factor in the credit analysis of revenue-backed non-sovereign government bonds?

A. 人均收入。
Per capita income

B. 税基广度。
Breadth of the tax base

C. 项目的偿债保障倍数。
Debt-service-coverage ratio of the project

正确答案 / Answer: C

收入支持债券主要依靠特定项目或设施的收入偿债，因此重点是项目现金流能否支付当期利息和应偿本金。偿债保障倍数把可用于偿债的现金流与债务服务额比较。人均收入和税基更直接用于分析依赖总体征税能力的一般责任债券。

Revenue-backed bonds rely primarily on cash flows from a specified project or facility. Credit analysis therefore asks whether those cash flows cover interest and required principal payments. The debt-service-coverage ratio compares cash available for debt service with required debt service. Per capita income and the tax base are more directly relevant to general-obligation bonds supported by taxing capacity.

错误选项为什么不成立 / Why the other options fail

A. 更适用于判断地区经济实力和一般税收偿债能力。

A. This is more relevant to local economic strength and tax-supported repayment capacity.

B. 税基广度主要关系到依靠征税偿债的一般责任债券。

B. The breadth of the tax base primarily informs analysis of tax-backed general-obligation debt.

CHAPTER 08 / 10 ORIGINAL QUESTIONS

证券化、抵押贷款与结构化分层

Securitization, mortgages and tranching

> 先追踪资产和现金流，再区分损失分配、偿还时间与提前还款风险。
> Trace assets and cash flows, then distinguish loss allocation, repayment timing and prepayment risk.

先做这些原题检验理解：Q8, Q45, Q71, Q41。
Start with these original checkpoints: Q8, Q45, Q71, Q41.

8.1 一笔贷款如何变成证券

8.1 How loans become securities

证券化把一组贷款或应收款的现金流组织起来，发行证券供投资者购买。原始资产卖方把资产转给特殊目的实体（SPE/SPV）；投资者承担的风险取决于资产、合同和分层安排。
Securitization packages cash flows from loans or receivables into securities. The originator or seller transfers assets to a special-purpose entity (SPE/SPV). Investor risk depends on collateral, contracts and tranche design.

角色 / Party	作用 / Role
卖方 / Seller or originator	提供并出售基础资产。 Creates or supplies and sells the collateral.
特殊目的实体 / SPE or SPV	依法拥有基础资产并发行相关证券。 Legally owns the collateral and issues the securities.
受托人 / Trustee	按本题的角色区分保管担保资产，代表证券持有人监督。Q8 Holds the collateral in the source question's role distinction and oversees investor interests. Q8
服务机构 / Servicer	收取借款人付款并执行贷款服务安排。 Collects borrower payments and services the loans.

Q11 的资产购买协议包含资产质量方面的陈述与保证（representations and warranties）。不要把购买协议直接等同于整项交易结构说明或所有信用增级文件。
The purchase agreement in Q11 contains representations and warranties about asset quality. It is not simply the complete transaction-structure description or the documentation of all credit enhancements.

8.2 过手利率、资产类别与两种分层

8.2 Pass-through rates, collateral types and two forms of tranching

本题所指的按揭转付证券将住宅按揭贷款现金流传递给投资者。其证券票息叫 pass-through rate；它通常反映服务费等扣减，不能直接等同于底层贷款的加权平均票息 WAC。Q9。
The mortgage pass-through security tested here passes residential mortgage cash flows to investors. Its coupon is the pass-through rate, generally reflecting deductions such as servicing fees; it is not simply the mortgage pool's weighted-average coupon (WAC). Q9.

类别 / Type	辨识方式 / Recognition
CLO	以贷款为主要抵押资产。 Collateral is primarily loans.
CBO	以债券为主要抵押资产。 Collateral is primarily bonds.

类别 / Type	辨识方式 / Recognition
Structured finance CDO	以 ABS 等结构化证券为主要抵押资产。Q24 Collateral includes structured securities such as ABS. Q24
信用分层 / Credit tranching	优先/次级结构改变损失承担顺序和信用风险。Q45 Senior/subordinated tranches allocate losses and credit risk. Q45
时间分层 / Time tranching	改变本金偿还时间和预期到期日。Q97 Tranches differ in principal repayment timing and expected maturities. Q97

信用分层问“谁先承担损失”；时间分层问“谁先收回本金”。同一证券化交易可以同时存在两种分层，不要因为出现 senior 字样就想到提前还款顺序。

Credit tranching asks who absorbs losses first; time tranching asks who receives principal first. A transaction may use both. The term senior alone does not identify a prepayment-timing rule.

8.3 信用卡 ABS 的循环期

8.3 The revolving period of credit-card ABS

信用卡应收款不是一组按固定摊销表逐期还清的普通按揭贷款。Q71 的 lockout/revolving period 内，收到的本金用于购买或补充应收款，而不是立即按比例全部返还投资者。之后才按交易安排进入本金偿还阶段。

Credit-card receivables do not behave like a fixed pool of fully amortizing mortgages. During the lockout or revolving period in Q71, principal collections are reinvested in receivables rather than immediately passed through in full. Principal payment begins later under the transaction terms.

8.4 利率变化如何改变按揭还款速度

8.4 How rates change mortgage repayment speed

情景 / Scenario	行为与风险 / Behavior and risk
利率下降 / Rates fall	借款人更有动力再融资和提前还款；收缩风险与再投资压力上升。 Refinancing and prepayment incentives rise, increasing contraction and reinvestment risk.
利率上升 / Rates rise	再融资吸引力下降，提前还款变慢；本金留得更久，展期风险上升。Q41 Refinancing becomes less attractive and prepayments slow, increasing extension risk. Q41

Non-recourse mortgage（无追索按揭）通常限制贷款人向担保物以外资产追偿。其他条件相同，它提高借款人在房屋价值低于贷款余额时战略性违约的诱因（Q32）。提前还款罚金限制或补偿提前偿还，通常对贷款人更有利（Q88）。

A non-recourse mortgage generally limits a lender’s claim beyond the collateral. All else equal, this increases the incentive for strategic default when the property is worth less than the loan balance (Q32). A prepayment penalty restricts or compensates early repayment and generally benefits the lender (Q88).

> 本章用三组反义词记忆：信用分层/时间分层；展期/收缩；有追索/无追索。再用 Q45、Q97、Q41、Q32 验证，不需要额外虚构例题。
>
> Remember three pairs: credit/time tranching, extension/contraction, and recourse/non-recourse. Test them with Q45, Q97, Q41 and Q32; no additional invented questions are needed.

参与者、合同与资产类别 / **Parties, contracts and collateral types**

Q008 | 原题 / Original question

证券化各方：卖方、SPE 与受托人

Securitization roles: seller, SPE, and trustee

来源：题库 PDF 第 70 页（固收第 2 页）；答案 PDF 第 8 页。

Source: question bank p. 70 (Fixed Income p. 2); solution PDF p. 8.

在证券化过程中，受托人最可能：

In the securitization process, the trustee most likely:

A. 出售基础抵押资产。
sells the underlying collateral.

B. 拥有基础抵押资产。
owns the underlying collateral.

C. 保管基础抵押资产。
holds the underlying collateral.

正确答案 / Answer: C

按本题的证券化角色划分，卖方将资产出售给 SPE，SPE 拥有资产，受托人代表投资者保管或控制相关抵押资产并监督约定。记忆时把“出售、拥有、保管”分配给不同角色。

Under the role distinctions used in this question, the seller transfers assets to the SPE, the SPE owns them, and the trustee holds the collateral for investors and oversees the relevant arrangements. Distinguish selling, owning, and holding the assets.

错误选项为什么不成立 / Why the other options fail

A. 出售资产的是资产卖方或发起人。

A. The seller or originator sells the collateral.

B. 资产所有权归特殊目的实体（SPE），不是本题所指的受托人角色。

B. The special purpose entity owns the assets; ownership is not the trustee role tested here.

Q011 | 原题 / Original question

资产购买协议中的陈述与保证

Representations and warranties in the asset purchase agreement

来源：题库 PDF 第 70 页（固收第 2 页）；答案 PDF 第 11 页。

Source: question bank p. 70 (Fixed Income p. 2); solution PDF p. 11.

在证券化交易中，抵押资产卖方与特殊目的实体之间的购买协议最可能提供：

In a securitization, the purchase agreement between the seller of the collateral and the special purpose entity most likely provides:

A. 对交易结构的描述。
a description of the transaction structure.

B. 关于资产质量的陈述。
representations about the quality of the assets.

C. 用于降低信用风险的增信措施文件。
documentation of enhancements used to reduce credit risk.

正确答案 / Answer: B

购买协议的重点之一，是卖方就出售资产作出的陈述与保证，例如资产是否符合约定的质量和资格标准。交易结构及增信安排的整体说明通常见于招股说明书。

A key function of the purchase agreement is to record the seller's representations and warranties about the assets sold, including agreed quality or eligibility characteristics. The prospectus typically describes the overall structure and credit-enhancement arrangements.

错误选项为什么不成立 / Why the other options fail

A. 交易结构的整体描述主要属于招股说明书的内容。

A. The overall transaction structure is described in the prospectus.

C. 增信安排的说明主要属于招股说明书的内容。

C. The prospectus documents the credit-enhancement arrangements.

Q009 | 原题 / Original question

抵押贷款利率与转付利率

Mortgage rate versus pass-through rate

来源：题库 PDF 第 70 页（固收第 2 页）；答案 PDF 第 9 页。

Source: question bank p. 70 (Fixed Income p. 2); solution PDF p. 9.

住宅抵押贷款支持证券（RMBS）的票息是：

The coupon of a residential mortgage-backed security is the:

A. 转付利率。
pass-through rate.

B. 加权平均票息率。
weighted average coupon rate.

C. 基础抵押贷款资产池的利率。
rate on the underlying pool of mortgages.

正确答案 / Answer: A

在本题讨论的抵押贷款转付证券中，投资者收到的票息率称为 pass-through rate。基础贷款支付的利息还要承担服务费及其他管理费，因此转付利率通常低于贷款池的加权平均票息率。

For the mortgage pass-through security tested here, the coupon paid to investors is the pass-through rate. Servicing and other administrative fees are deducted from interest on the underlying mortgages, so this rate is generally below the pool's weighted average coupon.

错误选项为什么不成立 / Why the other options fail

B. WAC 衡量基础贷款池的加权票息，尚未扣除相关费用。

B. WAC measures the underlying pool's weighted coupon before the relevant fees are deducted.

C. 贷款借款人支付的利率不是投资者扣费后收到的净票息率。

C. The rate paid by mortgage borrowers is not the net coupon received by security investors.

Q024 | 原题 / Original question

按基础资产区分 CDO、CLO 与 CBO

Classifying CDOs by their collateral

来源：题库 PDF 第 73 页（固收第 5 页）；答案 PDF 第 24 页。

Source: question bank p. 73 (Fixed Income p. 5); solution PDF p. 24.

下列哪一类担保债务凭证（CDO）最可能以资产支持证券作为支持资产？

Which of the following types of collateralized debt obligations (CDOs) are most likely backed by asset-backed Securities?

A. 结构化金融 CDO。
Structured finance CDOs

B. 担保贷款凭证（CLO）。
Collateralized loan obligations

C. 担保债券凭证（CBO）。
Collateralized bond obligations

正确答案 / Answer: A

看名称所指向的基础资产：CLO 主要以贷款为支持；CBO 主要以债券为支持；结构化金融 CDO 则以 ABS 等结构化金融证券为支持。

Classify by collateral: a CLO is primarily backed by loans, a CBO by bonds, and a structured-finance CDO by structured-finance securities such as ABS.

错误选项为什么不成立 / Why the other options fail

B. CLO 的典型基础资产是贷款。

B. The characteristic collateral of a CLO is loans.

C. CBO 的典型基础资产是债券。

C. The characteristic collateral of a CBO is bonds.

信用分层、时间分层与循环期 / **Credit tranches, time tranches and revolving periods**

Q045 | 原题 / Original question

信用分层：次级先吸收损失

Credit tranching: subordinated claims absorb losses first

来源：题库 PDF 第 77 页（固收第 9 页）；答案 PDF 第 45 页。

Source: question bank p. 77 (Fixed Income p. 9); solution PDF p. 45.

在资产证券化中，优先 / 次级结构最可能属于：

In a securitization, a senior/subordinated structure is most likely a form of:

A. 期限分层。
time tranching.

B. 信用分层。
credit tranching.

C. 提前偿还风险管理。
prepayment risk management.

正确答案 / Answer: B

优先 / 次级结构按损失承担顺序划分证券：次级档先吸收资产池信用损失，优先档获得保护。这是在不同档次之间重新分配信用风险。期限分层则主要调整本金偿还时间和预计到期日。

A senior/subordinated structure allocates credit losses by priority: subordinated tranches absorb losses before senior tranches. This redistributes credit risk across tranches. Time tranching instead primarily changes the timing of principal payments and expected maturities.

错误选项为什么不成立 / Why the other options fail

A. 期限分层关注现金流何时偿还，而非谁先承担信用损失。

A. Time tranching focuses on when principal is repaid, not who bears credit losses first.

C. 提前偿还风险管理主要涉及偿还速度与时点，不是优先 / 次级结构的定义。

C. Prepayment-risk management concerns payment speed and timing; it does not define seniority.

Q097 | 原题 / Original question

时间分层与信用分层的区别

Distinguish time tranching from credit tranching

来源：题库 PDF 第 88 页（固收第 20 页）；答案 PDF 第 97 页。

Source: question bank p. 88 (Fixed Income p. 20); solution PDF p. 97.

证券化中的时间分层最可能指下列哪方面存在差异？

In a securitization, time tranching most likely refers to differences in:

A. 违约风险。
default risk.

B. 预期到期时间。
expected maturities.

C. 基础抵押资产。
underlying collateral.

正确答案 / Answer: B

时间分层通过分配本金偿还的先后顺序，形成预期到期时间不同的证券层级；信用分层则分配违约损失的承担顺序。分层重新安排同一资产池现金流如何分配，并不要求更换基础资产。

Time tranching allocates principal repayments in different sequences, creating tranches with different expected maturities. Credit tranching instead allocates the order in which default losses are absorbed. Tranching rearranges distributions from an asset pool without requiring different collateral.

错误选项为什么不成立 / Why the other options fail

A. 违约损失承担顺序的差异属于信用分层。

A. Differences in the allocation of default losses describe credit tranching.

C. 同一资产池也可以设计不同时间层级，不必改变抵押资产。

C. A single collateral pool can support different time tranches; changing collateral is not required.

Q071 | 原题 / Original question

循环型 ABS：锁定期内本金补入新应收款

Revolving ABS: principal buys new receivables during the lockout period

来源：题库 PDF 第 83 页（固收第 15 页）；答案 PDF 第 71 页。

Source: question bank p. 83 (Fixed Income p. 15); solution PDF p. 71.

信用卡应收款资产支持证券最可能：

A credit card receivable asset-backed security most likely:

A. 面临较高的提前偿还风险。
faces high prepayment risk.

B. 以全额摊还贷款作为抵押品。
uses fully-amortizing loans as collateral.

C. 在锁定期内将收到的本金偿还款再投资。
reinvests principal repayments during the lockout period.

正确答案 / Answer: C

信用卡应收款通常是非摊还、循环性的资产池。锁定期或循环期内，收到的本金一般用于购买新的合格应收款，以维持资产池，而不是立即分配给投资者；利息及费用按结构支付。进入后续本金偿还阶段后，才按照交易安排向投资者返还本金。

Credit-card receivables generally form a revolving, non-amortizing pool. During the lockout or revolving period, principal collections are typically used to purchase new eligible receivables rather than immediately distributed to investors; interest and fees are handled according to the structure. Principal is returned to investors in the later repayment phase under the transaction terms.

错误选项为什么不成立 / Why the other options fail

A. 循环期内本金回款被用于补充资产池，不能直接等同于摊还型 MBS 的高提前偿还风险。

A. Because principal collections replenish the pool during the revolving period, they do not directly create the same high prepayment risk as amortizing MBS.

B. 信用卡应收款不是按固定计划逐期全额摊还的贷款。

B. Credit-card receivables are not fully amortizing loans with a fixed scheduled repayment path.

抵押贷款的提前还款与违约 / **Mortgage prepayment and default**

Q032 | 原题 / Original question

无追索权贷款与策略性违约

Non-recourse loans and strategic default

来源：题库 PDF 第 74 页（固收第 6 页）；答案 PDF 第 32 页。

Source: question bank p. 74 (Fixed Income p. 6); solution PDF p. 32.

其他条件相同，无追索权抵押贷款的策略性违约风险最可能：

All else being equal, the risk of a strategic default for a non-recourse mortgage is most likely:

A. 低于有追索权抵押贷款。
lower than for a recourse mortgage.

B. 等于有追索权抵押贷款。
the same as for a recourse mortgage.

C. 高于有追索权抵押贷款。
greater than for a recourse mortgage.

正确答案 / Answer: C

无追索权意味着贷款人通常只能追索抵押房产，不能再追索借款人的其他资产。当房产价值跌到贷款余额以下，借款人放弃抵押物的经济成本较低，策略性违约诱因更强。

Non-recourse generally limits the lender to the mortgaged property rather than the borrower's other assets. If the property falls below the loan balance, surrendering it can impose a smaller economic cost on the borrower, increasing the incentive for strategic default.

错误选项为什么不成立 / Why the other options fail

A. 不能追索其他资产削弱了对借款人策略性违约的约束，不会降低该风险。

A. The inability to pursue other assets weakens the deterrent to strategic default rather than reducing the risk.

B. 追索权改变了违约的经济后果，所以风险并非相同。

B. Recourse changes the economic consequences of default, so the risks are not the same.

Q041 | 原题 / Original question

抵押贷款：利率上升会延长本金回收时间

Mortgages: rising rates can delay principal repayment

来源：题库 PDF 第 76 页（固收第 8 页）；答案 PDF 第 41 页。

Source: question bank p. 76 (Fixed Income p. 8); solution PDF p. 41.

利率上升时，抵押贷款支持证券最可能增加哪一种风险？

When interest rates increase, mortgage-backed securities most likely exhibit increased:

A. 延期风险。
extension risk.

B. 收缩风险。
contraction risk

C. 再投资风险。
reinvestment risk.

正确答案 / Answer: A

利率上升后，借款人以更低利率再融资的动机减弱，提前还款通常放慢。MBS 投资者收回本金的时间因而延后，实际平均存续期可能超过原来预期，这叫延期风险。反过来，利率下降往往促使提前还款加快，形成收缩风险和较低利率下的再投资压力。

Higher rates reduce borrowers' incentives to refinance at lower rates, so prepayments typically slow. MBS investors receive principal later than expected and the realized average life may extend: this is extension risk. Falling rates tend to accelerate prepayments, creating contraction risk and reinvestment pressure at lower rates.

错误选项为什么不成立 / Why the other options fail

B. 收缩风险通常随降息和提前还款加快而增加。

B. Contraction risk usually rises when rates fall and prepayments accelerate.

C. 本题最直接的风险是延期；降息导致本金较快返还时，再投资压力通常更明显。

C. Extension is the direct effect here; reinvestment pressure is usually greater when lower rates return principal sooner.

Q088 | 原题 / Original question

提前还款罚金保护贷款人

Prepayment penalties protect the lender

来源：题库 PDF 第 86 页（固收第 18 页）；答案 PDF 第 88 页。

Source: question bank p. 86 (Fixed Income p. 18); solution PDF p. 88.

下列哪项按揭贷款特征最可能有利于贷款人？

Which of the following mortgage features most likely benefits the lender?

A. 无追索权贷款。
Non-recourse loan

B. 提前还款选择权。
Prepayment option

C. 提前还款罚金。
Prepayment penalty

正确答案 / Answer: C

利率下降时，借款人可能提前偿还旧贷款并以较低利率再融资，贷款人因此失去原有较高利息。提前还款罚金为贷款人提供补偿，并降低借款人立即提前还款的吸引力。

When rates fall, a borrower may repay the old mortgage and refinance at a lower rate, depriving the lender of the original higher interest income. A prepayment penalty compensates the lender and reduces the attraction of immediate refinancing.

错误选项为什么不成立 / Why the other options fail

A. 无追索权限制贷款人只能向抵押物求偿，通常更有利于借款人。

A. Non-recourse lending limits recovery to the collateral and generally benefits the borrower.

B. 提前还款选择权由借款人掌握，使贷款人现金流的金额和时间更不确定。

B. The borrower controls the prepayment option, which makes the lender's cash-flow timing and amounts less certain.

附录 A：公式速查

Appendix A. Formula reference

只在理解口径后使用本表。相同字母在不同题中可能指不同量，须以题干定义为准。

Use this reference after checking conventions. The same symbol may represent different quantities across questions; follow the question's definition.

每期票息 / Coupon per period

```
C = F x c / m
```

先统一付息频率。 / Align payment frequency.
对应原题 / Original questions: Q106, Q87

普通债券定价 / Plain-bond price

```
P = sum(C/(1+y/m)^t) + F/(1+y/m)^N
```

t = 1…N；N = 年数×m。 / N = years x m.
对应原题 / Original questions: Q106, Q87

即期利率定价 / Spot-rate price

```
P = sum[CF_t/(1+z_t)^t]
```

每笔现金流用匹配期限的利率。 / Match each cash flow with its spot rate.
对应原题 / Original questions: Q35, Q14

当期收益率 / Current yield

```
CY = Annual coupon / Price
```

忽略资本利得和再投资。 / Ignores capital gain and reinvestment.
对应原题 / Original questions: Q23

有效年收益率 / Effective annual yield

```
EAR = (1+APR_m/m)^m - 1
```

APR 与每期利率不同。 / APR is not the periodic rate.
对应原题 / Original questions: Q10

复利口径转换 / Compounding conversion

```
APR_n = n x [(1+APR_m/m)^(m/n)-1]
```

保持有效年收益相同。 / Preserve the same EAR.
对应原题 / Original questions: Q2, Q29

远期利率 / Forward rate

```
(1+z_(a+b))^(a+b) = (1+z_a)^a x (1+f_(a,b))^b
```

先画起点 a 与持续期 b。 / Identify start a and tenor b.
对应原题 / Original questions: Q59, Q73

修正久期 / Modified duration

```
D_mod = D_Mac / (1+y/m)
```

分母使用收益率。 / Use yield in the denominator.
对应原题 / Original questions: Q48, Q78

差分久期 / Difference duration

```
D ≈ (P_minus-P_plus)/(2 x P0 x delta_y)
```

冲击为单边幅度。 / delta_y is a one-sided shock.
对应原题 / Original questions: Q25, Q26, Q64, Q65, Q69, Q72

凸性 / Convexity

```
Conv ≈ (P_minus+P_plus-2P0)/(P0 x delta_y^2)
```

这里分母没有 2。 / No factor of 2 in this denominator.
对应原题 / Original questions: Q12

价格百分比变化 / Percentage price change

```
delta_P/P ≈ -D_mod x delta_y + 0.5 x Conv x delta_y^2
```

负凸性仍保留负号。 / Preserve negative convexity.
对应原题 / Original questions: Q4, Q75

金额久期 / Money duration

```
MoneyDur = D_mod x Full market value
```

使用全价或全市值。 / Use full price or full market value.
对应原题 / Original questions: Q54, Q84

基点价值 / PVBP

```
PVBP ≈ MoneyDur x 0.0001
```

单位与市值口径一致。 / Keep monetary units consistent.
对应原题 / Original questions: Q55

组合久期 / Portfolio duration

```
D_port = sum(w_i x D_i)
```

w_i 使用市值权重。 / Use market-value weights.
对应原题 / Original questions: Q56, Q78, Q86

净价全价 / Clean and full price

```
Full price = Clean price + Accrued interest
```

按净价报价、全价结算。 / Quote clean, settle full.
对应原题 / Original questions: Q67, Q68, Q84

票息终值 / Coupon future value

```
FV_coupon = sum[C_t x (1+r)^(H-t)]
```

持有期末票息指数为零。 / Last coupon has zero reinvestment time.
对应原题 / Original questions: Q66

持有期年化回报 / Horizon yield

```
r_H = [(P_sale+FV_coupon)/P_buy]^(1/H)-1
```

H 是实际持有期。 / H is the actual holding period.
对应原题 / Original questions: Q103

久期缺口 / Duration gap

```
Gap = D_Mac - H
```

零缺口对齐 MacDur。 / Zero gap matches MacDur.
对应原题 / Original questions: Q99

利息覆盖 / Interest coverage

```
EBIT coverage = EBIT / Interest expense
```

按题意区分 EBIT/EBITDA。 / Distinguish EBIT from EBITDA.
对应原题 / Original questions: Q51, Q21

预期损失 / Expected loss

```
EL = PD x (1-Recovery rate) x Exposure
```

回收率越高，其他条件相同损失越低。 / Higher recovery lowers loss, all else equal.
对应原题 / Original questions: Q30

贴现收益率 / Discount yield

```
d = (F-P)/F x Basis/Days
```

分母是兑付额 F。 / Use redemption value F.
对应原题 / Original questions: Q104

矩阵插值 / Matrix interpolation

```
y = yA + (T-TA)/(TB-TA) x (yB-yA)
```

先确认可比信用质量。 / Check comparable credit quality.
对应原题 / Original questions: Q60

> 利差速记：G = 政府债基准；I = 同币种掉期基准；Z = 整条即期曲线的恒定利差；OAS = 在模型中考虑嵌入期权后的利差。
> Spread reminder: G uses a government benchmark; I uses a same-currency swap benchmark; Z is a constant spread over the spot curve; OAS accounts for embedded options in a model.

附录 B：原资料提示与条件

Appendix B. Source notes and assumptions

以下15题在相应题目旁也有提示。题库答案与复算结果应分开看：原答案被保留，但文字瑕疵或缺失条件不应作为通用规则记忆。

The following 15 items also carry notes beside their solutions. Source answer keys are preserved, while wording defects or missing conditions should not be memorized as general rules.

Q004 | 答案 A / Answer A | p. 91

所附解答在中间一行将负数之和写成了正的 0.0577125，存在漏写负号；正确的价格变化率是 -0.0577125。原答案 A 及最终价格方向正确。

The supplied solution drops a minus sign in an intermediate equality and writes the sum as positive 0.0577125. The correct signed price change is -0.0577125. Its answer A and final price direction remain correct.

Q028 | 答案 B / Answer B | p. 21

本题按所附答案保留 B。Eurobond 历史上存在无记名形式，登记形式也受市场及规则影响；不要把本题“最可能”的答案记成“所有 Eurobond 必须记名”。

B is retained from the supplied key. Eurobonds have historically included bearer forms, and registration depends on market practices and rules; do not turn this “most likely” answer into a claim that every Eurobond must be registered.

Q035 | 答案 B / Answer B | p. 30

原表英文将 “3 years” 误拼为 “3yaers”，此处英文表格保留原文，中文按明确含义译为“3 年”。按原选项与所附解答，价格使用每 100 面值的口径。

The source table misspells “3 years” as “3yaers”; the English table preserves it and the Chinese translation gives its clear meaning. Consistent with the original options and supplied solution, the price is quoted per 100 of par.

Q040 | 答案 B / Answer B | p. 59

题目沿用“没有明确定义的内部收益率”的教材表述。可以针对假定现金流计算 YTM 或 YTC，但它们不能消除实际行权及现金流的不确定性。

The source uses the textbook phrase “no well-defined internal rate of return.” A YTM or YTC can still be calculated for assumed cash flows, but it does not remove uncertainty about exercise and realized cash flows.

Q044 | 答案 A / Answer A | p. 64

原选项 A 只写 key rate durations。准确理解应为“一组关键利率久期所反映的合计敏感度”，通常表述为其总和约等于有效久期。

Option A says only “key rate durations.” Read it as their aggregate sensitivity: the sum is approximately equal to effective duration under consistent assumptions.

Q051 | 答案 B / Answer B | p. 119

原题表格前有一个孤立的“<”符号，无对应的比较关系；本讲义将其视为排版残留，不纳入题干，表格数据保持原样。

A standalone “<” appears before the original table without a corresponding comparison. It is treated as a formatting remnant and excluded from the stem; all table data are preserved.

Q054 | 答案 A / Answer A | p. 77

原题仅写 Duration，未明确久期种类；按所附答案及选项，2.4 被作为修正久期使用。原英文选项 C 为“$24.000.”，疑似千位分隔符笔误，原文保留，不影响正确答案 A。

The stem states only “Duration.” The supplied answer and choices treat 2.4 as modified duration. Original option C reads “$24.000.”, apparently a separator typo; its wording is preserved and does not affect answer A.

Q059 | 答案 B / Answer B | p. 47

答案原页的推导有排版笔误，把部分增长因子写成“(1 + 1.035)^3”。正确应为“(1 + 0.035)^3”或“1.035^3”；原答案 B 及约 5.53% 的结果正确。

The source solution contains formula typos, including “(1 + 1.035)^3”. The correct factor is “(1 + 0.035)^3” or “1.035^3”. Its answer B and approximately 5.53% result remain correct.

Q062 | 答案 B / Answer B | p. 23

“only”是本题简化设定。现实中美国利率也可能通过汇率、英国利率、信用利差或发行人经营状况间接影响价格；不能将答案 B 理解为任何环境下都完全不受美国利率影响。

“Only” reflects the question’s simplified framework. In reality, US rates may indirectly affect sterling rates, exchange rates, credit spreads or the issuer’s business. Answer B should not be read as complete insulation from US rates in every setting.

Q083 | 答案 C / Answer C | p. 70

原题存在内部矛盾：在票息与YTM采用相同频率口径的通常定价下，债券1和3的票息等于YTM，应为平价；债券2票息低于YTM，应为折价，不是三者均为溢价。此处保留原题与原答案C，并按表中数据讲解。

The source is internally inconsistent. Under standard pricing with consistent coupon and YTM frequency conventions, Bonds 1 and 3 are at par because coupon equals YTM; Bond 2 is at a discount because coupon is below YTM. They are not all at a premium. The original wording and answer C are retained, and the explanation follows the table.

Q085 | 答案 C / Answer C | p. 65

“只有完全平坦才相等”是答案材料在标准附息债框架下的简化表述，不应当推广为所有数学情形的必要条件；单笔现金流等特殊情形可以例外。学习重点是两者的风险变量不同，不能仅凭“无期权”就默认相等。

The source's “only when flat” statement is a simplification for its standard coupon-bond framework, not a universal mathematical necessity; single-cash-flow cases can be exceptions. The key lesson is that the measures use different risk variables, so “option-free” alone does not justify treating them as equal.

Q092 | 答案 A / Answer A | p. 41

原答案选择A，并用108-107.43表示变动金额。按通常“新价格减旧价格”的定义，带符号变动是-$0.57，表示下跌；原选项未写负号，按变动绝对金额理解。题目只问债券价格变化，不能把它当作包含已收票息的总回报。

The source selects A and computes 108-107.43 as the amount changed. Under the usual new-price-minus-old-price convention, the signed change is -$0.57, a decrease. The unsigned option must be read as the absolute amount. This is a price change, not total return including coupons received.

Q099 | 答案 C / Answer C | p. 104

原题没有交代付息频率。原答案隐含每年付息（m=1）；通用关系是D_mac=D_mod×(1+y/m)。因此C依赖这一隐含口径，不能把7.4×1.09机械用于半年付息情形。

The question does not state coupon frequency. The supplied answer implicitly uses annual payments (m=1). The general relation is D_mac=D_mod×(1+y/m), so C relies on that convention; 7.4×1.09 should not be applied mechanically to a semiannual bond.

Q102 | 答案 B / Answer B | p. 105

此处的carrying value是本题恒定收益率收益分解口径，不是在判断各种会计分类下财报列示的账面金额；直接“售价减买价”仍可描述价格差，但不是本题所问的资本利得分项。

Here, carrying value refers to the constant-yield return-decomposition convention, not every possible financial-reporting measurement category. Sale price minus purchase price describes a price difference, but not the capital-gain component requested here.

Q106 | 答案 B / Answer B | p. 28

原答案B已复算确认。其对C的文字称“年付息、年折现、期初付息”得到122.53，但按该文字将12在第0、1、2年支付、100在第3年偿还，计算为123.5328，不能复现122.53。保留原选项数值，不把该干扰项解释当作正确公式。

Answer B has been independently verified. The source says C uses annual discounting and coupons at the beginning of each year to obtain 122.53. But paying 12 at times 0, 1, and 2 and repaying 100 at time 3 gives 123.5328, not 122.53. The original option is retained; its supplied rationale should not be learned as a valid formula.

附录 C：原题编号、答案与页码

Appendix C. Question, answer and page index

按原题编号排列，便于核对题库与答案文件。题目正文所在页是本讲义页码；原题 PDF 页码另列。

Sorted by original question number for cross-checking. Guide page locates the question in this document; source PDF page is shown separately.

原题 Q	答案 Key	知识章节 Chapter	讲义页 Guide p.	原题页 Source p.
001	C	01 债券是什么：条款、现金流与交易	10	69
002	C	02 债券估值与收益率：把时间和口径对齐	33	69
003	C	04 久期：选对指标，再算敏感度	62	69
004	A	05 凸性：为什么涨跌不对称	91	69
005	C	06 持有期回报、再投资风险与久期缺口	97	69
006	C	01 债券是什么：条款、现金流与交易	11	69
007	C	01 债券是什么：条款、现金流与交易	19	70
008	C	08 证券化、抵押贷款与结构化分层	123	70
009	A	08 证券化、抵押贷款与结构化分层	125	70
010	B	02 债券估值与收益率：把时间和口径对齐	34	70
011	B	08 证券化、抵押贷款与结构化分层	124	70
012	B	05 凸性：为什么涨跌不对称	90	70
013	C	03 利率曲线、远期利率与利差	45	71
014	B	02 债券估值与收益率：把时间和口径对齐	31	71
015	B	07 信用风险、偿债能力与清偿顺序	114	71
016	B	06 持有期回报、再投资风险与久期缺口	98	71
017	C	07 信用风险、偿债能力与清偿顺序	108	71
018	A	07 信用风险、偿债能力与清偿顺序	115	72
019	A	04 久期：选对指标，再算敏感度	66	72
020	A	07 信用风险、偿债能力与清偿顺序	109	72
021	B	07 信用风险、偿债能力与清偿顺序	118	72
022	B	01 债券是什么：条款、现金流与交易	20	72
023	B	02 债券估值与收益率：把时间和口径对齐	32	72
024	A	08 证券化、抵押贷款与结构化分层	126	73
025	B	04 久期：选对指标，再算敏感度	71	73
026	B	04 久期：选对指标，再算敏感度	72	73

原题 Q	答案 Key	知识章节 Chapter	讲义页 Guide p.	原题页 Source p.
027	B	04 久期：选对指标，再算敏感度	56	73
028	B	01 债券是什么：条款、现金流与交易	21	73
029	A	02 债券估值与收益率：把时间和口径对齐	35	74
030	B	07 信用风险、偿债能力与清偿顺序	117	74
031	B	02 债券估值与收益率：把时间和口径对齐	38	74
032	C	08 证券化、抵押贷款与结构化分层	130	74
033	A	04 久期：选对指标，再算敏感度	57	74
034	A	04 久期：选对指标，再算敏感度	58	74
035	B	02 债券估值与收益率：把时间和口径对齐	30	75
036	C	05 凸性：为什么涨跌不对称	85	75
037	C	05 凸性：为什么涨跌不对称	86	75
038	C	05 凸性：为什么涨跌不对称	87	75
039	B	03 利率曲线、远期利率与利差	46	76
040	B	04 久期：选对指标，再算敏感度	59	76
041	A	08 证券化、抵押贷款与结构化分层	131	76
042	B	03 利率曲线、远期利率与利差	50	76
043	C	01 债券是什么：条款、现金流与交易	22	76
044	A	04 久期：选对指标，再算敏感度	64	76
045	B	08 证券化、抵押贷款与结构化分层	127	77
046	B	04 久期：选对指标，再算敏感度	60	77
047	B	04 久期：选对指标，再算敏感度	61	77
048	B	04 久期：选对指标，再算敏感度	68	77
049	B	01 债券是什么：条款、现金流与交易	12	77
050	C	04 久期：选对指标，再算敏感度	69	78
051	B	07 信用风险、偿债能力与清偿顺序	119	78
052	A	04 久期：选对指标，再算敏感度	63	78
053	A	07 信用风险、偿债能力与清偿顺序	113	78
054	A	04 久期：选对指标，再算敏感度	77	79
055	A	04 久期：选对指标，再算敏感度	78	79
056	A	04 久期：选对指标，再算敏感度	80	79
057	C	07 信用风险、偿债能力与清偿顺序	111	79

原题 Q	答案 Key	知识章节 Chapter	讲义页 Guide p.	原题页 Source p.
058	C	07 信用风险、偿债能力与清偿顺序	120	79
059	B	03 利率曲线、远期利率与利差	47	80
060	A	02 债券估值与收益率：把时间和口径对齐	39	80
061	B	01 债券是什么：条款、现金流与交易	13	80
062	B	01 债券是什么：条款、现金流与交易	23	80
063	A	01 债券是什么：条款、现金流与交易	15	81
064	A	04 久期：选对指标，再算敏感度	73	81
065	B	04 久期：选对指标，再算敏感度	74	81
066	B	06 持有期回报、再投资风险与久期缺口	102	82
067	B	01 债券是什么：条款、现金流与交易	17	82
068	C	01 债券是什么：条款、现金流与交易	18	82
069	B	04 久期：选对指标，再算敏感度	75	82
070	A	01 债券是什么：条款、现金流与交易	14	82
071	C	08 证券化、抵押贷款与结构化分层	129	83
072	C	04 久期：选对指标，再算敏感度	76	83
073	A	03 利率曲线、远期利率与利差	48	83
074	B	07 信用风险、偿债能力与清偿顺序	110	83
075	B	05 凸性：为什么涨跌不对称	92	83
076	B	04 久期：选对指标，再算敏感度	67	84
077	B	01 债券是什么：条款、现金流与交易	24	84
078	A	04 久期：选对指标，再算敏感度	81	84
079	C	03 利率曲线、远期利率与利差	49	84
080	C	06 持有期回报、再投资风险与久期缺口	99	84
081	C	05 凸性：为什么涨跌不对称	88	85
082	C	02 债券估值与收益率：把时间和口径对齐	40	85
083	C	04 久期：选对指标，再算敏感度	70	85
084	B	04 久期：选对指标，再算敏感度	79	85
085	C	04 久期：选对指标，再算敏感度	65	86
086	B	04 久期：选对指标，再算敏感度	82	86
087	B	02 债券估值与收益率：把时间和口径对齐	29	86
088	C	08 证券化、抵押贷款与结构化分层	132	86

原题 Q	答案 Key	知识章节 Chapter	讲义页 Guide p.	原题页 Source p.
089	B	07 信用风险、偿债能力与清偿顺序	112	86
090	C	02 债券估值与收益率：把时间和口径对齐	36	87
091	C	06 持有期回报、再投资风险与久期缺口	100	87
092	A	02 债券估值与收益率：把时间和口径对齐	41	87
093	A	03 利率曲线、远期利率与利差	51	87
094	A	06 持有期回报、再投资风险与久期缺口	101	87
095	A	07 信用风险、偿债能力与清偿顺序	116	87
096	B	03 利率曲线、远期利率与利差	52	88
097	B	08 证券化、抵押贷款与结构化分层	128	88
098	B	01 债券是什么：条款、现金流与交易	16	88
099	C	06 持有期回报、再投资风险与久期缺口	104	88
100	B	05 凸性：为什么涨跌不对称	93	88
101	C	05 凸性：为什么涨跌不对称	94	88
102	B	06 持有期回报、再投资风险与久期缺口	105	89
103	C	06 持有期回报、再投资风险与久期缺口	103	89
104	B	02 债券估值与收益率：把时间和口径对齐	37	89
105	C	05 凸性：为什么涨跌不对称	89	89
106	B	02 债券估值与收益率：把时间和口径对齐	28	90

附录 D：复盘与掌握检查

Appendix D. Review and mastery checks

这不是新增题库，而是检查你能否解释已做原题的方法。每项都回到所列原题验证。

These are review prompts, not added practice questions. Use the cited originals to verify that you can explain the method.

主题 / Topic	掌握标准 / Mastery check	原题 / Originals
估值 / Valuation	能否同时对齐 N、C、每期 y，并说出溢价原因？ Can you align N, C and periodic y and explain why the bond is at a premium?	Q106, Q87
口径 / Convention	能否区分名义年率、有效年率、当期收益率？ Can you distinguish nominal annual yield, EAR and current yield?	Q2, Q10, Q23
远期 / Forward	能否画出起止时间并令两条路径终值相等？ Can you draw the interval and equate terminal values?	Q59, Q73, Q79
久期 / Duration	能否按冲击变量选久期，再检查单位？ Can you select duration by the shocked variable and check units?	Q25, Q48, Q55
组合 / Portfolio	能否逐只转换久期，再用市值加权？ Can you convert individual durations and market-value weight them?	Q56, Q78, Q86
凸性 / Convexity	能否保留负凸性符号，解释涨跌不对称？ Can you preserve negative convexity and explain asymmetry?	Q4, Q12, Q36
回报 / Return	能否分清卖价、票息终值和持有年数？ Can you distinguish sale price, coupon future value and years held?	Q66, Q92, Q103
信用 / Credit	能否分清违约概率、损失程度和债务顺位？ Can you separate default probability, loss severity and claim priority?	Q30, Q18, Q89
证券化 / Securitization	能否分清信用分层与时间分层、展期与收缩？ Can you distinguish credit/time tranching and extension/contraction?	Q45, Q97, Q41

最后一轮复习：把仍会做错或只能猜对的原题编号写下来，按五类错因归类。先修正方法，再重做同组题；不要只背 A、B、C。

For the final review, record the IDs you still miss or guess correctly and classify the reason. Correct the method, then repeat questions in the same group; do not memorize answer letters alone.